BS EN IEC 61851-1:2019

Incorporating corrigendum February 2020

BSI Standards Publication

Electric vehicle conductive charging system

Part 1: General requirements

National foreword

This British Standard is the UK implementation of EN IEC 61851-1:2019. It is identical to IEC 61851-1:2017. Together with BS EN 62752:2016, it supersedes BS EN 61851-1:2011 which is withdrawn.

The UK participation in its preparation was entrusted to Technical Committee PEL/69, Electric vehicles.

A list of organizations represented on this committee can be obtained on request to its secretary.

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Amendments/corrigenda issued since publication

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EUROPEAN STANDARD

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EUROPÄISCHE NORM

EN IEC 61851-1

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English Version

Electric vehicle conductive charging system - Part 1: General requirements (IEC 61851-1:2017)

Système de charge conductive pour véhicules électriques - Partie 1: Exigences générales (IEC 61851-1:2017)

Konduktive Ladesysteme für Elektrofahrzeuge - Teil 1: Allgemeine Anforderungen (IEC 61851-1:2017)

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Ref. No. EN IEC 61851-1:2019 E

European foreword

The text of document 69/436/FDIS, future edition 3 of IEC 61851-1, prepared by IEC/TC 69 "Electric road vehicles and electric industrial trucks" was submitted to the IEC-CENELEC parallel vote and approved by CENELEC as EN IEC 61851-1:2019.

The following dates are fixed:

• latest date by which the document has to be implemented at national level by publication of an identical national standard or by endorsement	(dop)	2020-01-05
• latest date by which the national standards conflicting with the document have to be withdrawn	(dow)	2022-07-05

This document supersedes EN 61851-1:2011.

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This document has been prepared under a mandate given to CENELEC by the European Commission and the European Free Trade Association, and supports essential requirements of EU Directive(s).

For the relationship with EU Directive(s) see informative Annex ZZ, which is an integral part of this document.

Endorsement notice

The text of the International Standard IEC 61851-1:2017 was approved by CENELEC as a European Standard without any modification.

In the official version, for Bibliography, the following notes have to be added for the standards indicated:

IEC 62053-21:2003	NOTE Harmonized as EN 62053-21:2003 (not modified)
ISO 4628-3:2016	NOTE Harmonized as EN ISO 4628-3:2016 (not modified)
IEC 60063:2015	NOTE Harmonized as EN 60063:2015 (not modified)
IEC 60068-2-2	NOTE Harmonized as EN 60068-2-2
IEC 60068-2-5:2010	NOTE Harmonized as EN 60068-2-5:2011 (not modified)
IEC 60068-2-6:2007	NOTE Harmonized as EN 60068-2-6:2008 (not modified)
IEC 60068-2-14:2009	NOTE Harmonized as EN 60068-2-14:2009 (not modified)
IEC 60068-2-27:2008	NOTE Harmonized as EN 60068-2-27:2009 (not modified)
IEC 60068-2-52:1996	NOTE Harmonized as EN 60068-2-52:1996 (not modified)
IEC 60068-2-53:2010	NOTE Harmonized as EN 60068-2-53:2010 (not modified)
IEC 60068-2-75	NOTE Harmonized as EN 60068-2-75
IEC 60364-6:2016	NOTE Harmonized as HD 60364-6:2016 (not modified)
IEC 60947-1:2007	NOTE Harmonized as EN 60947-1:2007 (not modified)

IEC 60947-1:2007/A1:2010	NOTE Harmonized as EN 60947-1:2007/A1:2011 (not modified)
IEC 60947-1:2007/A2:2014	NOTE Harmonized as EN 60947-1:2007/A2:2014 (not modified)
IEC 60947-6-1:2005	NOTE Harmonized as EN 60947-6-1:2005 (not modified)
IEC 61140	NOTE Harmonized as EN 61140
IEC 61439-1:2011	NOTE Harmonized as EN 61439-1:2011 (not modified)
IEC 61540	NOTE Harmonized as HD 639 S1
IEC 61558-1:2005	NOTE Harmonized as EN 61558-1:2005 (not modified)
IEC 61558-1:2005/A1:2009	NOTE Harmonized as EN 61558-1:2005/A1:2009 (not modified)
IEC 61558-2-4:2009	NOTE Harmonized as EN 61558-2-4:2009 (not modified)
IEC 61558-2-12:2011	NOTE Harmonized as EN 61558-2-12:2011 (not modified)
IEC 61558-2-16:2009	NOTE Harmonized as EN 61558-2-16:2009 (not modified)
IEC 61558-2-16:2009/A1:2013	NOTE Harmonized as EN 61558-2-16:2009/A1:2013 (not modified)
IEC 61851-21-2	NOTE Harmonized as EN 61851-21-2[1]
IEC 61980-1	NOTE Harmonized as EN 61980-1[2]
IEC 62262:2002	NOTE Harmonized as EN 62262:2002 (not modified)
ISO/IEC 15118 (series)	NOTE Harmonized as EN ISO 15118 (series)
ISO 13849-1:2015	NOTE Harmonized as EN ISO 13849-1:2015 (not modified)
ISO 15118-3	NOTE Harmonized as EN ISO 15118-3

[1] Under preparation. Stage at time of publication: FprEN 61851-21-2

[2] Under preparation. Stage at time of publication: prEN 61980-1

Annex ZA
(normative)

Normative references to international publications with their corresponding European publications

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

NOTE 1 Where an International Publication has been modified by common modifications, indicated by (mod), the relevant EN/HD applies.

NOTE 2 Up-to-date information on the latest versions of the European Standards listed in this annex is available here: www.cenelec.eu.

Publication	Year	Title	EN/HD	Year
IEC 60038 (mod)	-	IEC standard voltages	EN 60038	2011
IEC 60068-2-1	-	Environmental testing - Part 2-1: Tests - Test A: Cold	EN 60068-2-1	2007
IEC 60068-2-78	-	Environmental testing - Part 2-78: Tests - Test Cab: Damp heat, steady state	EN 60068-2-78	2013
IEC 60309-1	-	Plugs, socket-outlets and couplers for industrial purposes - Part 1: General requirements	EN 60309-1	1999
			+ A1 (mod)	2007
			+ A2	2012
IEC 60309-2	-	Plugs, socket-outlets and couplers for industrial purposes - Part 2: Dimensional interchangeability requirements for pin and contact-tube accessories	EN 60309-2	1999
			+ A1 (mod)	2007
			+ A2	2012
IEC 60364-4-41 (mod)	-	Low-voltage electrical installations - Part 4-41: Protection for safety - Protection against electric shock	HD 60364-4-41	2017
			+ A11	2017
IEC 60364-5-54	-	Low-voltage electrical installations - Part 5-54: Selection and erection of electrical equipment - Earthing arrangements and protective conductors	HD 60364-5-54	2011
			+ A11	2017
IEC 60529	2013[3]	Degrees of protection provided by enclosures (IP Code)	-	-

[3] Dated as no equivalent European standard exists.

Publication	Year	Title	EN/HD	Year
IEC 60664-1	2007	Insulation coordination for equipment within low-voltage systems - Part 1: Principles, requirements and tests	EN 60664-1	2007
IEC 60884-1	2002[3]	Plugs and socket-outlets for household and similar purposes -- Part 1: General requirements	-	-
IEC 60898	series	Electrical accessories - Circuit-breakers for overcurrent protection for household and similar installations	EN 60898	series
IEC 60898-1 (mod)	-	Electrical accessories - Circuit-breakers for overcurrent protection for household and similar installations - Part 1: Circuit-breakers for a.c. operation	EN 60898-1	2019
IEC 60947-2	-	Low-voltage switchgear and controlgear - Part 2: Circuit-breakers	EN 60947-2	2017
IEC 60947-3	-	Low-voltage switchgear and controlgear - Part 3: Switches, disconnectors, switch-disconnectors and fuse-combination units	EN 60947-3	2009
			+ A1	2012
			+ A2	2015
IEC 60947-4-1	-	Low voltage switchgear and controlgear – Part 4-1: Contactors and motorstarters – Electromechanical contactors and motor-starters	EN IEC 60947-4-1	2019
IEC 60947-6-2	-	Low-voltage switchgear and controlgear - Part 6-2: Multiple function equipment - Control and protective switching devices (or equipment) (CPS)	EN 60947-6-2	2003
			+ A1	2007
IEC 60950-1 (mod)	2005	Information technology equipment - Safety - Part 1: General requirements	EN 60950-1	2006
-	-		+ A11	2009
-	-		+ A12	2011
-	-		+ AC	2011
IEC 60990	-	Methods of measurement of touch current and protective conductor current	EN 60990	2016
IEC 61008-1 (mod)	-	Residual current operated circuit-breakers without integral overcurrent protection for household and similar uses (RCCBs) - Part 1: General rules	EN 61008-1	2012
			+ A1 (mod)	2014
			+ A2 (mod)	2014
			+ A11	2015
			+ A12	2017

Publication	Year	Title	EN/HD	Year
IEC 61009-1 (mod)	-	Residual current operated circuit-breakers with integral overcurrent protection for household and similar uses (RCBOs) - Part 1: General rules	EN 61009-1	2012
			+ A1 (mod)	2014
			+ A2 (mod)	2014
			+ A11	2015
			+ A12	2016
IEC 61180	-	High-voltage test techniques for low-voltage equipment - Definitions, test and procedure requirements, test equipment	EN 61180	2016
IEC 61316	1999	Industrial cable reels	EN 61316	1999
IEC/TS 61439-7	2014	Low-voltage switchgear and controlgear assemblies - Part 7: Assemblies for specific applications such as marinas, camping sites, market squares, electric vehicles charging stations	-	-
IEC 61508	series	Functional safety of electrical/electronic/programmable electronic safety-related systems	EN 61508	series
IEC 61558-1	-	Safety of power transformers, power supplies, reactors and similar products – Part 1: General requirements and tests	EN 61558-1[4]	—
IEC 61558-2-4	-	Safety of transformers, reactors, power supply units and similar products for supply voltages up to 1 100 V - Part 2-4: Particular requirements and tests for isolating transformers and power supply units incorporating isolating transformers	EN 61558-2-4	2009
IEC 61810-1	-	Electromechanical elementary relays - Part 1: General and safety requirements	EN 61810-1	2015
IEC 61851	series	Electric vehicle conductive charging system	EN IEC 61851	series
IEC 61851-23	2014	Electric vehicle conductive charging system - Part 23: DC electric vehicle charging station	EN 61851-23	2014
IEC 61851-24	2014	Electric vehicle conductive charging system - Part 24: Digital communication between a d.c. EV charging station and an electric vehicle for control of d.c. charging	EN 61851-24	2014
IEC 62196	series	Plugs, socket-outlets, vehicle connectors and vehicle inlets - Conductive charging of electric vehicles	EN 62196	series

[4] Under preparation. Stage at time of publication: FprEN 61558-1:2017

Publication	Year	Title	EN/HD	Year
IEC 62196-1 (mod)	2014	Plugs, socket-outlets, vehicle connectors and vehicle inlets - Conductive charging of electric vehicles - Part 1: General requirements	EN 62196-1	2014
IEC 62196-2	2016	Plugs, socket-outlets, vehicle connectors and vehicle inlets - Conductive charging of electric vehicles - Part 2: Dimensional compatibility and interchangeability requirements for a.c. pin and contact-tube accessories	EN 62196-2	2017
IEC 62196-3	2014	Plugs, socket-outlets, vehicle connectors and vehicle inlets - Conductive charging of electric vehicles - Part 3: Dimensional compatibility and interchangeability requirements for d.c. and a.c./d.c. pin and contact-tube vehicle couplers	EN 62196-3	2014
IEC 62262	-	Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)	EN 62262	2002
IEC 62423 (mod)	-	Type F and type B residual current operated circuit-breakers with and without integral overcurrent protection for household and similar uses	EN 62423	2012
IEC 62752	-	In-cable control and protection device for mode 2 charging of electric road vehicles (IC-CPD)	EN 62752	2016
ISO 17409	2015	Electrically propelled road vehicles - Connection to an external electric power supply - Safety requirements	EN ISO 17409	2017

Annex ZZ
(informative)

Relationship between this European standard and the safety objectives of Directive 2014/35/EU [2014 OJ L96] aimed to be covered

This European Standard has been prepared under a Commission's standardization request relating to harmonized standards in the field of the Low Voltage Directive, M/511, to provide one voluntary means of conforming to safety objectives of Directive 2014/35/EU of the European Parliament and of the Council of 26 February 2014 on the harmonization of the laws of the Member States relating to the making available on the market of electrical equipment designed for use within certain voltage limits [2014 OJ L96].

Once this standard is cited in the Official Journal of the European Union under that Directive, compliance with the normative clauses of this standard given in Table ZZ.1 confers, within the limits of the scope of this standard, a presumption of conformity with the corresponding safety objectives of that Directive, and associated EFTA regulations.

Table ZZ.1 — Correspondence between this European standard and Annex I of Directive 2014/35/EU [2014 OJ L96]

Safety objectives of Directive 2014/35/EU	Clause(s) / subclause(s) of this EN	Remarks / Notes
1. General Conditions		
(a) the essential characteristics, the recognition and observance of which will ensure that electrical equipment will be used safely and in applications for which it was made, shall be marked on the electrical equipment, or, if this is not possible, on an accompanying document	1 Scope 2 Normative References 3 Terms and Definitions 5 Classification 17 Marking and instructions	
(b) the electrical equipment, together with its component parts, shall be made in such a way as to ensure that it can be safely and properly assembled and connected	4 General requirement 6.3.1.2 Continuous continuity checking of the protective conductor 6.3.1.3 Verification that the EV is properly connected to the EV supply equipment 6.3.2.3 Intentional and unintentional disconnection of the vehicle connector and/or the EV plug 9 Conductive electrical interface requirements 10 Requirements for adaptors 11 Cable assembly requirements (incl. chapter12)	

Safety objectives of Directive 2014/35/EU	Clause(s) / subclause(s) of this EN	Remarks / Notes
(c) the electrical equipment shall be so designed and manufactured as to ensure that protection against the hazards set out in points 2 and 3 is assured, providing that the equipment is used in applications for which it was made and is adequately maintained	Details see points 2 and 3	
2. Protection against hazards arising from the electrical equipment		
(a) persons and domestic animals are adequately protected against the danger of physical injury or other harm which might be caused by direct or indirect contact	8 Protection against electric shock 9 Conductive electrical interface requirements 13.4 IP Degree 13.6 Touch current 15 Automatic reclosing of protective devices	
(b) temperatures, arcs or radiation which would cause a danger, are not produced	6.3.2.3 Intentional and unintentional disconnection of the vehicle connector and/or the EV plug 9 Conductive electrical interface requirements 13 EV supply equipment constructional requirements and tests 14 Overload and short-circuit protection	
(c) persons, domestic animals and property are adequately protected against non-electrical dangers caused by the electrical equipment which are revealed by experience	13 EV supply equipment constructional requirements and tests	
(d) the insulation is suitable for foreseeable conditions	13.5 Insulation resistance 13.7 Dielectric withstand voltage	
3. Protection against hazards which may be caused by external influences on the electrical equipment		
(a) meets the expected mechanical requirements in such a way that persons, domestic animals and property are not endangered	4 General requirements 13.11 Mechanical strength	

Safety objectives of Directive 2014/35/EU	Clause(s) / subclause(s) of this EN	Remarks / Notes
(b) is resistant to non-mechanical influences in expected environmental conditions, in such a way that persons, domestic animals and property are not endangered	13.3 Clearances and creepage distances 13.4 IP degrees 13.6 Touch current 13.7 Dielectric withstand voltage 13.8 Temperature rise 13.9 Damp heat functional test 13.10 Minimum temperature functional test	
(c) does not endanger persons, domestic animals and property in foreseeable conditions of overload	11 Cable assembly requirements 14 Overload and short-circuit protection 14.2 Overload protection of the cable assembly 14.3 Short-circuit protection of the charging cable	

WARNING 1 — Presumption of conformity stays valid only as long as a reference to this European standard is maintained in the list published in the Official Journal of the European Union. Users of this standard should consult frequently the latest list published in the Official Journal of the European Union.

WARNING 2 — Other Union legislation may be applicable to the product(s) falling within the scope of this standard.

IEC 61851-1

Edition 3.0 2017-02

INTERNATIONAL STANDARD

NORME INTERNATIONALE

colour inside

Electric vehicle conductive charging system – Part 1: General requirements

Système de charge conductive pour véhicules électriques – Partie 1: Exigences générales

INTERNATIONAL ELECTROTECHNICAL COMMISSION

COMMISSION ELECTROTECHNIQUE INTERNATIONALE

ICS 43.120

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ELECTRIC VEHICLE CONDUCTIVE CHARGING SYSTEM –

Part 1: General requirements

FOREWORD

1) The International Electrotechnical Commission (IEC) is a worldwide organization for standardization comprising all national electrotechnical committees (IEC National Committees). The object of IEC is to promote international co-operation on all questions concerning standardization in the electrical and electronic fields. To this end and in addition to other activities, IEC publishes International Standards, Technical Specifications, Technical Reports, Publicly Available Specifications (PAS) and Guides (hereafter referred to as "IEC Publication(s)"). Their preparation is entrusted to technical committees; any IEC National Committee interested in the subject dealt with may participate in this preparatory work. International, governmental and non-governmental organizations liaising with the IEC also participate in this preparation. IEC collaborates closely with the International Organization for Standardization (ISO) in accordance with conditions determined by agreement between the two organizations.
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International Standard IEC 61851-1 has been prepared by IEC technical committee 69: Electric road vehicles and electric industrial trucks.

This third edition cancels and replaces the second edition published in 2010. It constitutes a technical revision.

This edition includes the following significant technical changes with respect to the previous edition:

a) The contents of IEC 61851-1:2010 have been re-ordered. Numbering of clauses has changed as new clauses were introduced and some contents moved for easy reading. The following lines give an insight to the new ordering in addition to the main technical changes.
b) All requirements from IEC 61851-22 have been moved to this standard, as work on IEC 61851-22 has ceased.

c) Any requirements that concern EMC have been removed from the text and are expected to be part of the future version of 61851-21-2[1].

d) Clause 4 contains the original text from IEC 61851-1:2010 and all general requirements from Clause 6 of IEC 61851-1:2010.

e) Clause 5 has been introduced to provide classifications for EV supply equipment.

f) Previous general requirements of Clause 6 have been integrated into Clause 4. Clause 6 contains all Mode descriptions and control requirements. Specific requirements for the combined use of AC and DC on the same contacts are included.

g) Clause 9 is derived from previous Clause 8. Adaptation of the description of DC accessories to allow for the DC charging modes that have only recently been proposed by industry and based on the standards IEC 61851-23, IEC 61851-24 as well as IEC 62196-1, IEC 62196-2 and IEC 62196-3. Information and tables contained in the IEC 62196 series standards have been removed from this standard.

h) Clause 10 specifically concerns the requirements for adaptors, initially in Clause 6.

i) Clause 11 includes new requirements for the protection of the cable.

j) Specific requirements for equipment that is not covered in the IEC 62752 remain in the present document.

k) Previous Clause 11 is now treated in Clauses 12 to 13. The requirements in 61851-1 cover the EV supply equipment of both mode 2 and mode 3 types, with the exception in-cable control and protection devices for mode 2 charging of electric road vehicles (IC-CPD) which are covered by IEC 62752.

l) Clause 14 gives requirements on automatic reclosing of protection equipment.

m) Clause 16 gives requirements for the marking of equipment and the contents of the installation and user manual. This makes specific mention of the need to maintain coherence with the standards for the fixed installation. It also contains an important text on the markings for temperature ratings.

n) Annex A has been reviewed to introduce complete sequences and tests and to make the exact cycles explicit. Annex A in this edition supersedes IEC TS 62763 (Edition 1).

o) Annex B is normative and has requirements for proximity circuits with and without current coding.

p) Previous Annex C has been removed and informative descriptions of pilot function and proximity function implementations initially in Annex B are moved to Annex C.

q) New informative Annex D describing an alternative pilot function system has been introduced.

r) Dimensional requirements for free space to be left around socket-outlets used for EV energy supply are given in the informative Annex E.

s) The inclusion of protection devices within the EV supply equipment could, in some cases, contribute to the protection against electric shock as required by the installation. This is covered by the information required for the installation of EV supply equipment in Clause 16 (Marking).

The text of this standard is based on the following documents:

FDIS	Report on voting
69/436/FDIS	69/469/RVD

Full information on the voting for the approval of this standard can be found in the report on voting indicated in the above table.

This publication has been drafted in accordance with the ISO/IEC Directives, Part 2.

1 Under preparation.

A list of all parts of the IEC 61851 series, under the general title *Electric vehicle conductive charging system* can be found on the IEC website.

In this standard, the following print types are used:

– *test specifications and instructions regarding application of Part 1: italic type.*

– notes: smaller roman type.

The committee has decided that the contents of this publication will remain unchanged until the stability date indicated on the IEC website under "http://webstore.iec.ch" in the data related to the specific publication. At this date, the publication will be

- reconfirmed,
- withdrawn,
- replaced by a revised edition, or
- amended.

IMPORTANT – The 'colour inside' logo on the cover page of this publication indicates that it contains colours which are considered to be useful for the correct understanding of its contents. Users should therefore print this document using a colour printer.

INTRODUCTION

This standard is the first part of the IEC 61851 series of standards that gives the general requirements for the supply[2] of electric energy to Electric road vehicles[3]. It is to be noted that the vehicle and the EV supply equipment[2] make up a complete system that is covered by a number of IEC and ISO standards.

IEC 61851 covers the mechanical, electrical, communications, EMC and performance requirements for EV supply equipment used to charge electric vehicles, including light electric vehicles.

IEC 61851 is divided into several parts as follows:

– *Part 1: General Requirements*,
  This document gives the general requirements that serve as a basis for all the subsequent standards in the series. It includes the requirements for AC EV supply equipment.

– *Part 21-1*[4]*: Electric vehicle onboard charger EMC requirements for conductive connection to an AC/DC supply*. This part will cover requirements for EMC onboard the vehicle.

– *Part 21-2*[5]*: EMC requirements for OFF board electric vehicle charging systems*. This part will cover all requirements for AC and DC EV supply equipment. EMC requirements for wireless power transfer systems (WPT) will not be included.

– *Part 23: DC electric vehicle charging station* (2014).This part covers the requirements for DC charging stations both permanently wired and cable and plug connected.

– *Part 24: Digital communication between a d.c. EV charging station and an electric vehicle for control of d.c. charging* (2014). This part provides the requirements for communication between the vehicle and the DC charging stations of Part 23.

IEC 61851-3 subseries is under development and is intended to cover EV supply equipment with a DC output not exceeding 120 V where reinforced or double insulation or class III is used as the principal means of protection against electric shock (information on scope as available on 3/2016).

– *Part 3-1: Electric vehicles conductive power supply system – Part 3-1: General Requirements for Light Electric Vehicles (LEV) AC and DC conductive power supply systems.*

– *Part 3-2: Electric vehicles conductive power supply system – Part 3-2: Requirements for Light Electric Vehicles (LEV) DC off-board conductive power supply systems.*

– *Part 3-3: Electric vehicles conductive power supply system – Part 3-3: Requirements for Light Electric Vehicles (LEV) battery swap systems.*

– *Part 3-4: Electric vehicles conductive power supply system – Part 3-4: Requirements for Light Electric Vehicles (LEV) communication.*

– *Part 3-5: Electric vehicles conductive power supply system – Part 3-5: Requirements for Light Electric Vehicles communication – Pre-defined communication parameters.*

– *Part 3-6: Electric vehicles conductive power supply system – Part 3-6: Requirements for Light Electric Vehicles communication –Voltage converter unit.*

– *Part 3-7: Electric vehicles conductive power supply system – Part 3-7: Requirements for Light Electric Vehicles communication – Battery system.*

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2 The term “supply or electric energy” is used to designate energy flow to and from the electric vehicle. The term “charging” used in the title is also used to designate such energy flow.

3 The reader is advised to refer to the definitions clause 3 for this and all subsequent terms that are used in this document.

4 Under preparation.

5 Under preparation.

Documents directly related to the present document:

- ISO 17409:2015, *Electrically propelled road vehicles – Connection to an external electric power supply – Safety requirements.*

  This document gives requirements for electric vehicle that is to be connected to the EV supply equipment. It covers all the classes of vehicles that are in the scope of ISO/TC 22/SC 37.

- IEC 62752:2016, *In-cable control and protection device for mode 2 charging of electric road vehicles (IC-CPD).*

  This product standard gives the requirements for Mode 2 cable assemblies that include supplementary protective and control devices that allow the safe connection of a vehicle to a mains socket-outlet of an installation.

- ISO/IEC 15118 (all parts*), Road vehicles — Vehicle to grid communication interface*

  This series of documents gives the description and the requirements for high level data communication between the EV and the EV supply equipment.

Requirements for wireless power transfer systems are given in IEC 61980-1.

ELECTRIC VEHICLE CONDUCTIVE CHARGING SYSTEM –

Part 1: General requirements

1 Scope

This part of IEC 61851 applies to EV supply equipment for charging electric road vehicles, with a rated supply voltage up to 1 000 V AC or up to 1 500 V DC. and a rated output voltage up to 1 000 V AC. or up to 1 500 V DC.

Electric road vehicles (EV) cover all road vehicles, including plug-in hybrid road vehicles (PHEV), that derive all or part of their energy from on-board rechargeable energy storage systems (RESS).

This standard also applies to EV supply equipment supplied from on-site storage systems (e.g. buffer batteries).

The aspects covered in this standard include:

- the characteristics and operating conditions of the EV supply equipment;
- the specification of the connection between the EV supply equipment and the EV;
- the requirements for electrical safety for the EV supply equipment.

Additional requirements may apply to equipment designed for specific environments or conditions, for example:

- EV supply equipment located in hazardous areas where flammable gas or vapour and/or combustible materials, fuels or other combustible, or explosive materials are present;
- EV supply equipment designed to be installed at an altitude of more than 2 000 m;
- EV supply equipment intended to be used on board on ships;

Requirements for electrical devices and components used in EV supply equipment are not included in this standard and are covered by their specific product standards.

EMC requirements for EV supply equipment are expected to be covered in the future IEC 61851-21-2[6].

Requirements for bi-directional energy transfer are under consideration and are not in this edition of IEC 61851-1.

This standard does not apply to:

- safety aspects related to maintenance;
- charging of trolley buses, rail vehicles, industrial trucks and vehicles designed primarily for use off-road;
- equipment on the EV;
- EMC requirements for equipment on the EV while connected, which are covered in IEC 61851-21-1;
- Charging RESS off board of the EV;

6 Under consideration.

- DC EV supply equipment that relies specifically on double/reinforced insulation or class III protection against electric shock. See IEC 61851-23 or the future IEC 61851-3 series.

The IEC 61851 series covers all EV supply equipment with the exception of in-cable control and protection devices for mode 2 charging of electric road vehicles (IC-CPD) which are covered by IEC 62752.

2 Normative references

The following documents are referred to in the text in such a way that some or all of their content constitutes requirements of this document. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

IEC 60038, *IEC standard voltages*

IEC 60068-2-1, *Environmental testing – Part 2-1: Tests – Test A: Cold*

IEC 60068-2-78, *Environmental testing – Part 2-78: Tests – Test Cab: Damp heat, steady state*

IEC 60309-1, *Plugs, socket-outlets and couplers for industrial purposes – Part 1: General requirements*

IEC 60309-2, *Plugs, socket-outlets and couplers for industrial purposes – Part 2: Dimensional interchangeability requirements for pin and contact-tube accessories*

IEC 60364-4-41, *Low-voltage electrical installations – Part 4-41: Protection for safety – Protection against electric shock*

IEC 60364-5-54, *Low-voltage electrical installations – Part 5-54: Selection and erection of electrical equipment – Earthing arrangements and protective conductors*

IEC 60529, *Degrees of protection provided by enclosures (IP Code)*

IEC 60664-1:2007, *Insulation coordination for equipment within low-voltage systems – Part 1: Principles, requirements and tests*

IEC 60884-1, *Plugs and socket-outlets for household and similar purposes – Part 1: General requirements*

IEC 60898 (all parts), *Circuit-breakers for overcurrent protection for household and similar installations*

IEC 60898-1, *Electrical accessories – Circuit-breakers for overcurrent protection for household and similar installations – Part 1: Circuit-breakers for a.c. operation*

IEC 60947-2, *Low-voltage switchgear and controlgear – Part 2: Circuit-breakers*

IEC 60947-3, *Low-voltage switchgear and controlgear – Part 3: Switches, disconnectors, switch-disconnectors and fuse-combination units*

IEC 60947-4-1, *Low voltage switchgear and controlgear – Part 4-1: Contactors and motor-starters – Electromechanical contactors and motor-starters*

IEC 60947-6-2, *Low-voltage switchgear and controlgear – Part 6-2: Multiple function equipment – Control and protective switching devices (or equipment) (CPS)*

IEC 60950-1:2005, *Information technology equipment – Safety – Part 1: General requirements*

IEC 60990, *Methods of measurement of touch current and protective conductor current*

IEC 61008-1, *Residual current operated circuit-breakers without integral overcurrent protection for household and similar uses (RCCBs) – Part 1: General rules*

IEC 61009-1, *Residual current operated circuit-breakers with integral overcurrent protection for household and similar uses (RCBOs) – Part 1: General rules*

IEC 61180, *High-voltage test techniques for low-voltage equipment – Definitions, test and procedure requirements, test equipment*

IEC 61316:1999, *Industrial cable reels*

IEC TS 61439-7:2014, *Low-voltage switchgear and controlgear assemblies – Part 7: Assemblies for specific applications such as marinas, camping sites, market squares, electric vehicles charging stations*

IEC 61508 (all parts), *Functional safety of electrical/electronic/programmable electronic safety-related systems*

IEC 61558-1, *Safety of power transformers, power supplies, reactors and similar products – Part 1: General requirements and tests*

IEC 61558-2-4, *Safety of transformers, reactors, power supply units and similar products for supply voltages up to 1 100 V – Part 2-4: Particular requirements and tests for isolating transformers and power supply units incorporating isolating transformers*

IEC 61810-1, *Electromechanical elementary relays – Part 1: General and safety requirements*

IEC 61851 (all parts), *Electric vehicle conductive charging system*

IEC 61851-23:2014, *Electric vehicle conductive charging system – Part 23: DC electric vehicle charging station*

IEC 61851-24:2014, *Electric vehicle conductive charging system – Part 24: Digital communication between a d.c. EV charging station and an electric vehicle for control of d.c. charging*

IEC 62196 (all parts), *Plugs, socket-outlets, vehicle connectors and vehicle inlets – Conductive charging of electric vehicles*

IEC 62196-1:2014, *Plugs, socket-outlets, vehicle connectors and vehicle inlets – Conductive charging of electric vehicles – Part 1: General requirements*

IEC 62196-2:2016, *Plugs, socket-outlets, vehicle connectors and vehicle inlets – Conductive charging of electric vehicles – Part 2: Dimensional compatibility and interchangeability requirements for a.c.pin and contact-tube accessories*

IEC 62196-3:2014, *Plugs, socket-outlets, vehicle connectors and vehicle inlets – Conductive charging of electric vehicles – Part 3: Dimensional compatibility and interchangeability requirements for d.c. and a.c./d.c. pin and contact-tube vehicle couplers*

IEC 62262, *Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)*

IEC 62423, *Type F and type B residual current operated circuit-breakers with and without integral overcurrent protection for household and similar uses*

IEC 62752, *In-Cable Control and Protection Device for mode 2 charging of electric road vehicles (IC-CPD)*

ISO 17409:2015, *Electrically propelled road vehicles – Connection to an external electric power supply – Safety requirements*

3 Terms and definitions

For the purposes of this document, the following terms and definitions apply.

ISO and IEC maintain terminological databases for use in standardization at the following addresses:

- IEC Electropedia: available at http://www.electropedia.org/
- ISO Online browsing platform: available at http://www.iso.org/obp

3.1 Electric supply equipment

3.1.1
EV supply equipment
equipment or a combination of equipment, providing dedicated functions to supply electric energy from a fixed electrical installation or supply network to an EV for the purpose of charging

EXAMPLE 1: For Mode 3 case B, the EV supply equipment consists of the charging station and the cable assembly.

EXAMPLE 2: For Mode 3 case C, the EV supply equipment consists of the charging station with its cable assembly.

3.1.2
AC EV supply equipment
EV supply equipment that supplies alternating current to an EV

3.1.3
DC EV supply equipment
EV supply equipment that supplies direct current to an EV

3.1.4
EV charging system
complete system including the EV supply equipment and the EV functions that are required to supply electric energy to an EV for the purpose of charging

3.1.5
EV charging station
stationary part of EV supply equipment connected to the supply network

Note 1 to entry: For case C, the cable assembly is part of the EV charging station.

3.1.6
DC EV charging station
EV charging station that supplies direct current to an EV

3.1.7
AC EV charging station
EV charging station that supplies alternating current to an EV

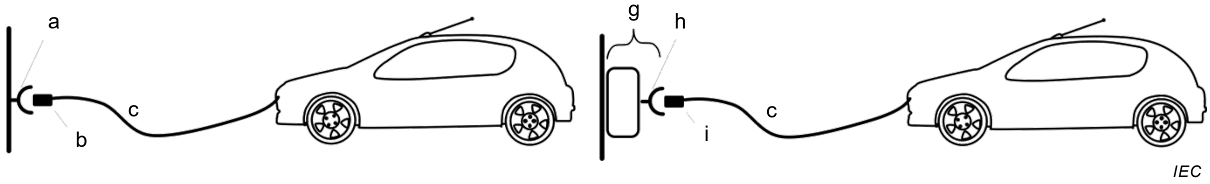
3.1.8
charging
all functions necessary to condition voltage and/or current provided by the AC or DC supply network to assure the supply of electric energy to the RESS

3.1.9
charging mode
method for connection of an EV to the supply network to supply energy to the vehicle

Note 1 to entry: Mode 1, mode 2, mode 3 and mode 4 are described in Clause 6.

3.1.10
case A
connection of an EV to the supply network with a plug and cable permanently attached to the EV

SEE: Figure 1

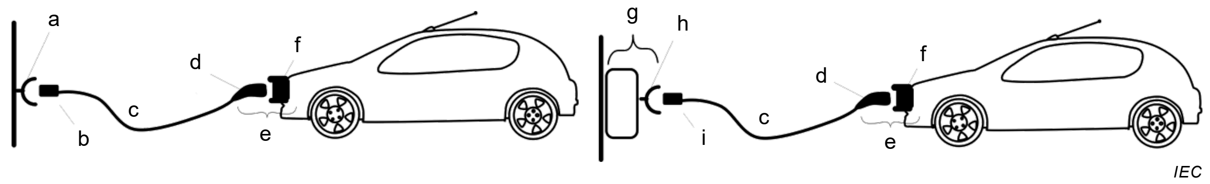


Note 1 to entry: The cable assembly is part of the vehicle.

Figure 1 – Case A connection

3.1.11
case B
connection of an EV to a supply network with a cable assembly detachable at both ends

SEE: Figure 2

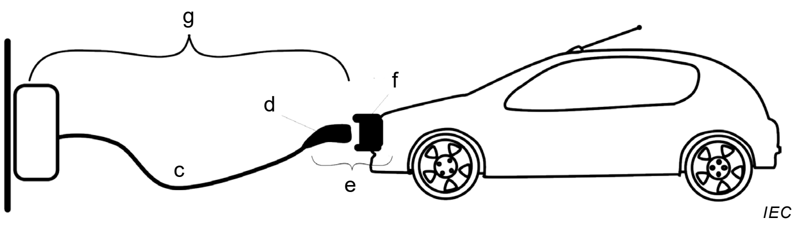


Note 1 to entry: The detachable cable assembly is not part of the vehicle or the charging station.

Figure 2 – Case B connection

3.1.12
case C
connection of an EV to a supply network utilizing a cable and vehicle connector permanently attached to the EV charging station.

SEE: Figure 3



Note 1 to entry: The cable assembly is part of the EV charging station.

Figure 3 – Case C connection

Key for Figures 1 to 3

(a) Socket-outlet
(b) Plug
(c) Cable
(d) Vehicle connector
(e) Vehicle coupler
(f) Vehicle inlet
(g) Charging station
(h) EV socket-outlet
(i) EV plug

3.2 Insulation

3.2.1
basic insulation
insulation of hazardous-live-parts which provides basic protection

[SOURCE: IEC 60050-826:2004, 826-12-14]

3.2.2
direct contact
electrical contact of persons or animals with live parts

[SOURCE: IEC 60050-195:1998, 195-06-03]

3.2.3
double insulation
insulation comprising both basic insulation and supplementary insulation

[SOURCE: IEC 60050-826:2004, 826-12-16]

3.2.4
conductive part
part which can carry electric current

[SOURCE: IEC 60050-195:1998, 195-01-06]

3.2.5
exposed conductive part
conductive part of electrical equipment, which can be touched and which is not normally live, but which can become live when basic insulation fails

Note 1 to entry: A conductive part of electrical equipment which can only become live through contact with an exposed-conductive-part which has become live is not considered to be an exposed-conductive-part itself.

[SOURCE: IEC 60050-441:1998, 442-01-21]

3.2.6
hazardous-live-part
live part which, under certain conditions, can give a harmful electric shock

[SOURCE: IEC 60050-195:1998, 195-06-05]

3.2.7
fault protection
protection against shock under single-fault conditions

[SOURCE: IEC 60050-195:1998/AMD1:2001, 195-06-02]

3.2.8
insulation
all the materials and parts used to insulate conductive elements of a device, or a set of properties which characterize the ability of insulation to provide its function

[SOURCE: IEC 60050-151:2001, 151-15-41, modified and 151-15-42, modified]

3.2.9
live part
conductor or conductive part intended to be energized in normal operation, including a neutral conductor, but by convention not a PEN conductor or PEM conductor or PEL conductor

Note 1 to entry: This concept does not necessarily imply a risk of electric shock.

[SOURCE: IEC 60050-195:1998, 195-02-19]

3.2.10
reinforced insulation
insulation of hazardous-live-parts which provides a degree of protection against electric shock equivalent to double insulation

Note 1 to entry: Reinforced insulation may comprise several layers which cannot be tested singly as basic insulation or supplementary insulation.

[SOURCE: IEC 60050-195:1998, 195-06-09]

3.2.11
supplementary insulation
independent insulation applied in addition to basic insulation for fault protection

[SOURCE: IEC 60050-826:2004, 826-12-15]

3.3 Functions

3.3.1
control pilot conductor
insulated conductor incorporated in a cable assembly, which, together with the protective conductor, is part of the control pilot circuit

3.3.2
control pilot circuit
circuit designed for the transmission of signals or communication between the EV and the EV supply equipment

Note 1 to entry: for Mode 2 it is between the EV and the ICCB or IC-CPD.

3.3.3
control pilot function
function used to monitor and control the interaction between the EV and the EV supply equipment

3.3.4
control pilot function controller
CPFC
device in the EV supply equipment and the EV responsible for the control pilot function and the generation of the PWM signal

Note 1 to entry: This note applies to the French language only.

3.3.5
proximity function
electrical or mechanical means to indicate the insertion state of the vehicle connector in the vehicle inlet to the EV and/or to indicate the insertion state of the plug in the socket-outlet of the EV charging station

3.4 Vehicle

3.4.1
electric vehicle
EV
(electric road vehicle)
any vehicle propelled by an electric motor drawing current from an RESS, intended primarily for use on public roads

Note 1 to entry: In this standard, these terms refer only to those vehicles that can be charged from an external electrical source.

[SOURCE: ISO 17409]

3.4.2
plug in hybrid electric road vehicle
PHEV
electrical vehicle that can charge the rechargeable electrical energy storage device from an external electric source and also derives part of its energy from another on-board source

Note 1 to entry: This note applies to the French language only.

3.4.3
rechargeable energy storage system
RESS
system that stores energy for delivery of electric energy and which is rechargeable

EXAMPLE: batteries, capacitors

Note 1 to entry: This note applies to the French language only.

[SOURCE: ISO 17409]

3.5 Cords, cables and connection means

3.5.1
adaptor
portable accessory constructed as an integral unit incorporating both a plug portion and one socket-outlet portion

[SOURCE: IEC 60050-442:1998, 442-03-19, modified]

3.5.2
cable assembly
assembly consisting of flexible cable or cord fitted with a plug and/or a vehicle connector, that is used to establish the connection between the EV and the supply network or an EV charging station

Note 1 to entry: A cable assembly can be detachable or be a part of the EV or the EV charging station.

Note 2 to entry: A cable assembly can include one or more cables, with or without a fixed jacket, which can be in a flexible tube, conduit or wire way.

[SOURCE: IEC 62196-1:2014, 3.1, modified]

3.5.3
cable management system
one or more devices that is intended to protect a cable assembly from mechanical damage and/or to facilitate its handling

EXAMPLE: Cable suspension device.

3.5.4
cord extension set
assembly consisting of a flexible cable or cord fitted with plug and a portable socket-outlet or connector which can match each other

Note 1 to entry: The cord is called an “adapter cord” when the plug and socket-outlet do not match.

Note 2 to entry: A Mode 1, Mode 2 or Mode 3 cable assembly is not considered as a cord extension set.

[SOURCE: IEC 60050-461:2008, 461-06-17, modified]

3.5.5
in-cable control box
ICCB
device incorporated in the Mode 2 cable assembly, which performs control functions and safety functions

Note 1 to entry: The “ICCB” is called “function box” in IEC 62752.

Note 2 to entry: This note applies to the French language only.

3.5.6
IC-CPD
Mode 2 cable assembly that complies with IEC 62752

3.5.7
EV socket-outlet
specific socket-outlet intended to be used as part of EV supply equipment and defined in the IEC 62196 series

3.5.8
EV plug
specific plug intended to be used as part of EV supply equipment and defined in the IEC 62196 series

3.5.9
plug
accessory having contacts designed to engage with the contacts of a socket-outlet, also incorporating means for the electrical connection and mechanical retention of flexible cables or cords

[SOURCE: IEC 60050-442:1998, 442-03-01, modified – The word "pins" has been replaced by "contacts" in the definition.]

3.5.10
socket-outlet
accessory having socket-contacts designed to engage with the contacts of a plug and having terminals for the connection of cables or cords

[SOURCE: IEC 60050-442:1998, 442-03-02, modified – The word "pins" has been replaced by "contacts" in the definition.]

3.5.11
standard plug and socket-outlet
plug and socket-outlet which meets the requirements of any IEC and/or any national standard that provides interchangeability by standard sheets, excluding the specific EV accessories as defined in the IEC 62196 series

Note 1 to entry: The IEC 60309-1, IEC 60309-2 and IEC 60884-1 and IEC TR 60083, define standard plugs and socket-outlets.

3.5.12
vehicle coupler
electric vehicle coupler
means of enabling the connection at will of a flexible cable to an electric vehicle

Note 1 to entry: It consists of two parts: a vehicle connector and a vehicle inlet.

[SOURCE: IEC 62196-1:2014, 3.3]

3.5.13
vehicle connector
electric vehicle connector
part of a vehicle coupler integral with, or intended to be attached to the cable assembly

[SOURCE: IEC 62196-1:2014, 3.3.1]

3.5.14
vehicle inlet
electric vehicle inlet
part of a vehicle coupler incorporated in, or fixed to, the electric vehicle

[SOURCE: IEC 62196-1:2014, 3.3.2]

3.5.15
connecting point
point where one electric vehicle is connected to the fixed installation

Note 1 to entry: The connecting point for Modes 1 and 2 is the point where one electric vehicle is connected to the fixed installation or supply network.

Note 2 to entry: The connecting point for Modes 3 and 4 is the point where one electric vehicle is connected to the EV charging station

Note 3 to entry: The connecting point for Mode 1, Mode 2 and Mode 4, connected by a cable and plug is the standard socket-outlet.

Note 4 to entry: The connecting point for Mode 3 and permanently connected Mode 4 is the EV socket-outlet (case A and case B) or the EV connector (case C).

3.5.16
interlock
device or combination of devices that prevents the power contacts of a socket-outlet/vehicle connector from becoming live before it is in proper engagement with a plug/vehicle inlet, and which either prevents the plug/vehicle connector from being withdrawn while its power contacts are live or makes the power contacts dead before separation

3.5.17
retaining means
device (e.g. mechanical or electromechanical) that holds a plug or vehicle connector in position when it is in proper engagement, and prevents its unintentional withdrawal

[SOURCE: IEC 62196-1:2014, 3.9]

3.5.18
latching device
part of the interlock mechanism provided to hold a plug in the socket-outlet or vehicle connector in the vehicle inlet to prevent its intentional or unintentional withdrawal

EXAMPLE See standard sheets 2-II and 2-IIId in IEC 62196-2:2014 and 3-IIIc in IEC 62196-3:2014.

[SOURCE: IEC 62196-1:2014, 3.10, modified]

3.5.19
locking mechanism
means intended to reduce the likelihood of tampering with, or an unauthorised removal, of the accessories

[SOURCE: IEC 62196-1:2014, 3.11, modified]

3.5.20
vehicle adaptor
portable accessories constructed as an integral unit incorporating both vehicle inlet portion and vehicle connector portion

3.6 Service and usage

3.6.1
indoor use
intended for operation under normal ambient conditions in a building

[SOURCE: IEC 60050-151:2001, 151-16-06, modified – The term "indoor" has been replaced by "indoor use".]

3.6.2
outdoor use
capable of operating under specific range of outdoor conditions

[SOURCE: IEC 60050-151:2001, 151-16-05, modified – The term "outdoor" has been replaced by "outdoor use".]

3.6.3
equipment for locations with restricted access
equipment accessible to all persons who are authorized to have access to the location (e.g. equipment located in private housing, private parking areas or similar places)

[SOURCE: IEC 60050-195:1998, 195-04-04, modified]

3.6.4
equipment for locations with non-restricted access
equipment accessible for all persons, e.g. access available in a public area

3.6.5
portable equipment
cord and plug connected equipment, cable assembly, adaptors or other accessories that are capable to be carried by one person and designed and intended to be carried within the EV

3.6.6
mobile equipment
electric equipment which is moved while in operation or which can easily be moved from one place to another while connected to the supply.

[SOURCE: IEC 60050-826:2004, 826-16-04]

3.6.7
stationary equipment
equipment or electric equipment not provided with a carrying handle and having such a mass that it cannot easily be moved

Note 1 to entry: Stationary equipment is intended to be either permanently connected to the supply network or connected to the supply network by a cable and plug.

[SOURCE: IEC 60050-826:2004, 826-16-06, modified]

3.6.8
ground mounted
equipment with a part intended to be embedded or attached to the ground

3.6.9
permanently connected EV supply equipment
EV supply equipment that can only be connected to, or disconnected from, the AC or DC supply network by the use of a tool

[SOURCE: IEC 60050-151:2001/AMD2:2014, 151-11-29, modified]

3.6.10
user
party who will specify, purchase, use and/or operate the EV supply equipment, or someone acting on their behalf

[SOURCE: IEC 61439-1:2011, 3.10.3, modified]

3.7 General terms

3.7.1
supply network
any source of electric energy (eg. mains or electric grid, distributed energy resources (DER), battery bank, PV installation, generator, etc.)

3.7.2
protective conductor
conductor provided for purposes of safety, for example protection against electric shock

EXAMPLE Examples of a protective conductor include a protective bonding conductor, protective earthing conductor and an earthing conductor when used for protection against electric shock.

[SOURCE: IEC 60050-826:2004, 826-13-22, modified]

3.7.3
protective earthing conductor
protective grounding conductor (US, CA)
equipment grounding conductor (US, CA)
protective conductor provided for protective earthing

Note 1 to entry: In the U.S.A. and in CA, the term "ground" is used instead of "earth".

[SOURCE: IEC 60050-195:1998, 195-02-11, modified – The third term and Note 1 have been added.]

3.7.4
earthing terminal
grounding terminal (US, CA)
terminal provided on equipment or on a device and intended for the electric connection with the earthing arrangement

[SOURCE: IEC 60050-195:1998, 195-02-31]

3.7.5
protective earthing
protective grounding (US, CA)
earthing a point or points in a system or in an installation or in equipment for purposes of electrical safety

[SOURCE: IEC 60050-195:1998/AMD1:2001, 195-01-11]

3.7.6
residual current device
RCD
mechanical switching device designed to make, carry and break currents under normal service conditions and to cause the opening of the contacts when the residual current attains a given value under specified conditions

Note 1 to entry: A residual current device can be a combination of various separate elements designed to detect and evaluate the residual current and to make and break current.

[SOURCE: IEC 60050-442:1998, 442-05-02]

3.7.7
leakage current
electric current in an unwanted conductive path under normal operating conditions

[SOURCE: IEC 60050-195:1998, 195-05-15]

3.7.8
switching device
device designed to make or break the current in one or more electric circuits

[SOURCE: IEC 60050-441:1984, 441-14-01]]

3.7.9
mechanical switching device
switching device designed to close and open one or more electric circuits by means of separable contacts

[SOURCE: IEC 60050-441:1984, 441-14-02, modified – The note has been removed.]

3.7.10
touch current
electric current passing through a human body or through an animal body when it touches one or more accessible parts of an electrical installation or electrical equipment

[SOURCE: IEC 60050-826:2004, 826-11-12]

4 General requirements

The EV supply equipment shall be so constructed that an EV can be connected to the EV supply equipment so that in normal conditions of use, the energy transfer operates safely, and its performance is reliable and minimises the risk of danger to the user or surroundings.

Compliance is checked by meeting all of the relevant requirements of the IEC 61851 series.

Unless otherwise stated all tests indicated in this document are type tests.

Unless otherwise stated, all tests required by this standard may be conducted on separate samples. They may be done on the same samples at the manufacturer's agreement.

Unless otherwise stated, each test is conducted once.

Unless otherwise specified, all tests shall be carried out in a draught-free location and at an ambient temperature of 20°± 5 °C.

The EV supply equipment shall be rated for one or more of standard nominal voltages and frequencies as given in IEC 60038.

NOTE In the following countries, national standards or regulations provide different requirements for frequency: JP.

Assemblies for EV supply equipment shall comply with IEC TS 61439-7 with the exceptions or additions as indicated in Clause 13.

The standard applies to equipment that is designed to be used at an altitude up to 2 000 m.

For equipment designed to be used at altitudes above 2 000 m, it is necessary to take into account the reduction of the dielectric strength and the cooling effect of the air. Electrical equipment intended to operate under these conditions shall be designed or used in accordance with an agreement between manufacturer and user.

5 Classification

5.1 Characteristics of power supply and output

5.1.1 Characteristics of power supply input

The EV supply equipment shall be classified according to the supply network system that it is intended to be connected to:

– EV supply equipment connected to AC supply network;
– EV supply equipment connected to DC supply network.

The EV supply equipment shall be classified according to the electric connection method:

– Plug and cable connected;
– Permanently connected.

5.1.2 Characteristics of power supply output

The EV supply equipment shall be classified according to the type of current the EV supply equipment delivers:

– AC EV supply equipment;
– DC EV supply equipment;
– AC and/or DC EV supply equipment.

5.2 Normal environmental conditions

The EV supply equipment shall be classified according to the environmental conditions and use:

– indoor use;
– outdoor use.

NOTE The conditions that define indoor and outdoor use are given in 7.1.1 of IEC 61439-1:2011.

5.3 Special environmental conditions

The EV supply equipment may be classified according to their suitability for use in special environmental conditions other than those specified in this document, if declared so by the manufacturer.

5.4 Access

The EV supply equipment shall be classified according to the location they are intended for:

– equipment for locations with restricted access;
– equipment for locations with non-restricted access.

5.5 Mounting method

The EV supply equipment shall be classified according to the type of mounting:

a) stationary equipment;
   – mounted on walls, poles or equivalent positions:
      • flush mounted;
      • surface mounted.
   – pole/column/pipe-mounted
   – floor mounted
   – ground mounted.
b) non stationary equipment
   – portable equipment;
   – mobile equipment.

NOTE More than one classification can apply.

5.6 Protection against electric shock

The equipment shall be classified according to the protection against electric shock:

– class I equipment;
– class II equipment;
– class III equipment.

NOTE Class I, Class II and Class III descriptions can be found in IEC 61140:2016, 7.3, 7,4 and 7.5.

5.7 Charging modes

The EV supply equipment shall be classified according to 6.2:

– Mode 1;
– Mode 2;
– Mode 3;
– Mode 4.

NOTE EV supply equipment with multiple outputs can be classified as supporting more than one mode.

6 Charging modes and functions

6.1 General

Clause 6 describes the different charging modes and functions for energy transfer to EVs.

6.2 Charging modes

6.2.1 Mode 1

Mode 1 is a method for the connection of an EV to a standard socket-outlet of an AC supply network, utilizing a cable and plug, both of which are not fitted with any supplementary pilot or auxiliary contacts.

The rated values for current and voltage shall not exceed:

– 16 A and 250 V AC, single-phase,
– 16 A and 480 V AC, three-phase.

EV supply equipment intended for Mode 1 charging shall provide a protective earthing conductor from the standard plug to the vehicle connector.

Current limitations are also subject to the standard socket-outlet ratings described in 9.2.

NOTE 1 In the following countries, Mode 1 charging is prohibited by national codes: US, IL, UK

NOTE 2 In the following countries, Mode 1 charging is prohibited in public areas by national codes: IT

NOTE 3 In the following countries, Mode 1 charging without integral ground fault leakage interruption is not permitted: CA

NOTE 4 In the following countries, plugs and socket-outlets according to IEC 60309-2 are recommended for Mode 1 connections for more than 8 A (2 kVA): CH.

NOTE 5 In the following countries, for EV supply equipment equipped with a plug for household and similar use, repeated continuous loads of long duration, shall be limited to 6 A: DK

NOTE 6 In the following countries, EV supply equipment equipped with a plug for household and similar use, if the charging cycle can exceed 2 hours, the maximum rated current is 10 A: NO

NOTE 7 In the following countries, EV supply equipment equipped with a plug for household and similar use, if the charging cycle can exceed 2 hours, the maximum rated current is 8 A: FR

NOTE 8 In the following countries, Mode 1 cable assembly without PCRD shall not be used but only Mode 1 cables with a PCRD: DE

– (Due to Article 14 in the constitutional law of Germany which frames the preservation of status quo of existing electrical installations it cannot be ensured that fixed electrical installations at all times provide an RCD in Germany).

6.2.2 Mode 2

Mode 2 is a method for the connection of an EV to a standard socket-outlet of an AC supply network utilizing an AC EV supply equipment with a cable and plug, with a control pilot function and system for personal protection against electric shock placed between the standard plug and the EV.

The rated values for current and voltage shall not exceed:

– 32 A and 250 V AC single-phase;
– 32 A and 480 V AC three-phase.

Current limitations are also subject to the standard socket-outlet ratings described in 9.2.

EV supply equipment intended for Mode 2 charging shall provide a protective earthing conductor from the standard plug to the vehicle connector.

Mode 2 equipment that is destined to be mounted on a wall but is detachable by the user, or to be used in a shock resistant enclosure shall use protection equipment as required by IEC 62752.

NOTE 1 In the following countries, Mode 2 charging is limited by national codes to 250 V maximum: US, CA.

NOTE 2 In the following countries, Mode 2 is not allowed in public areas: IT.

NOTE 3 In the following countries, Mode 2 shall not exceed 16 A and not exceed 250 V AC in single phase systems: CH.

NOTE 4 In the following countries, the use of IEC 60309-2 accessories is recommended for Mode 2 connections for more than 8 A (2 kVA): CH.

NOTE 5 In the following countries, for EV supply equipment equipped with a plug for household and similar use, repeated continuous loads of long duration, shall be limited to 6 A: DK

NOTE 6 In the following countries, EV supply equipment equipped with a plug for household and similar use, if the charging cycle can exceed 2 hours, the maximum rated current is 8 A: FR

NOTE 7 In the following countries, EV supply equipment equipped with a plug for household and similar use, if the charging cycle can exceed 2 hours, the maximum rated current is 10 A: NO

NOTE 8 In the following countries, the use of IEC 60309-2 accessories is recommended for Mode 2 connections for more than 10 A: IT

6.2.3 Mode 3

Mode 3 is a method for the connection of an EV to an AC EV supply equipment permanently connected to an AC supply network, with a control pilot function that extends from the AC EV supply equipment to the EV.

EV supply equipment intended for Mode 3 charging shall provide a protective earthing conductor to the EV socket-outlet and/or to the vehicle connector.

6.2.4 Mode 4

Mode 4 is a method for the connection of an EV to an AC or DC supply network utilizing a DC EV supply equipment, with a control pilot function that extends from the DC EV supply equipment to the EV.

Mode 4 equipment may be either permanently connected or connected by a cable and plug to the supply network.

EV supply equipment intended for Mode 4 charging shall provide a protective earthing conductor or protective conductor to the vehicle connector.

Additional requirements for DC EV supply equipment are given in IEC 61851-23.

6.3 Functions provided in Mode 2, 3 and 4

6.3.1 Mandatory functions in Modes 2, 3, and 4

6.3.1.1 General

The following control pilot functions shall be provided by the EV supply equipment:

- Continuous continuity checking of the protective conductor according to 6.3.1.2;
- Verification that the EV is properly connected to the EV supply equipment according to 6.3.1.3;
- Energization of the power supply to the EV according to 6.3.1.4;
- De-energization of the power supply to the EV according to 6.3.1.5;
- Maximum allowable current according to 6.3.1.6.

Compliance is checked by inspection and test where applicable.

If EV supply equipment can supply more than one vehicle simultaneously, it shall ensure that the control pilot function performs the above functions independently at each connecting point.

NOTE The control pilot functions can be achieved using a PWM signal and a control pilot as described in Annex A or by any other non PWM system that provides the same results and is compatible with Annex A. An example is given in informative Annex D.

EV supply equipment designed for Mode 2 or Mode 3, using the control pilot conductor and utilizing accessories according to IEC 62196-2, shall be provided with control pilot function according to Annex A.

6.3.1.2 Continuous continuity checking of the protective conductor

While charging in Mode 2, the electrical continuity of the protective earthing conductor between the ICCB and the respective EV contact shall be continuously monitored by the ICCB.

While charging in Mode 3, the electrical continuity of the protective earthing conductor between the EV charging station and the respective EV contact shall be continuously monitored by the EV supply equipment.

While charging in Mode 4, the electrical continuity of the protective conductor between the EV charging station and the respective EV contact shall be continuously monitored by the EV supply equipment.

The EV supply equipment shall disconnect the supply to the EV in case of:

- loss of electrical continuity of the protective conductor (i.e. open control pilot circuit), within 100 ms.
- incapacity to verify the continuity of the protective conductor (e.g. short circuit between pilot wire and protective conductor), within 3 s.

6.3.1.3 Verification that the EV is properly connected to the EV supply equipment

The EV supply equipment shall be able to determine that the EV is properly connected to the EV supply equipment.

Proper connection is assumed when the continuity of the control pilot circuit is detected.

6.3.1.4 Energization of the power supply to the EV

The EV socket-outlet or the vehicle connector shall not be energized unless the control pilot function between EV supply equipment and EV has been established correctly with signal states allowing energization.

The presence of such states does not imply that energy will be transferred between the EV supply equipment and the EV as this may be subject to other external conditions, e.g. energy management system.

If the EV requests ventilation, the EV supply equipment shall only energize the system if such ventilation is provided by the installation or the premises.

6.3.1.5 De-energization of the power supply to the EV

If the control pilot signal is interrupted the power supply to the EV shall be interrupted according to 6.3.1.2.

If the control pilot signal status no longer allows energization, the power supply to the EV shall be interrupted but the control pilot signalling may remain in operation.

6.3.1.6 Maximum allowable current

A means shall be provided to inform the EV of the value of the maximum current it is allowed to draw. The value of the maximum current permitted shall be transmitted and shall not exceed any of the following:

- the rated output current of the EV supply equipment,
- the rated current of the cable assembly.

NOTE The cable assembly includes Mode 2 and Mode 3 cable assemblies.

The transmitted value may change, without exceeding the maximum allowed current, to adapt to power limitations, e.g. for load management.

The EV supply equipment may interrupt the energy supply if the current drawn by the EV exceeds the transmitted value.

6.3.2 Optional functions for Modes 2, 3 and 4

6.3.2.1 General

The optional functions that are implemented shall be indicated in the manual and shall fulfil the requirements of 6.3.2.

Other functions can be provided.

Compliance is checked by inspection and test where applicable.

6.3.2.2 Ventilation during supply of energy

EV supply equipment can exchange information with installation regarding the request and presence for ventilation.

NOTE 1 Ventilation requirements might be subject to local or national regulation or standard.

NOTE 2 In the following countries national regulations may require ventilation for some situations for indoor charging under certain conditions: CA.

NOTE 3 This is mainly a requirement for indoor charging.

6.3.2.3 Intentional and unintentional disconnection of the vehicle connector and/or the EV plug

A mechanical or electromechanical means shall be provided to prevent intentional and unintentional disconnection under load of the vehicle connector and/or plug according to IEC 62196-1.

NOTE 1 IEC 62196-1:2014 defines three levels of prevention of disconnection as given in the definitions 3.9, 3.10 and 3.11 in the order of increasing constraints, as follows:

– Unintentional disconnection prevention -> retaining means,

– Unintentional and intentional disconnection prevention -> latching device,

– Unintentional, intentional disconnection and tampering prevention -> locking mechanism.

NOTE 2 In the following countries a means to prevent unintentional disconnect of the EV accessories is required: CA, US.

6.3.2.4 Mode 4 using the combined charging system

The combined charging system as described in Annex CC of IEC 61851-23:2014 and ISO 17409 shall be so designed that:

- AC chargeable EVs with a basic vehicle inlet do not require any means to protect the EV against DC voltage at the inlet.
- AC EV supply equipment does not require any means to be self-protected against DC voltage coming from the EV.

NOTE 1 Standard Sheets for configuration EE and FF of IEC 62196-3 gives dimensions, rating and relevant mechanical functions of the interfaces EE and FF for DC. EV charging system C is described in Annex CC of IEC 61851-23:2014.

For DC charging, digital communication shall be established between the vehicle and the DC EV charging station that validates the DC energy transfer. The DC supply to the vehicle shall not be connected until such complete validation from the vehicle is achieved.

A combined interface extends the use of a basic interface for AC and DC charging.

DC charging can be achieved by using separate and additional DC power contacts to supply DC energy to the EV or by using power contacts placed at the position of the AC power contacts of a basic interface, if the vehicle connector and the vehicle inlet are both suitable for DC.

The basic portion of the combined vehicle inlet can be used with a basic connector for AC charging only or with a combined connector having separate contacts for AC or DC charging.

AC and DC power transfer shall not occur through the combined interface at the same time.

The combined interface used for DC charging shall only be used with the “Combined Charging System” described in Annex CC of IEC 61851-23:2014.

Analysis and design of the EV supply equipment using a basic interface for DC shall apply a risk analysis according to IEC 61508 (all parts) applying a severity level of at least S2 for the function preventing the risk of unintended DC voltage output.

NOTE 2 Such analysis includes the effect of vehicle faults on EV supply equipment.

7 Communications

7.1 Digital communication between the EV supply equipment and the EV

Digital communication is optional for Modes 1, 2 and 3.

For Mode 4 the digital communication as described in IEC 61851-24 shall be provided to allow the EV to control the EV supply equipment.

NOTE 1 Digital communication as described in ISO/IEC 15118 series is also called high level communication.

NOTE 2 Annex D also provides information on digital communication.

7.2 Digital communication between the EV supply equipment and the management system

Telecommunication network or telecommunication port of the EV supply equipment, connected to the telecommunication network, if any, shall comply with the requirements for connection to telecommunication networks according to Clause 6 of IEC 60950-1:2005.

8 Protection against electric shock

8.1 Degrees of protection against access to hazardous-live-parts

The different parts of the EV supply equipment as mentioned shall fulfil the following requirements:

- IP ratings for enclosures shall be at least IPXXC;
- vehicle connector when mated with vehicle inlet: IPXXD;
- plug mated with socket-outlet: IPXXD;
- vehicle connector intended for Mode 1 use, not mated: IPXXD;
- vehicle connector intended for Mode 2 use, not mated: IPXXB and fulfilling the following:

  Minimum opening of the contact equal to the clearance according to IEC 60664-1 considering overvoltage category 2 (e.g. the value given in IEC 60664-1 for 230 V/400 V is 2,5 kV rated impulse voltage withstand that implies 1,5 mm separation of contacts) and inhibits the charging and warns the user in case of welded contact.
- vehicle connector and EV socket-outlet intended for Mode 3 use, not mated: IPXXB provided it is associated directly upstream with a mechanical switching device (see also 12.3) and fulfilling one of the following:
  a) minimum opening of the contact equal to the clearance according to IEC 60664-1 considering overvoltage category 3 (e.g. the value given in IEC 60664-1 for 230 V/400 V is 4 kV rated impulse voltage withstand that implies at least 3 mm separation of contacts);
  b) presence of monitoring of the switching contacts associated with a means to operate another mechanical switching device providing isolating function upstream the above in case of fault of operation of the switching device upstream the accessory;
  c) presence of shutters on live entry hole of the socket-outlets or connectors for case C.

NOTE 1 In the following countries, IPXXB is required for Mode 1: JP, SE, CH, AT, DE, BE, FI

NOTE 2 In the following countries options a and b are only acceptable if both options are used together: NL, IT

NOTE 3 In the following countries the option b alone is not allowed: BE, CH

NOTE 4 In following countries socket-outlets are equipped with shutters where they are mandatory in residential and public locations: FR, UK

NOTE 5 In following countries standard socket-outlets are equipped with shutters where they are mandatory in residential and public locations: DK

NOTE 6 In the following countries, for installations in dwellings and for 16 A applications, Wiring Rules prescribe the use of socket-outlets with shutters: ES

NOTE 7 In the following countries national regulations require shutters or equivalent protection methods with equivalent safety levels. For example: installation heights, blocking objects against touch ability, interlocking, locking cover etc. SE

NOTE 8 In the following countries the use of software controlled means cannot be used to control isolating devices: UK

Compliance is checked by inspection and measurement.

8.2 Stored energy

8.2.1 Disconnection of plug connected EV supply equipment

For plug connected EV supply equipment, where the connection pins are accessible after unplugging, one second after disconnecting the standard plug from the standard socket-outlet, the voltage between any combination of accessible contacts of the standard plug shall be less than or equal to 60 V DC or the stored charge available shall be less than 50 μC.

Compliance is checked by inspection and by test with the EV disconnected according to 2.1.1.5 of IEC 60950-1:2005.

NOTE Requirements for the EV are specified in ISO 17409.

8.2.2 Loss of supply voltage to permanently connected EV supply equipment

The voltage between power lines or power lines and protective earthing conductor, when measured at the input supply terminals of the EV supply equipment, shall be less than or equal to 60 V DC or the stored energy shall be less than or equal to 0,2 J within 5 seconds after disconnecting the power supply voltage to the EV supply equipment.

Compliance is checked by inspection and by test with no EV connected to the EV supply equipment according to 2.1.1.7 of IEC 60950-1:2005.

8.3 Fault protection

Fault protection shall consist of one or more protective measures as permitted according to IEC 60364-4-41:

- automatic disconnection of supply;
- double or reinforced insulation;
- electrical separation if limited to the supply of one item of current-using equipment;
- extra low-voltage (SELV and PELV).

Electric separation is fulfilled if there is one electrically separated circuit for each EV.

Compliance is checked by inspection.

8.4 Protective conductor

The protective earthing conductor and the protective conductor shall be of sufficient rating in accordance with requirements of IEC TS 61439-7.

NOTE In the following countries, the size and rating of the protective earthing conductor are determined by national codes and regulations: CA, US, JP

For Modes 1, 2 and 3, a protective earthing conductor shall be provided between the AC supply input earthing terminal of the EV supply equipment and the EV.

Mode 4 EV supply equipment shall provide either:

a) a protective earthing conductor from the input earthing terminal of the AC supply network to the EV

   or

b) a protective conductor from the EV supply equipment to the EV if fault protection is based on electric separation.

For Modes 3 and 4 permanently connected EV supply equipment, protective earthing conductors shall not be switched.

Compliance is checked by inspection.

8.5 Residual current protective devices

EV supply equipment can have one or more connecting points to supply energy to EVs.

Where connecting points can be used simultaneously and are connected to a common input terminal of the EV supply equipment, they shall have individual protection incorporated in the EV supply equipment.

If the EV supply equipment has more than one connecting point that cannot be used simultaneously then such connecting points can have common protection devices.

EV supply equipment that includes an RCD and that does not use the protective measure of electrical separation shall comply with the following:

- The connecting point of the EV supply equipment shall be protected by an RCD having a rated residual operating current not exceeding 30 mA;
- RCD(s) protecting connecting points shall be at least type A;
- RCDs shall comply with one of the following standards: IEC 61008-1, IEC 61009-1, IEC 60947-2 and IEC 62423;
- RCDs shall disconnect all live conductors.

NOTE 1 This applies to single-phase or three-phase connecting points.

Where the EV supply equipment is equipped with a socket-outlet or vehicle connector for AC use in accordance with IEC 62196 (all parts), protective measures against DC fault current shall be taken. The appropriate measures shall be:

- RCD type B or
- RCD Type A and appropriate equipment that ensures the disconnection of the supply in case of DC fault current above 6 mA.

NOTE 2 An example of appropriate equipment that ensures the disconnection of the supply in case of DC fault is expected in the future IEC 62955[7].

NOTE 3 The RCDs or the appropriate equipment that ensure the disconnection of the supply in case of DC fault can be provided inside the EV supply equipment, in the installation or in both places.

NOTE 4 Selectivity can be maintained between the RCD protecting a connecting point and an RCD installed upstream when required for service purposes.

NOTE 5 In the following countries, as the installations are equipped with RCD of Type AC, a means for the protection of fault current with a performance at least equal to Type A in addition to appropriate equipment that ensures the disconnection of the supply in case of DC fault current above 6 mA is provided by the EV supply equipment: JP.

NOTE 6 In the following country, the use of a system of protection is required that is intended to interrupt the electric circuit to the load when:

– a fault current to earth (ground) exceeds some predetermined value that is less than that required to operate the overcurrent protective device of the supply circuit,

– the earthing (grounding) path becomes open circuited or becomes an excessively high impedance, or a path to earth (ground) is detected on an isolated (ungrounded) system: US, CA.

7 Under preparation.

NOTE 7 In the following countries the use of a RCD of type AC for EVs connected by Mode 1 to domestic installations is allowed: JP, SE, UK, CA.

NOTE 8 In some countries, a device which measures leakage current using a frequency sensitive network and trips at predefined levels of leakage current up to 20 mA, based upon the frequency is required: US, CA.

NOTE 9 A PRCD as described in IEC 61540 or IEC 62335, can be used to improve protection for connection to existing AC power supply networks in Mode 1.

8.6 Safety requirements for signalling circuits between the EV supply equipment and the EV

Any circuit for signalling, which extends beyond the EV supply equipment enclosure for connection with the EV (e.g. control pilot circuit), shall be extra low voltage (SELV or PELV) according to IEC 60364-4-41.

8.7 Isolating transformers

Isolating transformers (excluding safety isolating transformers used for signalling) shall comply with the requirements of IEC 61558-1 and IEC 61558-2-4.

Compliance is checked by inspection.

9 Conductive electrical interface requirements

9.1 General

Clause 9 provides a description of the conductive electrical interface requirements.

9.2 Functional description of standard accessories

Standard accessories used for EV supply equipment shall be in accordance with IEC 60309-1, IEC 60309-2 or IEC 60884-1 or the national standard. Standard accessories that are intermateable with interfaces described in the IEC 60320 series shall not be used for EV supply equipment.

Compliance is checked by inspection.

NOTE 1 In the following countries, IEC 60884-1, IEC 60884-2-5 and all other parts of the IEC 60884 series of standards are not indispensable and not applicable. National standards apply to plugs and socket-outlets for household use: UK.

Socket-outlets and plugs designed for household and similar use might not be designed for extended current draw or continuous use at maximum rated currents and might be subject to national regulations and standards for supply of energy to an EV.

NOTE 2 In the following countries the branch circuit overcurrent protection is based on 125 % of the rated current: US, CA.

NOTE 3 In the following countries, specific requirements may apply for the use of such accessories for EV energy supply: US.

NOTE 4 In the following countries EV supply equipment equipped with a plug for household and similar use the maximum rated current is 8 A, if the charging cycle can exceed 2 hours: FR.

NOTE 5 In the following countries, EV supply equipment equipped with a plug for household and similar use, if the charging cycle can exceed 2 hours, the maximum rated current is 10 A: NO.

NOTE 6 In the following countries, for EV supply equipment equipped with a plug for household and similar use, repeated continuous loads of long duration, shall be limited to 6 A: DK.

NOTE 7 In the following countries, plugs and socket-outlets, according to IEC 60309-2 are recommended for Mode 1 and Mode 2 connections for more than 8 A (2 kVA): CH.

NOTE 8 In the following country, 20 A socket-outlet is recommended for AC 200 V: JP.

9.3 Functional description of the basic interface

General requirements and ratings shall be in accordance with the requirements specified in IEC 62196-1.

The basic interface is specified in 6.5 of IEC 62196-1:2014.

The following contacts are indicated:

- up to three phases (L1, L2, L3);
- neutral (N);
- protective conductor (PE);
- control pilot (CP);
- proximity contact (PP).

It may be used either for single-phase or for three-phase or both.

Ratings and requirements for the use of the basic interface shall be in accordance with the requirements specified in IEC 62196-2.

9.4 Functional description of the universal interface

General requirements and ratings shall be in accordance with the requirements specified in IEC 62196-1.

The universal interface is specified in 6.4 and Table 2 of IEC 62196-1:2014.

9.5 Functional description of the DC interface

General requirements and ratings shall be in accordance with the requirements specified in IEC 62196-1.

The DC interface, configurations and ratings are specified in 6.6 and Table 4 of IEC 62196-1:2014.

Ratings and requirements for the use of DC interface shall be in accordance with the requirements specified in IEC 62196-3.

9.6 Functional description of the combined interface

The combined interface is specified in 6.7 and Table 5 of IEC 62196-1:2014.

General requirements and ratings shall be in accordance with the requirements specified in IEC 62196-1.

Ratings and requirements for the use of the combined interface with alternating current shall be in accordance with the requirements specified in IEC 62196-2.

Ratings and requirements for the use of the combined interface with direct current shall be in accordance with the requirements specified in IEC 62196-3.

9.7 Wiring of the neutral conductor

Where accessories according to IEC 62196 are used for three phase supply the neutral conductor shall always be wired to the accessories.

Where accessories according to IEC 62196 are used for single phase supply, the terminals L (L1) and N (Neutral) shall always be wired.

Compliance is checked by inspection.

NOTE In the following countries certain AC supply networks do not provide a neutral conductor: US, BE, IT, CA, JP, CH, NO.

Wiring instructions shall be provided in the manual (see 16.1).

10 Requirements for adaptors

Vehicle adaptors shall not be used to connect a vehicle connector to a vehicle inlet.

Adaptors between the EV socket-outlet and the EV plug shall only be used if specifically designated and approved by the vehicle manufacturer or by the EV supply equipment manufacturer and in accordance with national requirements, if any (see 16.2).

Such adaptors shall comply with the requirements of this standard, and the other relevant standards governing either the EV plug or EV socket-outlet portions of the adaptor. The adaptors shall be marked to indicate the specific conditions of use allowed by the manufacturer, e.g. IEC 62196 series.

Such adaptors shall not allow transitions from one mode to another.

NOTE 1 In the following countries, the use of adaptors from standard socket-outlets to Mode 3 vehicle cable assembly, that maintain the overall safety requirements of this standard is allowed: IT, SE, BE, CH.

NOTE 2 In the following countries, the use of adaptors from standard socket-outlets to cable assembly with a basic or universal interface that maintains the overall safety requirements of this standard is allowed: FR, IT.

NOTE 3 In the following countries, for Mode 1 and Mode 2, a short cord extension set of less than 30 cm in length and with a standard plug and no Mode change, is permitted for use on the EV supply equipment: SE.

NOTE 4 In the following countries, the use of adapter cable assembly with a plug according to IEC 60884 and a socket-outlet according to IEC 60309, with no Mode change, of less than 30 cm and an overcurrent protection of 8 A in the plug portion, is allowed: CH.

11 Cable assembly requirements

11.1 General

The cable assembly shall be provided with a cable that is suitable for the application.

NOTE IEC 62893 (under consideration) is a standard for some types of cables for the EV supply equipment.

Cable assemblies shall not allow transitions from one mode to another. This does not concern Mode 2 cable assembles that are constructed according to IEC 62752.

11.2 Electrical rating

For case C, the voltage and current ratings of the cable assembly shall be compatible with the rating of the EV supply equipment.

For accessories requiring current coding according to Annex B and IEC 62196-2, the maximum value of the current coding as indicated in Clause B.2 shall be in accordance with the current rating of the cable assembly.

Cables used with accessories according to IEC 62196-2 for Mode 3 case B, shall have a minimum withstand I^2t value of 75 000 A^2s.

Compliance is checked by inspection.

NOTE 1 IEC 62893 standard for EV charging cables is under consideration.

NOTE 2 In following countries, specific cable types for cable assemblies are required by national regulations: US (type cable EV, EVJ families), JP (VCT, etc.).

NOTE 3 The I^2t value can be evaluated according to 434.5.2 of IEC 60364-4-43:2008.

11.3 Dielectric withstand characteristics

Dielectric withstand characteristics of the cable assembly shall be as indicated for the EV supply equipment in 12.7.

For Class I equipment: between live part and earth with test voltage for Class I equipment;

For Class II equipment: between live part and exposed conductive parts with test voltage for Class II equipment.

11.4 Construction requirements

A cable assembly shall be so constructed that it cannot be used as a cord extension set.

NOTE As in IEC 62196-1, plugs and connectors are designed not to match each other.

A cable assembly may include one or more cables, which may be in a flexible tube, conduit or wire way.

The cable may be fitted with an earth-connected metal shielding.

The cable insulation shall be wear resistant and maintain flexibility over the full temperature range required by the classification of the EV supply equipment.

Compliance is checked by inspection.

11.5 Cable dimensions

The maximum cable length shall be in accordance with the national codes if any.

NOTE 1 In the following countries, the overall length of the EV supply cable does not exceed 7,5 m unless equipped with a cable management system as required by national codes and regulations: US.

NOTE 2 In the following countries, the overall length of the EV supply cable does not exceed 5 m unless equipped with a cable management system as required by national codes and regulations: CH.

Compliance is checked by inspection.

11.6 Strain relief

The strain relief of the cable in the vehicle connector, EV plug or in the standard plug shall be as specified in the relevant product standard (e.g. IEC 62196-1, IEC 60309-1 or IEC 60884-1).

For case C the strain relief at the EV supply equipment shall be in accordance with the requirements in IEC 62196-1.

11.7 Cable management and storage means for cables assemblies

For case C EV supply equipment, a storage means shall be provided for the vehicle connector when not in use.

For case C EV supply equipment the lowest point of the vehicle connector when stored shall be located at a height between 0,5 m and 1,5 m above ground level.

NOTE In the following countries, requirements related to people with disabilities are covered by national regulations: US.

For case C EV charging stations with cables of more than 7,5 m, a cable management system shall be provided. The free cable length shall not exceed 7,5 m when not in use.

Prevention of overheating of cables or cable assemblies used in stored or partially stored position shall be ensured.

Compliance is checked according to Clause 22 of IEC 61316:1999 for cable reel storage.

12 EV supply equipment constructional requirements and tests

12.1 General

The control means and the protection means in Mode 2 EV supply equipment that is intended to be used both as stationary equipment and as portable equipment shall comply with IEC 61851-1 and with IEC 62752.

For case C EV supply equipment, the output cable assembly is considered part of the assembly for testing purpose.

Electric devices and components of EV supply equipment shall comply with their relevant standards. The tests of devices and components shall be carried out with the specimen, or any movable part of it, placed in the most unfavourable position that can occur in normal use.

For extreme environment or other special service conditions, see IEC TS 61439-7.

NOTE 1 In the following countries, there are other requirements to be met for EV supply equipment: JP, US,CA.

NOTE 2 In the following countries, as exception to 10.2.101 and 10.2.102 of IEC TS 61439-7:2014, the product shall fulfil a minimum requirement of IP XXB after the test: SE.

12.2 Characteristics of mechanical switching devices

12.2.1 General

Switching devices within EV supply equipment intended to supply the connecting points shall comply with their relevant standards, with at least the characteristics as given in 12.2.

12.2.2 Switch and switch-disconnector

Switches and switch-disconnectors shall comply with IEC 60947-3.

For AC applications, switches and switch-disconnectors shall have a rated current, at a utilization category of at least AC-22A, not less than the rated current of the circuit that they are intended to operate in.

For DC applications, switches and switch-disconnectors shall have a rated current, at a utilization category of at least DC-21A, not less than the rated current of the circuit that they are intended to operate in.

Compliance is checked by inspection.

NOTE In the following countries, national standards or regulations provide different requirements: JP.

12.2.3 Contactor

Contactors shall comply with IEC 60947-4-1.

For AC applications, contactors shall have a rated current, at a utilization category of at least AC-1, not less than the rated current of the circuit that they are intended to operate in.

For DC applications, contactors shall have a rated current, at a utilization category of at least DC-1, not less than the rated current of the circuit that they are intended to operate in.

Compliance is checked by inspection.

NOTE In the following countries, national standards or regulations provide different requirements: JP.

12.2.4 Circuit-breaker

Circuit breakers, if any, shall comply with IEC 60898-1 or IEC 60947-2 or IEC 61009-1.

Compliance is checked by inspection.

NOTE In the following countries, national standards or regulations provide different requirements: JP.

12.2.5 Relays

Relays used to switch the main current path shall comply with IEC 61810-1 with the following minimum characteristics:

- 50 000 cycles,
- contact category: CC 2.

12.2.6 Inrush current

AC EV supply equipment shall withstand the inrush current according to 8.2.2 of ISO 17409:2015.

The following values are specified in ISO 17409:

- After closing the contactor in the EV supply equipment at the peak value of the supply voltage, the EV supply equipment shall be able to withstand 230 A peak within the duration of 100 μs.
- During the next second the EV supply equipment shall be able to withstand 30 A (rms).

Compliance to this requirement can be verified by test on the complete EV supply equipment or on the individual switching device according IEC TS 61439-7.

NOTE An example of a test is given in 9.8 of IEC 62752:2016.

The protection means shall be selected not to trip for inrush current.

12.2.7 Residual direct current monitoring device (RDC MD)

This will be covered in the future IEC 62955 (under consideration).

12.3 Clearances and creepage distances

The clearances and creepage distances in the EV supply equipment, installed as intended by the manufacturer, shall be in accordance with the requirements specified in IEC 60664-1.

Parts of the EV supply equipment directly connected to the public AC supply network shall be designed according to overvoltage category IV.

Permanently connected EV supply equipment shall be designed according to a minimum overvoltage category III except for the socket-outlet or the vehicle connector in case C where a minimum overvoltage category II applies.

EV supply equipment supplied through a cable and plug shall be designed according to a minimum overvoltage category II.

Equipment that is intended to be used under the conditions of a higher overvoltage category shall include appropriate overvoltage protective device (see 4.3.3.6 of IEC 60664-1:2007).

12.4 IP degrees

12.4.1 Degrees of protection against solid foreign objects and water for the enclosures

Enclosures of the EV supply equipment shall have an IP degree, according to IEC 60529 as follows:

- indoor use: at least IP41;
- outdoor use: at least IP44.

Compliance is checked by test in accordance with IEC 60529.

The minimum IP degree for socket-outlets and the vehicle connectors shall be in accordance with their appropriate standards.

IPX4 may be obtained by the combination of the socket-outlet or connector and the lid or cap, EV supply equipment enclosure or EV enclosure.

12.4.2 Degrees of protection against solid foreign objects and water for basic, universal and combined and DC interfaces

The minimum IP degrees for ingress of objects and liquids shall be:

- Indoor use:
  - – vehicle connector when mated with vehicle inlet: IP21;
  - – EV plug mated with EV socket-outlet: IP21;
  - – vehicle connector for case C when not mated: IP21;
  - – vehicle connector for case B when not mated: IP24.
- Outdoor use:
  - – vehicle connector when mated with vehicle inlet: IP44;
  - – EV plug mated with EV socket-outlet: IP44;
  - – vehicle connector when not mated: IP24;
  - – vehicle connector for case B when not mated: IP24;
  - – socket-outlet when not mated: IP24.

Compliance is checked by test in accordance with IEC 60529.

IPX4 may be obtained by the combination of the socket-outlet or connector and the lid or cap, EV supply equipment enclosure or EV enclosure.

NOTE In the following countries, the UL articulated finger probe is used according to national regulations: US, CA.

12.5 Insulation resistance

The insulation resistance measured with a 500 V DC voltage applied between all inputs/outputs connected together (power source included) and the accessible parts shall be:

- for a class I EV supply equipment: R > 1 MΩ;
- for a class II EV supply equipment: R > 7 MΩ.

For this test all extra low voltage (ELV) circuits shall be connected to the accessible parts during the test. The measurement of insulation resistance shall be carried out with the protective impedances disconnected, and after applying the test voltage for the duration of 1 min and immediately after the damp heat continuous test of IEC 60068-2-78, test Ca, at 40 °C ± 2 °C and 93 % relative humidity for four days.

The conditioning test for the insulation test and the touch current can be avoided if the conditioning for test of 12.9 followed by test of 12.5, 12.6 and final test of 12.9, are conducted sequentially in that order.

12.6 Touch current

The touch current between any AC supply network poles and the accessible metal parts connected with each other, and with a metal foil covering insulated external parts, is measured in accordance with IEC 60990 and shall not exceed the values indicated in Table 1.

The touch current shall be measured within one hour after the damp heat continuous test of IEC 60068-2-78, test Ca, at 40 °C± 2 °C and 93 % relative humidity for four days, with the electric vehicle charging station connected to AC supply network in accordance with IEC 60990.

The test voltage shall be 1,1 times the maximum rated voltage.

Table 1 – Touch current limits

	Class I	**Class II**
Between any network poles and the accessible metal parts connected with each other and a metal foil covering insulated external parts	3,5 mA	0,25 mA
Between any network poles and the metal inaccessible parts normally non-activated (in the case of double insulation)	Not applicable	3,5 mA
Between inaccessible and accessible parts connected with each other and a metal foil covering insulated external parts (additional insulation)	Not applicable	0,5 mA

This test shall be made when the EV supply equipment is functioning with a resistive load at rated output power.

Circuitry that is connected through a fixed resistance or referenced to earth (for example, proximity function and control pilot function) are disconnected before this test.

The equipment is fed through an isolating transformer or installed in such a manner that it is isolated from the earth.

12.7 Dielectric withstand voltage

12.7.1 AC withstand voltage

The dielectric withstand voltage, at power frequency of 50 Hz or 60 Hz, shall be applied for 1 min as follows:

1) For a class I EV supply equipment.

   (Un + 1 200 V) (r.m.s.) in common mode (all circuits in relation to the exposed conductive parts) and differential mode (between each electrically independent circuit and all other exposed conductive parts or circuits) as specified in 5.3.3.2 of IEC 60664-1:2007.

   NOTE 1 Un is the nominal line to neutral voltage of the neutral-earthed supply system.

2) For a class II EV supply equipment.

   2 times (Un +1 200 V) (r.m.s). in common mode (all circuits in relation to the exposed conductive parts) and differential mode (between each electrically independent circuit and all other exposed conductive parts or circuits) as specified in 5.3.3.2.3 of IEC 60664-1:2007.

3) For both class I and class II AC EV supply equipment where the insulation between the AC supply network and the extra low voltage circuit is double or reinforced insulation, 2 times (Un + 1 200 V) (r.m.s.) shall be applied to the insulation.

Alternatively the test can be carried out using a DC voltage equal to the AC peak values.

For this test, all the electrical equipment shall be connected, except those items of apparatus which, according to the relevant specifications, are designed for a lower test voltage; current-consuming apparatus (e.g. windings, measuring instruments, voltage surge suppression devices) in which the application of the test voltage would cause the flow of a current, shall be disconnected. Such apparatus shall be disconnected at one of their terminals unless they are not designed to withstand the full test voltage, in which case all terminals may be disconnected.

NOTE 2 For test voltage tolerances and the selection of test equipment, see IEC 61180.

12.7.2 Impulse dielectric withstand (1,2 μs/50 μs)

The dielectric withstand of the power circuits at impulse test shall be tested according to IEC 60664-1.

The impulse voltage shall be applied to live parts and exposed conductive parts.

The test shall be carried out in accordance with the requirements of IEC 61180.

Parts of the EV supply equipment directly connected to the public AC supply network shall be tested according to overvoltage category IV.

Permanently connected EV supply equipment shall be tested according to an overvoltage category III except for the socket-outlet or the vehicle connector in case C where an overvoltage category II applies.

EV supply equipment supplied through a cable and plug shall be tested according to an overvoltage category II.

12.8 Temperature rise

EV supply equipment shall comply with IEC TS 61439-7.

12.9 Damp heat functional test

Following the conditioning defined below, the EV supply equipment is deemed to pass the test, if, it passes the normal sequences test according to A.4.7 of Annex A. The precision of the timing does not need to be verified.

Conditioning:

– For indoor units, 6 cycles of 24 h each to a damp heat cycling test according to IEC 60068-2-30 (Test Db) at (40±3) °C and relative humidity of 95 %;
– For outdoor units, two 12 day periods, with each period consisting of 5 cycles of 24 h each to a damp heat cycling test according to IEC 60068-2-30 (Test Db) at (40±3) °C and relative humidity of 95 %.

12.10 Minimum temperature functional test

The EV supply equipment shall be pre-conditioned in accordance with IEC 60068-2-1, test Ab, at the minimum operating temperature (either -5 °C for indoor, -25 °C outdoor or lower values declared by the manufacturer ± 3 K) for (16 ± 1) h.

The EV supply equipment is deemed to pass the test, if, immediately after the pre-conditioning, it passes the sequences test according to A.4.7 of Annex A while at the minimum operating temperature. The precision of the timing does not need to be verified.

12.11 Mechanical strength

For Mode 2 EV supply equipment the minimum degree of protection of the external enclosure against mechanical impact shall be IK08 according to IEC 62262.

After the test, the samples shall show that:

– the IP degree according to 12.5 is not impaired;
– no part has moved, loosened, detached or deformed to the extent that any safety functions are impaired;
– the test did not cause a condition that results in the equipment not complying with the strain relief requirements, if applicable;
– the test did not result in a reduction of creepage and clearance between uninsulated live parts of opposite polarity, uninsulated live parts and accessible dead or grounded metal below the minimum acceptable values;
– the test did not result in any other evidence of damage that could increase the risk of fire or electric shock.

13 Overload and short-circuit protection

13.1 General

Where connecting points can be used simultaneously and are intended to be supplied from the same input line, they shall have individual protection incorporated in the EV supply equipment.

If the EV supply equipment presents more than one connecting point then such connecting points may have common overload protection means and may have common short-circuit protection means, if those protection means provide the required protection for each of the connecting points (e.g. the common protection device shall have a rating no higher than the lowest rating of the connecting points).

NOTE 1 Such a configuration might have an impact on availability, which could be resolved by adequate load management (eg. load sharing).

If the EV supply equipment presents more than one connecting point that cannot be used simultaneously then such connecting points can have common protection means.

Such overcurrent protective devices shall comply with IEC 60947-2, IEC 60947-6-2 or IEC 61009-1 or with the relevant parts of IEC 60898 series or IEC 60269 series.

NOTE 2 In the following countries, the methods of protection against overcurrent and overvoltage are in accordance with national codes: US, JP,CA.

NOTE 3 In the following countries, the branch circuit overcurrent protection is based upon 125 % of the equipment rating: US,CA.

NOTE 4 In the following countries, EV charging is considered a continuous load and is limited to 80 % of the branch circuit fuse or circuit breaker rating by National rules: US, CA.

NOTE 5 Protection devices can be provided inside the EV supply equipment, in the fixed installation or in both places.

NOTE 6 In the following countries, the equipment earthing path complies with the test requirement in national standard: JP.

13.2 Overload protection of the cable assembly

The EV charging stations or Mode 2 EV supply equipment shall provide overload protection for all cases for all intended cable conductor sizes if not provided by the upstream supply network.

The overload protection may be provided by a circuit breaker, fuse or combination thereof.

If overload protection is provided by a means other than a circuit breaker, fuse or combination thereof, such means shall trip within 1 min if the current exceeds 1,3 times the rated current of the cable assembly.

13.3 Short-circuit protection of the charging cable

The EV charging stations or Mode 2 EV supply equipment shall provide short-circuit current protection for the cable assembly if not provided by the supply network.

In case of short-circuit, the value of I^2t at the EV socket-outlet of the Mode 3 charging station shall not exceed 75 000 A^2s.

In case of short-circuit, the value of I^2t at the vehicle connector (Case C) of the Mode 3 charging station shall not exceed 80 000 A^2s.

NOTE 1 This can either be achieved by integrating the appropriate short-circuit protection device in the EV charging station or by providing the relevant information in the installation manual.

NOTE 2 Protection against short-circuit can be provided inside the EV charging station, in the fixed installation or in both places.

NOTE 3 The value of 80 000 A^2s is identical to that indicated to ISO 17409:2015.

The real value of the prospective short-circuit current is evaluated at the point where the cable assembly is connected.

14 Automatic reclosing of protective devices

The automatic or remote reclosing of protective devices after tripping in the EV supply equipment shall only be possible in case the following requirement is fulfilled:

- the socket-outlet shall not be mated to a plug. This shall be checked by the EV supply equipment.

For automatic or remote reclosing automatic reclosing devices (ARDs) with an assessment means may be used.

The EV supply equipment may close the contactor during an automatic or remote reset cycle to establish conductivity between the protection device and the socket-outlet.

By this procedure the EV supply equipment can check the circuit up to the socket-outlet to be free of fault current.

NOTE In the following countries automatic reclosing of protection means is not allowed: DK, UK, FR, CH.

For case C the EV supply equipment shall not provide automatic or remote reclosing of protective devices.

15 Emergency switching or disconnect (optional)

Emergency switching or disconnect equipment shall be used either to disconnect the supply network from EV supply equipment or to disconnect the socket-outlet(s) or the cable assembly(ies) from the supply network.

Such equipment shall be installed in accordance with national rules.

Such equipment may be part of the supply network or either the EV charging station or the Mode 2 supply equipment.

NOTE In the following countries emergency disconnecting means are provided in an accessible location for some EV supply equipment rated more than 60 A or more than 150 V to ground according to national rules: US, CA.

16 Marking and instructions

16.1 Installation manual of EV charging stations

The installation manual of EV charging stations shall indicate the classification as given in Clause 5.

The EV supply equipment manufacturer shall state the interface characteristics specified in Clause 5 of IEC TS 61439-7:2014 in the manual where applicable. Wiring instructions shall be provided.

If protective devices are included in the EV charging station, the manual shall indicate the characteristics of those protection devices explicitly describing the type and rating. The information may be provided in a detailed electric diagram.

If the protective devices are not in the EV charging station, the manual shall indicate all information necessary for the installation of external protection explicitly describing the type and rating of the devices to be used.

It is recommended that the installation manual be made available to future customers.

If the EV charging station has more than one connection of the equipment to the AC supply network, and does not have individual protection for each connecting point to the vehicles, then the installation manual shall indicate that each connection of the equipment to the AC supply network requires individual protection.

The installation manual shall indicate if the optional function for ventilation is supported by the charging station (6.3.2.2).

The installation manual shall indicate ratings or other information that denote special (severe or unusual) environmental conditions of use, see 5.3.

16.2 User manual for EV supply equipment

User information shall be provided by the manufacturer on the EV supply equipment or in a user’s manual.

Such information shall state:

- which adaptors or conversion adapters are allowed to be used, or
- which adaptors or conversion adapters are not allowed to be used, or
- that adaptors or conversion adapters are not allowed to be used,

and

- that cord extension sets are not allowed to be used.

The user manual shall include information about national usage restrictions.

16.3 Marking of EV supply equipment

The EV supply equipment manufacturer shall provide each EV supply equipment with one or more labels, marked in a durable manner and located in a place such that they are visible and legible during installation and maintenance:

a) EV supply equipment manufacturer's name, initials, trade mark or distinctive marking;
b) type designation or identification number or any other means of identification, making it possible to obtain relevant information from the EV supply equipment manufacturer;
c) “Indoor Use Only", or the equivalent, if intended for indoor use only;

The EV supply equipment manufacturer shall provide each EV supply equipment with one or more labels, marked in a durable manner and located in a place such that they are visible and legible during installation:

d) means of identifying date of manufacture;
e) type of current;
f) frequency and number of phases in case of alternating current;
g) rated voltage (input and output if different);
h) rated current (input and output if different) and the ambient temperature used to determine the rated current;
i) degree of protection;
j) all necessary information relating to the special declared classifications, characteristics and diversity factor(s), severe or unusual environmental conditions of use, see 5.3.

NOTE In the following countries the special environmental conditions are required to be marked: US, CA.

Compliance is checked by inspection and by 16.5.

16.4 Marking of charging cable assemblies case B

Cable assemblies for Mode 1 Case B or Mode 3 Case B shall be marked in a durable manner with the following information:

a) manufacturer's name or trade mark;
b) type designation or identification number or any other means of identification, making it possible to obtain relevant information from the manufacturer;

c) rated voltage;

d) rated current;

e) number of phases.

f) degree of protection

NOTE In the following countries all Mode 1 cable assemblies without a PRCD shall bear the following safety information: “Shall not be used in Germany": DE.

Marking for the entire cable assembly shall be provided in a clear manner by a label or equivalent means.

Compliance is checked by inspection and by 16.5.

16.5 Durability test for marking

Marking made by moulding, pressing, engraving or similar, including labels with a laminated plastic covering, shall not be submitted to the following test.

The markings required by this standard shall be legible with corrected vision, durable and visible during use.

Compliance is checked by inspection and by rubbing the marking by hand for 15 s with a piece of cloth soaked with water and again for 15 s with a piece of cloth soaked with petroleum spirit.

NOTE The petroleum spirit is defined as a solvent hexane with a content of aromatics of maximum 0,1 % in volume, a kauributanol value of 29, an initial boiling point of 65 °C, a final boiling point of 69 °C and a density of approximately 0,68 g/cm3.

After the test, the marking shall be legible to normal or corrected vision without additional magnification. It shall not be easily possible to remove marking plates and they shall show no curling.

Annex A
(normative)

Control pilot function through a control pilot circuit using a PWM signal and a control pilot wire

A.1 General

Annex A describes the control pilot function through a control pilot circuit using pulse width modulation (PWM) for Mode 2, Mode 3 and Mode 4.

Two types of control pilot function are possible: simplified (A.2.3) and typical (A.2.2).

Annex A describes the circuit parameters and sequencing of events for these control pilot functions. The parameters indicated in this Annex A have been chosen in order to ensure the interoperability of systems with those designed according to the standard SAE J1772.

Additional requirements for implementation in Mode 4 system are described in IEC 61851-23.

Annex A is applicable to EV supply equipment and EVs using a control pilot function based on a PWM signal over the control pilot circuit.

A.2 Control pilot circuit

A.2.1 General

Figures A.1 and A.2 illustrate an electric equivalent circuit of the control pilot circuit. The EV supply equipment shall set the duty cycle of the PWM control pilot signal to indicate the maximum current according to Table A.7. The indicated maximum current transmitted shall not exceed the value according to 6.3.1.6.

The EV supply equipment may open the switching device that energizes the EV if the EV draws a higher current than the PWM signal (duty cycle) indicates. In this case, the EV supply equipment shall respect the following conditions:

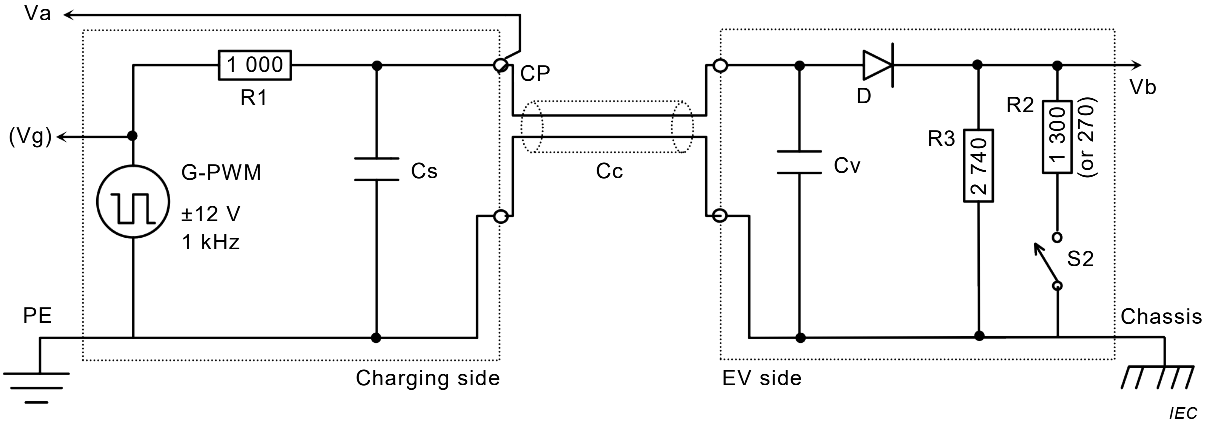
- the allowed response time of the EV, according to Table A.6 (e.g. sequence 6).
- the current tolerance related to the duty cycle generated by the EV supply equipment (1 percentage point).
- the tolerances of the current measurement used in the EV supply equipment itself.

NOTE Total tolerances might be higher than 15 %.

The control pilot circuit shall be designed in accordance with Figures A.1 or A.2 with the values defined in Table A.2, Table A.3 and Table A.4.

The functionality of the control pilot circuit shall follow the requirements defined in Table A.4, Table A.6, Table A.7 and Table A.8.

A.2.2 Typical control pilot circuit



Key

G-PWM	PWM signal generator for pilot function
Va	Pilot wire voltage measured at EV supply equipment output
Vg	Internal voltage of PWM signal generator
R1,Cs	as defined in Table A.2
R2,R3,Cv,D	as defined in Table A.3
Vb	EV measurement of voltage, duty cycle and frequency
CP	control pilot contact
Chassis	vehicle chassis connection

NOTE The references of the components R2 and R3 have been interchanged with respect to IEC 61851-1:2010.

Figure A.1 – Typical control pilot circuit (equivalent circuit)

The EV supply equipment communicates by setting the duty cycle of a PWM signal or a continuous DC voltage signal (Table A.7).

The EV supply equipment may change the duty cycle of the PWM signal at any time.

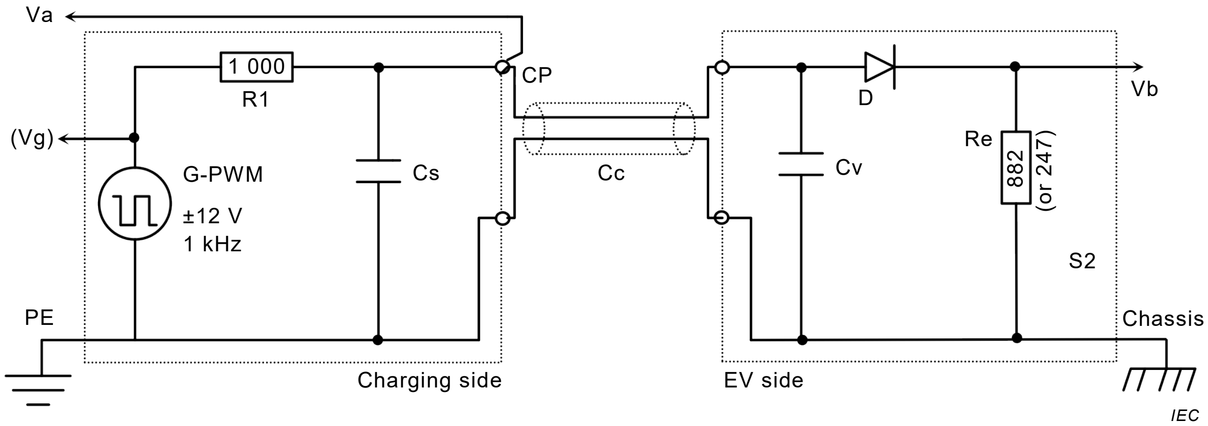
The EV responds by applying a resistive load to the positive half-wave to the control pilot circuit.

For further information about the PWM signal see also Table A.2, Table A.3 and Table A.4.

EVs using typical control pilot circuit (Figure A.1) shall be able to create state B and use it according to the sequences specified in Table A.6.

EV using a typical control pilot circuit shall determine the maximum current from EV supply equipment from the duty cycle of the PWM signal (Table A.8).

A.2.3 Simplified control pilot circuit



Key

G-PWM	PWM signal generator for pilot function
Va	Pilot wire voltage measured at EV supply equipment output
Vg	Internal voltage of PWM signal generator
R1,Cs	as defined in Table A.2
Re,Cv,D	as defined in Table A.3
Vb	EV measurement of voltage, duty cycle and frequency
CP	control pilot contact
Chassis	vehicle chassis connection

Figure A.2 – Simplified control pilot circuit (equivalent circuit)

An EV using the simplified control pilot circuit shall limit itself to single phase charging and shall not draw a current of more than 10 A.

EV supply equipment that supports an EV using the simplified control pilot shall modulate the PWM signal in the same manner as done for EVs using the typical control pilot circuit.

EVs using simplified control pilot circuit (Figure A.2) are not able to create state B.

An EV using the simplified control pilot circuit can measure the duty cycle.

The designer of an EV using the simplified control pilot should be aware that the EV supply equipment can open its switching device, if the EV supply equipment indicates less current (by the duty cycle) than the EV draws (see A 2.1).

It is not recommended to use the simplified control pilot circuit for new EV design.

NOTE In some countries simplified control pilot circuit is not allowed: US, CH.

A.2.4 Additional components and high frequency signals

Digital communication as described in ISO/IEC 15118 series may be carried out over the control pilot conductor. Additional components can be needed to couple this high-frequency signal onto the control pilot signal.

Additional components required for signal coupling shall not deform the control pilot signal beyond the limits defined in Tables A.2 and A.4.

Conformity is tested according to A.4.6.

The maximum inductance of the control pilot circuit of the EV supply equipment is limited to 1 mH (see Table A.3).

The maximum inductance of the control pilot circuit of the EV is limited to 1 mH (see Table A.2).

NOTE ISO 15118-3 provides guidance on selection of components that provide damping.

The additional signal for digital communication shall have a frequency of at least 148 kHz.

The voltage of the high frequency signal (used for digital communication) shall be in accordance with the values given in Table A.1.

Table A.1 – Maximum allowable high frequency signal voltages on control pilot conductor and the protective conductor

Frequency kHz	**Max Peak/Peak voltage** V
148-249	0,4
250-499	0,6
500-1 000	1,2
> 1 000	2,5

One further capacitive (max of 2 000 pF) branch (on the vehicle and on the EV supply equipment) can be used for detection of the high frequency signals, provided the resistance/impedance to ground is higher than 10 kΩ. Such capacitive/resistive branch would typically be used for signal inputs and automatic signal voltage control (refer to Table A.1).

A.3 Requirements for parameters and system behaviour

The control pilot circuit parameters shall be in accordance with Table A.2 and Table A.3 and are shown in Figures A.1 and A.2.

Table A.2 – Control pilot circuit parameters and values for the EV supply equipment

Parameter[a]	Symbol	Min. value	Typical value	Max. value	Unit	Remark
Generator open circuit positive voltage [c]	Voch	11,4	12	12,6	V	
Generator open circuit negative voltage[c]	Vocl	-12,6	-12	-11,4	V	
Frequency generator output	Fo	980	1 000	1 020	Hz	
Pulse width [b c]	Pwo	Per Table A.7 -5µs	Per Table A.7	Per Table A.7 + 5 µs	µs	
Rise time (10 % to 90 %) [c]	Trg	-		2	µs	Design value for the oscillator.
Fall time (90 % to 10 %)[c]	Tfg	-		2	µs	Design value for the oscillator.
Settling time to 95 % steady state [c]	Tsg	-		3	µs	Design value for the oscillator.
Equivalent source resistance	R1	970	1 000	1 030	Ω	970 Ω to 1 030 Ω 1 %equivalent resistors commonly recommended
EV supply equipment capacitance [d]	Cs	300	-	1 600	pF	
Cable capacitance	Cc	-	-	1 500	pF	Case B (cord set)
Optional series (damped) inductance [e]	Lse	-	-	1	mH	Maximum value allowed on off board EV supply equipment

[a] Tolerances to be maintained over the full useful life and under environmental conditions as specified by the manufacturer.

[b] At 0 V crossing of the 12 V signal.

[c] At point Vg as indicated on Figure A.1 and Figure A.2.(to be measured at output – open circuit).

[d] For Mode 3 Case C and for a Mode 2 cord set the max equivalent capacitance is total of Cc + Cs.

[e] Damped inductance. Nominal values of additional components such as L and required damping (R-damp)) used for high-frequency signal, are defined in Table A.11 of ISO 15118-3:2015.

EV pilot circuit values and parameters as indicated on Figures A.1 and A.2 are given in Table A.3.

Table A.3 – EV control pilot circuit values and parameters and values for the EV

Parameter	Symbol	Min. value	Typical value	Max. value	Unit
Permanent resistor value (Figure A.1)	R3	2 658	2 740	2 822	Ω
Switched resistor value for vehicles not requiring ventilation (Figure A.1)	R2 State Cx	1 261	1 300	1 339	Ω
Switched resistor value for vehicles requiring ventilation (Figure A.1)	R2 State Dx	261,9	270	278,1	Ω
Equivalent total resistor value no ventilation (Figure A.2)	Re State Cx	856	882	908	Ω
Equivalent total resistor ventilation required (Figure A.2)	Re State Dx	239	246	253	Ω
Diode (D) voltage drop (2,75 – 10 mA, -40 °C to + 85 °C)	Vd	0,55	0,7	0,85	V
Reverse recovery time	Tr	-	-	200	ns
Total equivalent input capacitance [a]	Cv	-	-	2 400	pF
Optional additional series (damped) inductance [b]	Lsv	-		1	mH

[a] For Mode 3 Case A the max equivalent capacitance is total of Cc + Cv. Cc is given in Table A.2.

[b] Damped inductance. Stray and additional components such as RDamp, L used for high-frequency signal, are defined in Table A.11 of ISO 15118-3:2015.

Value ranges shall be maintained over full useful life and under design environmental conditions.

1 % tolerance resistors are commonly recommended for this application.

Table A.4 indicates the pilot voltage range based on components values in Tables A.2 and A.3. It incorporates an increased voltage margin for Va to allow for measurement tolerances of the EV supply equipment.

Table A.4 – System states detected by the EV supply equipment

Va [a] Lower Level V	Va [a] Nominal V	Va [a] Higher Level V	PWM status [b]	System state	EV connected to the EV supply equipment	S2 [d]	EV ready to receive energy [e]	EV supply equipment ready to supply energy [f]	Remark
11	12	13	Off	A1	no	N/A	No	Not Ready	Vb = 0 V
11	12	13	On	A2 [g]			No	Ready	
10		11	on or off	Ax or Bx [h]	no/yes	open	No	State Dependent	
8	9	10	Off	B1	yes	open	No	Not Ready	Re = R3 = 2,74 kΩ detected
8	9	10	On	B2 [g]			No	Ready	
7		8	On or off	Bx or Cx [h]		open/close	State dependent		
5	6	7	Off	C1		close	Yes	Not Ready	Re = 882 Ω detected EV does not require charging area ventilation
5	6	7	On	C2 [c,g]			Yes	Ready	
4		5	on or off	Cx or Dx [h]			Yes	State dependent	
2	3	4	Off	D1			Yes	Not Ready	Re = 246 Ω detected EV requires charging area ventilation
2	3	4	On	D2 [c,g]			Yes	Ready	
1	N/A	2	On or off	Dx or E [h]		Open or closed	State dependent		
-1	0	1	Off	E	N/A	N/A	N/A	Not Ready	
-10		-1	On or Off	Invalid [c]	N/A	N/A	N/A		Fault in control circuit [c]
-11		-10	Off	F or invalid	N/A	N/A	N/A	Not Ready	
-13	-12	-11	Off	F	N/A	N/A	N/A	Not Ready	
-11		-10	On	x2 or Invalid [h]	No/Yes	open/close	State dependent		[c]
-13	-12	-11	On	x2 [c]	N/A	N/A	State dependent		Low side of PWM signal [c]
Voltage values, Va, as indicated in the table are informative, actual values to be tested according to Clause A.4.									

[a] All voltages are measured after stabilization period. It is recommended for the EV supply equipment to use Vg as reference for the measurement of Va.

[b] PWM status “on” describes a generated square wave voltage of +/-12 V.
PWM status “off” describes a steady state DC voltage.

[c] The EV supply equipment shall check PWM signal low state of -12 V, diode presence, at least once before the closing of the supply switching device on the EV supply equipment.

[d] S2 = Switch in the EV (see Figure A.1).

[e] EV ready to receive energy = EV ready for power transfer by closing S2 contacts.

[f] EV supply equipment ready to supply energy = ready –> PWM status “on”, not ready –> PWM status “off”.

[g] Negative voltage range tolerances of the PWM signal are defined by “Low side of PWM -signal” row (last row).

[h] A control pilot circuit defines its own trigger level to separate the states inside this voltage range. It is recommended to use different trigger level depending on the direction of the state change to include hysteresis behaviour.

There is no undefined voltage range, for the PWM signal, between the system states.

The state is valid if it is within the above values. The state detection shall be noise resistant, e.g. against EMC and high frequency data signals on the control pilot circuit.

For reliable detection of a state, it is recommended to apply averaging of the measurement over several milliseconds or PWM cycles.

The EV supply equipment shall verify that the EV is properly connected by verifying the presence of the diode in the control pilot circuit, before energizing the system. This shall be done at the transition from x1 to x2 or at least once during state x2, before closing the supply switching device. Presence of the diode is detected if the low side of the PWM-signal is within the voltage range defined in Table A.4.

The EV supply equipment shall open or close the supply switching device within the time indicated in Table A.6.

Compliance is tested as in Clause A.4.

The state changes between A, B, C and D are caused by the EV or by the user.

The state changes between state x1 and x2 are created by the EV supply equipment.

A change between states x1 and x2 indicates an availability (x2) or unavailability (x1) of power supply to the EV.

Table A.5 – State behaviour

States	Description	Behaviour
x1 [a]	The EV supply equipment will not deliver the energy, e.g.: • due to the lack of available power in the grid for supply • the EV supply equipment has intentionally stopped due to intermittent or other power supply limitations.	If energy is available, the EV supply equipment will change to x2 [b] according to sequence 3.1 or 3.2 in Table A.6. The EV may use this transition as a trigger to start or resume charging.
State E	This state is generally caused by an error condition, for example: • No power to the EV supply equipment (e.g. AC voltage outage). • Short circuit between control pilot and the protective conductor. This state shall not intentionally be used by the EV supply equipment for signalling, except the work around in A.5.3.	The EV supply equipment unlatches / unlocks the socket-outlet at maximum of 30 s, if any.
State F	This state is intentionally set by the EV supply equipment to signal a fault condition, for example: • maintenance of the EV supply equipment is needed	The EV supply equipment unlatches / unlocks the socket-outlet at maximum of 30 s, if any.

NOTE 1 In case of a power outage and if the EV supply equipment has a backup battery, the EV supply equipment can stay in state x1. After the drainage of the battery, the EV supply equipment will enter state E.

NOTE 2 In case of state F and if the EV supply equipment is able to unlatch / unlock the socket-outlet via user interaction (e.g. authorisation) there is no need to unlatch / unlock within 30 s according to sequence 12 in Table A.6.

[a] State x1 can be referred to state A1 or state B1 or state C1 or state D1.

[b] State x2 can be referred to state B2 or state C2 or state D2.

After changing to state F and while the reason for changing to state F persists, an EV supply equipment with permanently attached cable (case C) shall:

– remain in state F, or
– remain in state F for at least 300 ms and then change to state x1 (and stays there), in order to detect if an EV is connected.

If the failure is not recovered after disconnecting the vehicle connector, the EV supply equipment shall:

– remain in or change to state F, or
– remain in state x1, if the EV supply equipment provides an indicator (e.g. a display) which shows “not available”.

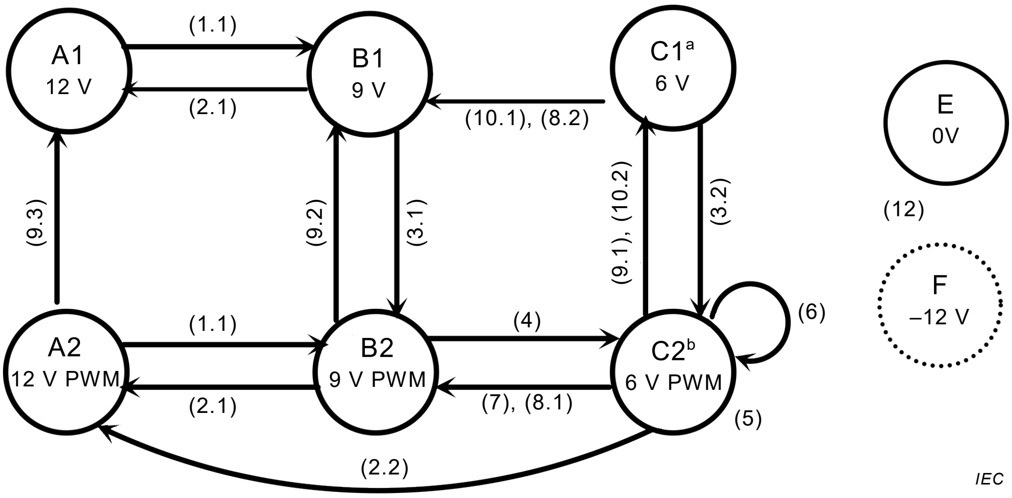
In the absence of a fault condition in the EV supply equipment, the EV supply equipment shall not use the state F in order to signal that the EV supply equipment will not deliver the energy to the EV. Instead, this shall be done by the state x1.

A transition from state E or state F to any other state (x1 or x2) is allowed.

If the EV is connected to the EV supply equipment which does not use 5 % duty cycle, and authentication (e.g. RFID identification, payment, etc.) is needed, the control pilot signal shall stay at x1 as long as the energy is not allowed to be supplied. In case, no authentication is needed, the system may go to state x2.

In case EV supply equipment requires authentication to supply power, a change from states CX or DX to state BX shall not lead to loss of authentication. This means that no repeated authentication shall be needed.

See Figures A.3 and A.4 for state diagrams.



Numbers in brackets refer to the sequence reference in Table A.6.

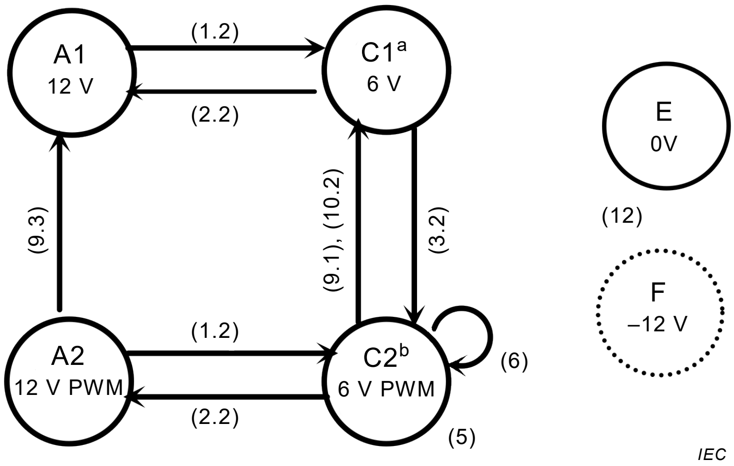
A change from any state to state Ax, E or F may take place at any time.

a Can be state D1 (3V).

b Can be state D2 (3V PWM).

Not all state changes and sequences described in Table A.6 are shown in this figure, e.g. a change from any state to state Ax, state E or state F may take place at any time.

Figure A.3 – State diagram for typical control pilot (informative)



Numbers in brackets refer to the sequence reference in Table A.6.

Not all state changes and sequences described in Table A.6 are shown in this figure, e.g. a change from any state to state Ax, state E or state F may take place at any time.

[a] Can be state D1,(3V).

[b] Can be state D2,(3V PWM).

NOTE Simplified pilot is not supported in SAE J1772:2016.

Figure A.4 – State diagram for simplified control pilot (informative)

Table A.6 indicates the principle sequences and transitions from one state to another with the timing requirements where applicable. Some transitions that may take place are not indicated in the table.

If the EV supply equipment or the EV changes to a new state within the timing indicated for that sequence, the new sequence is initiated and replaces the previous sequence.

Table A.6 – List of sequences

Table A.6: Sequence 1.1 Plug-in (With S2)

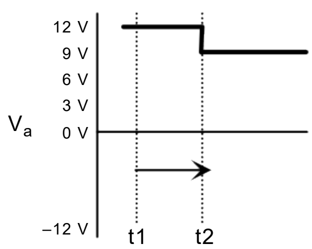
Example diagrams	State or transition	Conditions	Timing
1.1  AC supply remains off S2 remains open AC current remains off Trigger: n/a	A1	(1) EV not connected +12 V	$(t_2\text{-}t_1)$ = No max[a]
	A1→B1 or A2→B2 (not shown in image)	(2) The cable assembly is connected to the vehicle and to the EV supply equipment, +9 V.	

Table A.6: Sequence 1.2 Plug-in (w/o S2 or S2 always in close position)

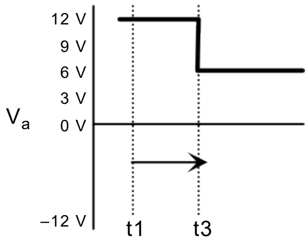
Example diagrams	State or transition	Conditions	Timing
1.2  AC supply remains off S2 already or always closed AC current remains off Trigger: n/a	A1	(1) EV not connected +12 V	$(t_3\text{-}t_1)$ = No max[a]
	A1→C1/D1 or A2→C2/D2 (not shown in image)	(3) The cable assembly is connected to the vehicle and to the EV supply equipment NOTE 1 This sequence indicates EV operates in simplified control pilot function. NOTE 2 t2 does not exist in this sequence. NOTE 3 In case this sequence 1.2, the EV supply equipment can assume EV operates in simplified control pilot and may not follow the current limitation indication by PWM. To detect if the EV operates in simplified control pilot, it is recommended for the EV supply equipment to start in state A1. Simplified pilot is not supported in SAE J1772:2016.	

Table A.6: Sequence 2.1 Unplug at state Bx

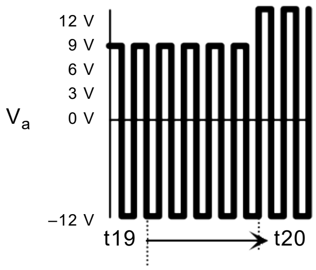
Example diagrams	State or transition	Conditions	Timing
2.1  AC supply remains off S2 remains open AC current remains off Trigger: n/a	B2→A2 or B1→A1 (not shown in image)	(19) Plug disconnected from the EV supply equipment or vehicle connector disconnected from the vehicle inlet. (20) EV not connected The EV supply equipment shall allow removal of the plug automatically, at a maximum of 5 s, when entering state A (case A or B) unless the locking was initiated through user interaction (e.g. authorization). Then unlatching / unlocking can be done only by using the adequate user interaction or both. In case A, EV with attached cable, a switch may be added on the control pilot circuit, on the EV side (cable, plug, vehicle), to simulate the EV disconnection (state A).	$(t_{20}\text{-}t_{19})$ = No max[a]

Table A.6: Sequence 2.2 Unplug at state Cx, Dx

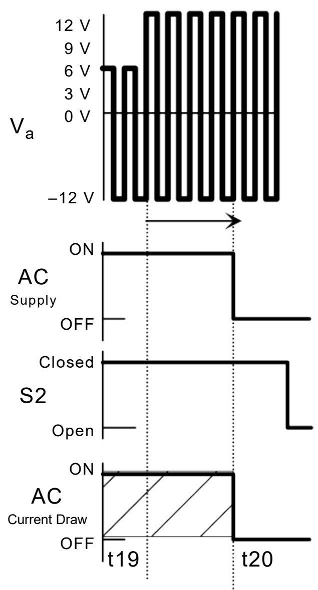
Example diagrams	State or transition	Conditions	Timing
2.2 	C2, D2 → A2	(19) In case of a failure, if – the control pilot circuit is broken or – the plug disconnected from the EV supply equipment under load or – the vehicle connector disconnected from the vehicle inlet under load, the EV supply equipment shall open its switching device. The EV shall open its S2, if any.	$(t_{20}-t_{19})$ = Max 100 ms from t_{19} Max 3 s
	C2, D2 → A2	(19) In case of normal operation, – the plug disconnected from the EV supply equipment not under load or – the vehicle connector disconnected from the vehicle inlet not under load, the EV supply equipment shall open its switching device. The EV shall open its S2, if any.	$(t_{20}-t_{19})$ = Max 100 ms from t_{19} Max 3 s
Trigger: C2 (or D2) → A2			
Trigger: C1 (or D1) → A1	C2,D2→ A2 or C1,D1→ A1	(19) EV not connected The EV supply equipment shall allow removal of the plug automatically, at a maximum of 5 s, when entering state A (case A or B) unless the locking was initiated through user interaction (e.g. authorization). Then unlatching / unlocking can be done only by using the adequate user interaction or both. In case A, EV with attached cable, a switch may be added on the pilot line, on the EV side (cable, plug, vehicle), to simulate the EV disconnection (state A), an EV that uses this, needs to make sure that the load is below 1 A.	

Table A.6: Sequence 3.1 EV supply equipment power available (state B)

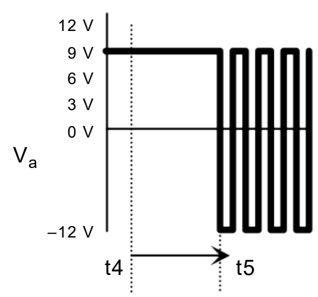
Example diagrams	State or transition	Conditions	Timing
3.1  AC supply remains off S2 remains open AC current remains off Trigger: EV supply equipment is able to supply energy	B1→B2	(5) The EV supply equipment is now able to supply power, and indicates the maximum current by the PWM duty cycle. The EV may detect this transition from B1 to B2, e.g. for wake-up. NOTE 4 This sequence can take place in the beginning of a charging session or to resume a charging session.	(t_5-t_4) = No Max[a]

Table A.6: Sequence 3.2 EV supply equipment power available (state C)

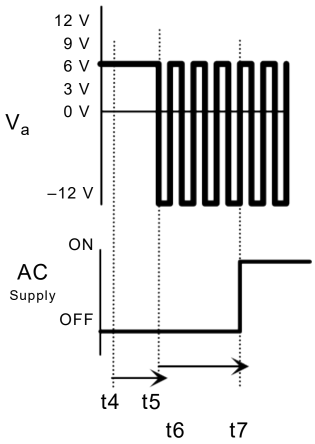
Example diagrams	State or transition	Conditions	Timing
3.2  S2 remains closed AC current remains off Trigger t4: EV supply equipment is able to supply energy Trigger t5: C1 →C2 or D1→D2	C1→C2 Or D1→D2 (not shown in image)	(5) The EV supply equipment is now able to supply energy, and indicates the maximum current by the PWM duty cycle.	$(t_5\text{-}t_4)$ = No Max[a]
		(6) The EV is ready to receive energy. (7) EV supply equipment energizes the system. If state D2 is detected, the supply will close only if ventilation requirements are met. If the duty cycle is equivalent to 5 % the EV Supply Equipment may not energize the system without digital communication (see Table A.8). NOTE 5 Maximum values for the inrush current of the EV are defined in ISO 17409.	$(t_6\text{-}t_5)$ = 0 s $(t_7\text{-}t_6)$ = Max 3 s

Table A.6: Sequence 4 EV ready to charge

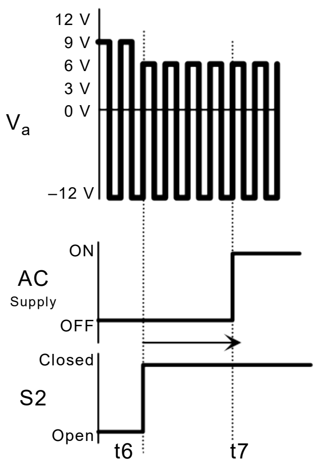
Example diagrams	State or transition	Conditions	Timing
4  AC current remains off Trigger: B2→C2,D2	B2→C2,D2	(6) The EV is ready to receive energy. (7) EV supply equipment energizes the system, unless the EV supply equipment changes within the 3 s timing to another state (e.g. C1).	$(t_7\text{-}t_6)$ = Max 3 s
	C2,D2	If state D2 is detected, the supply will close only if ventilation requirements are met. In case of an EV asks for ventilation delay, ventilation command turns on after transition from state C2 to state D2 in 3 s. in case the EV supply equipment does not have ventilation, it shall open its switching devices and may change to state x1. NOTE 6 In case of 5 % duty cycle, the amount of current is indicated by the digital communication, the EV supply equipment may close the supply switching device only after the authorization given by digital communication. NOTE 7 Maximum values for the inrush current of the EV are defined in ISO 17409.	

Table A.6: Sequence 5 EV starts charging

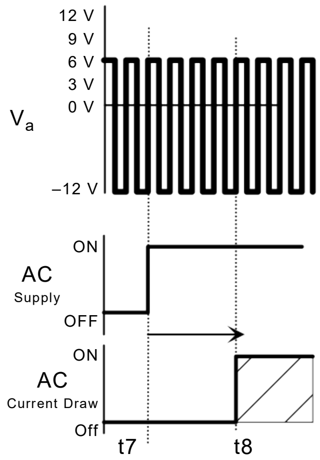
Example diagrams	State or transition	Conditions	Timing
5  S2: remains closed Trigger: AC supply to EV	C2,D2	(8) Charge current drawn by the EV within the limit indicated by the duty cycle of the PWM signal given in Table A.8. NOTE 8 Maximum values for the inrush current of the EV are defined in ISO 17409.	$(t_8\text{-}t_7)$ = No Min, No Max[a]

Table A.6: Sequence 6 Current change

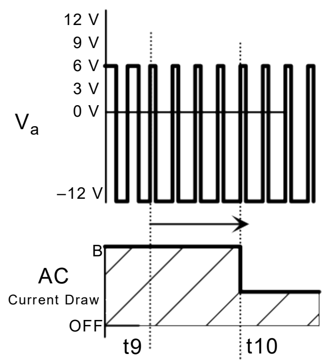
Example diagrams	State or transition	Conditions	Timing
6  AC supply remains available S2 remains closed Trigger: PWM duty cycle change	C2,D2	(9) EV supply equipment indicates an adjustment to the maximum AC line current. Such a change may originate from the grid, by manual settings or automatic changes calculated by the EV supply equipment. The EV supply equipment may change the PWM duty cycle at any time to any valid PWM duty cycle. In normal operation, during the 5 s allowed adjustment time (t10-t9), the EV supply equipment shall not initiate a new sequence 6 for changing the PWM.	Max 10 s From the instant when the EV supply equipment gets the status and responds by adjusting the duty cycle
		(10) The EV shall adjust its maximum current drawn to be equal or below the maximum current indicated by the PWM duty cycle.	$(t_{10}\text{-}t_9)$ = Max 5 s

Table A.6: Sequence 7 EV stops charging

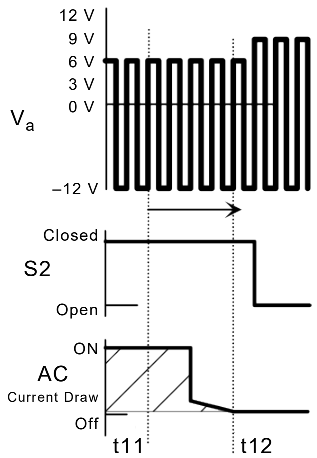
Example diagrams	State or transition	Conditions	Timing
7  AC supply remains available Trigger: n/a	C2, D2	(11) In normal operation an EV shall decrease the current draw to minimum (less than 1 A) before opening S2. During a non-normal operation (emergency) the EV may open S2 immediately.	$(t_{12}-t_{11})$ = No Max[a]
	C2,D2 → B2	(12) The EV opens S2. NOTE 9 SAE J1772:2016 does not specify any minimum current draw before opening S2.	

Table A.6: Sequence 8.1 EV supply equipment responds to EV opens S2 (with PWM)

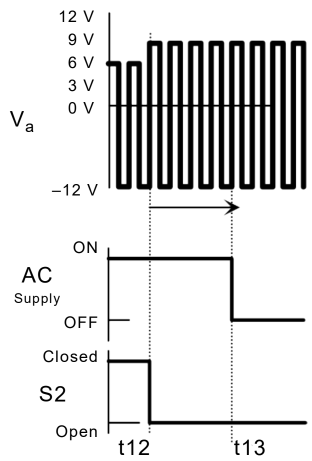
Example diagrams	State or transition	Conditions	Timing
8.1  AC current draw: off and remains off Trigger: C2/D2 → B2	B2	(13) The EV supply equipment shall open its switching device responding to a state change from state C2/D2 to state B2. (During a non-normal operation (emergency), the switching device might have to open under load). NOTE 10 SAE J1772:2016 defines a max. time of 3 s.	$(t_{13}-t_{12})$ = Max 100 ms

Table A.6: Sequence 8.2 EV supply equipment responds to EV opens S2 (w/o PWM)

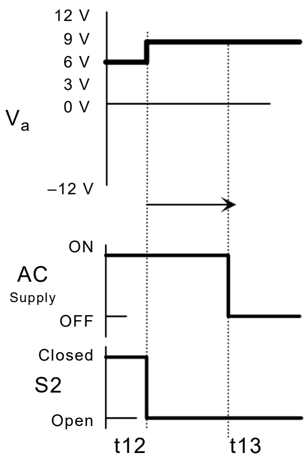
<table>
<tr><th>Example diagrams</th><th>State or transition</th><th>Conditions</th><th>Timing</th></tr>
<tr><td>8.2

AC current draw: off and remains off
Trigger: C1/D1 → B1</td><td>B1</td><td>(13) The EV supply equipment shall open its switching device responding to a state change from state C1/D1 to state B1.
An EV using the simplified pilot circuit is not able to generate this sequence.
Simplified pilot is not supported in SAE J1772:2016.</td><td>$(t_{13}-t_{12})$ =
Max 100 ms</td></tr>
</table>

Table A.6: Sequence 9.1 EV supply equipment requests to stop charging

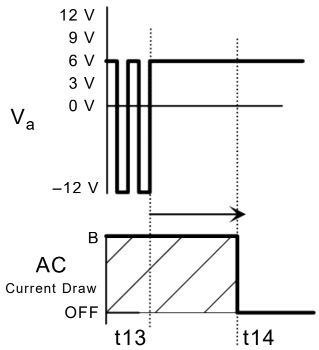
<table>
<tr><th>Example diagrams</th><th>State or transition</th><th>Conditions</th><th>Timing</th></tr>
<tr><td rowspan="3">9

AC supply remains on
S2 remains closed
Trigger: C2,D2→C1,D1</td><td>C2,D2→C1,D1</td><td>(13) EV supply equipment may change to state x1 in order to indicate the EV to stop the current draw.</td><td rowspan="2">$(t_{14}-t_{13})$ =
Max 3 s</td></tr>
<tr><td>C1,D1</td><td>(14) The EV shall respond to the steady state PWM, and stops the current draw.</td></tr>
<tr><td colspan="3">The EV supply equipment may open its switching device if the vehicle load current exceeds the maximum current indicated by PWM according to Table A.8.</td></tr>
</table>

Table A.6: Sequence 9.2 EV supply equipment stops PWM at state B

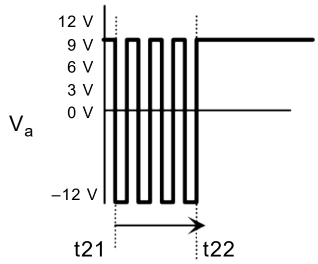
Example diagrams	State or transition	Conditions	Timing
9.2  AC supply remains off S2 remains open AC current draw remains zero Trigger: n/a	B2 → B1	(22) EV supply equipment may stop the PWM at any time. No action by the EV needs to take place. In case sequence 3.1 will follow sequence 9.2, the EV supply equipment shall wait at least 3 s. NOTE 11 This sequence might interfere with time triggered charging or preconditioning of the EV.	$(t_{22}\text{-}t_{21})$ = No Max[a]

Table A.6: Sequence 9.3 EV supply equipment stops PWM at state A

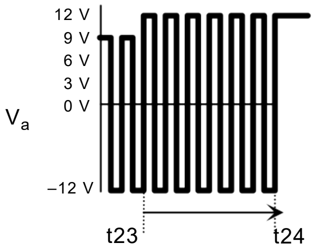
Example diagrams	State or transition	Conditions	Timing
9.3  AC supply remains off S2 remains open AC current draw remains zero Trigger B2/C2/D2 → A2	A2 → A1	(23) EV supply equipment may stop the PWM at any time. (24) No action by the EV needs to take place. EV disconnected. NOTE 12 SAE J1772:2016 requires a turn-off time less than 2 s.	$(t_{24}\text{-}t_{23})$ = No Max[a]

Table A.6: Sequence 10.1 EV responds to stop charging request

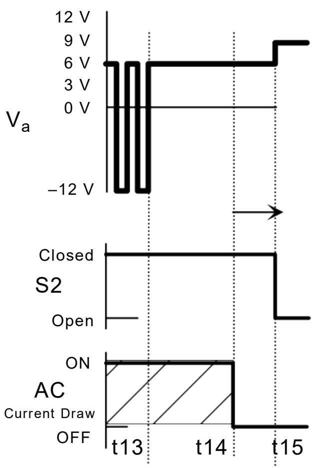
Example diagrams	State or transition	Conditions	Timing
10.1  AC supply remains on Trigger:EV stops current draw		This sequence follows seq. 9.1. and the EV responds to the steady state by stopping the current draw.	
	C1,D1→B1	(15) The EV shall open S2.	$(t_{15}\text{-}t_{14})$ = Max 3 s
	This sequence shall be followed by sequence 8.2 An EV using the simplified pilot circuit is not able to generate this sequence Simplified pilot is not supported in SAE J1772:2016.		

Table A.6: Sequence 10.2 EV does not respond to a stop charging request

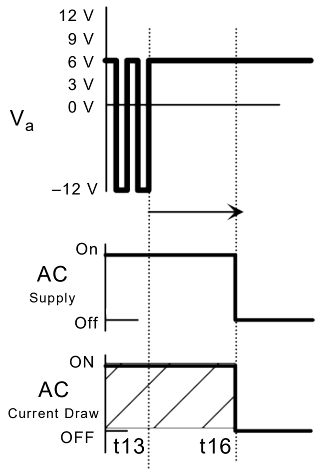
Example diagrams	State or transition	Conditions	Timing
10.2  S2 remains closed Trigger: C2/D2 → C1,D1		This sequence follows seq. 9.1. but the EV does not respond to the steady state and does not stop the current draw, contrary to seq. 9.1.	
	C1,D1	(16) The EV supply equipment may open its switching device under load. (Timer starts upon the PWM change)	$(t_{16}-t_{13}) =$ Min 6 s
	NOTE 14 t_{15} does not exist due to any change of S2 in this sequence. Simplified pilot is not supported in SAE J1772:2016.		

Table A.6: Sequence 11 EV signal to the EV supply equipment

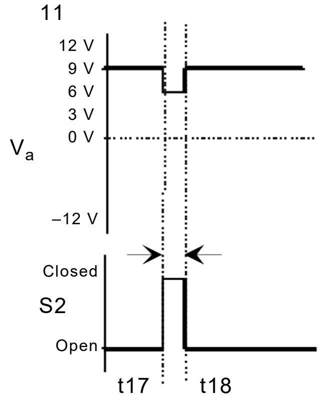
Example diagrams	State or transition	Conditions	Timing
11  AC supply remains off AC current draw remains off Trigger: see ISO/IEC 15118	Bx→Cx/Dx →Bx	(17, 18) A transition from state Bx to Cx or Dx and Cx or Dx to Bx. The EV supply equipment shall not move to state F due to sequence 11. This sequence is optional, and shall be used only with digital communication (ISO/IEC 15118 series). This sequence may be used by the EV in order to signal the EV supply equipment (e.g. wakeup the digital modem). EV shall not draw current during this sequence.	$(t_{18}-t_{17}) =$ Min 200 ms, Max 3 s

Table A.6: Sequence 12 State caused by error or fault condition

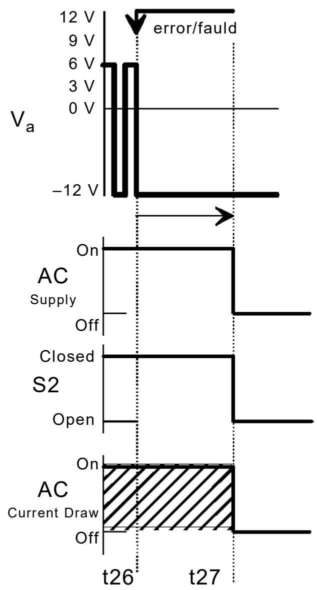
Example diagrams	State or transition	Conditions	Timing
 Trigger: any state → E or F	XX→F XX→E (not shown in image)	Changing from any state to state E, the EV supply equipment switching device shall be open.	$(t_{27}\text{-}t_{26})$ = Max 3 s
		EV shall open S2, if any.	$(t_{27}\text{-}t_{26})$ = Max 3 s
		The EV supply equipment unlatches / unlocks the socket-outlet if any. (Timing not shown on diagram). NOTE 13 In case B and if the cable assembly belongs to the EV supply equipment owner, the unlatching / unlocking is at the EV supply equipment owner's discretion.	Max 30 s
If locking is used, the EV supply equipment shall latch/lock the plug connected to the socket-outlet before closing its switching device, according to IEC 62196 series.			
• Va – Voltage of the control pilot at the socket-outlet or at the vehicle connector (as shown in Figure A.1 and Figure A.2). • AC supply – Status of the relays/contactor at the EV supply equipment (EV supply equipment is ready to supply energy). • S2 – Switching contacts terminals of the EV switch. • AC current draw – EV can draw electric energy.			
[a] The indication "No Max" implies that the delay time has no constraints and may depend on external influences and the conditions existing on the EV supply equipment or the EV.			

Table A.7 – PWM duty cycle provided by EV supply equipment

Maximum current I_{av}	**Nominal control pilot duty cycle D_N**	**Description**
I_{av} = 0 A	D_N = 0 %	Continuous -12 V, EV supply equipment not available; state F
	D_N = 100 %	No current available – state x1 (see Table A.5)
Maximum current is indicated via digital communication	D_N = 5 %	A duty cycle of 5 % indicates that digital communication is required and shall be established between the EV supply equipment and EV before enabling energy supply. If digital communication cannot be established, the EV supply equipment shall: • stay in 5 % duty cycle or • change to x1 (100 % duty cycle) for at least 3 s or • change to x1 (100 % duty cycle) for at least 3 s and then change to a duty cycle between 10 % and 96 %.
6 A ≤ I_{av} ≤ 51 A	$D_N = I_{av}$ / 0,6 A	10 % ≤ D_N ≤ 85 %
51 A < I_{av} ≤ 80 A	$D_N = (I_{av}$ / 2,5 A + 64)	85 % < D_N ≤ 96 %

NOTE Duty cycle tolerances are indicated in Table A.2.

Table A.8 – Maximum current to be drawn by vehicle

Control pilot duty cycle D_{in} at EV plug (in case A) or vehicle inlet	**Maximum current I_{max}.to be drawn by vehicle**	**Description**
Duty cycle < 3 %	0 A	Drawing current not allowed
3 % ≤ D_{in} ≤ 7 %	As indicated by digital communication	A duty cycle of 5 % indicates that digital communication according to ISO/IEC 15118 series or IEC 61851-24 is required and shall be established between the EV supply equipment and the EV before energy supply. Drawing current is not allowed without digital communication. Digital communication may also be used with other duty cycles.
7 % < D_{in} < 8 %	0 A	Drawing current not allowed
8 % ≤ D_{in} < 10 %	6 A	
10 % ≤ D_{in} ≤ 85 %	D_{in} × 0,6 A	
85 % < D_{in} ≤ 96 %	$(D_{in}$ – 64)× 2,5 A	
96 % < D_{in} ≤ 97 %	80 A	
97 % < D_{in} ≤ 100 %	0 A	Drawing current not allowed.
If the PWM signal is between 8 % and 97 % and there is a digital communication established, the maximum current shall not exceed the current indicated by the lower value of either the PWM signal or the digital communication. In 3-phase systems, the duty cycle value indicates the current limit per each phase. The EV supply equipment can start at any valid value of the duty cycle, and may change during power supply. The EV shall detect the frequency. In case the frequency is outside of (1 ± 5 %) kHz, the EV should not charge. For EVs using the simplified control pilot this is not applicable. The value given as “Maximum current (I_{max}) to be drawn by vehicle” does not cover either inrush current or leakage current such as current flowing to y-capacitors.		

A.4 Test procedures

A.4.1 General

This Clause A.4 describes tests for immunity of EV supply equipment to wide tolerances on the control pilot circuit and the presence of high frequency data signals on the control pilot circuit. The EV supply equipment is designed to be in conformity with the parameters as defined in Clauses A.2 and A.3. However, it is necessary for the EV supply equipment to be tolerant to slight parameter changes (due for example to poor contacts or leakage on the control pilot circuit) in order to ensure reliable supply of energy to EVs under most conditions.

A.4.2 Constructional requirements of the EV simulator

Testing is done using an EV simulator on the control pilot circuit that allows the testing in normal operation and at the tolerance limits allowed for the voltage and including the imposition of a high frequency signal on the control pilot circuit. The test scheme described in Clause A.4 allows the testing of the EV supply equipment when in normal operation and when subjected to high frequency imposed signals on the control pilot circuit.

An EV simulator shall have the possibility of testing the EV supply equipment with all three possible resistor values as indicated in Table A.9 with the following values for the other components.

- Cv_{test} shall use the maximum value from Table A.3 (including the 1 000 pF of the generator);
- Lsv_{test} shall use the maximum allowed value from Table A.3;
- Cc_{test} shall use the maximum value from Table A.2;
- The high frequency test signal shall be injected at the EV supply equipment socket-outlet for cases A and B, and at the vehicle coupler for case C;
- The diode shall conform with the specifications in Table A.3;
- The 9 test resistor values shall be within a tolerance of 0,2 % of the value indicated in Table A.9.

Table A.9 – Test resistance values

Test resistor	Minimum value	Nominal Value	Maximum Value
R3$_{test}$ (Ω)	1 870	2 740	4 610
R2 $_{test}$ (Ω) state Cx	909	1 300	1 723
R2 $_{test}$ (Ω) state Dx	140	270	448
This table is not applicable to values used on vehicles (see Table A.3).			

NOTE An example of a test setup is described in A.4.10, Figure A.8.

A.4.3 Test procedure

The proper function of the EV supply equipment shall be tested under the following conditions.

A sine wave generator with an impedance of 50 Ω is connected to the control pilot circuit via a 1 000 pF capacitor, as shown in Figure A.8.

The output amplitude of the sine wave generator has to be adjusted in such a way that the high frequency voltage component on the control pilot conductor is 2,5 V peak-peak at 1 MHz, measured at the EV socket-outlet (case A and B) or the vehicle connector (case C or a Mode 2 cable assembly) in state B at the beginning of each sequence.

The frequency of the sine wave generator shall sweep through the frequency range from 1 MHz to 30 MHz with a logarithmic step width of 4 % and a holding time of 0,5 s.

Unless otherwise specified, input voltage from power supply shall be the rated value, within the range of its tolerance.

Unless otherwise specified, the tests shall be carried out in a draught-free location and at an ambient temperature of (20 ±5) °C.

NOTE The measure of the control pilot wire will take place on the EV supply equipment, socket-outlet or plug, in case A and case B, and on the EV coupler in case C.

A.4.4 Oscillator frequency and generator voltage test

$R2_{test}$ (state Cx), $R2_{test}$ (state Dx) and $R3_{test}$ shall be at the nominal value for this test.

The frequency shall be within ±0,5 % of 1 000 Hz at state B2 and C2 and D2 (if ventilation supported).

Frequency and voltage shall be measured at the contacts CP and PE of the EV socket-outlet (in case A and case B) or of the vehicle connector (in case C or Mode 2 cable assembly).

The precision of measurements of voltages for this test shall be better than ± 0,5 %.

The voltage measured at the EV supply equipment output shall be as given in Table A.10.

Table A.10 – Parameters of control pilot voltages

	Minimum voltage	Maximum voltage
In state A1 and positive part of PWM signal in state A2	11,4	12,6
In state B1 and positive part of PWM signal in state B2	8,37	9,59
In state C1 and positive part of PWM signal in state C2	5,47	6,53
Negative part of PWM signal in states A2 and B2	-12,6	-11,4

The internal resistor of the EV supply equipment (R1_calc) value is calculated by the formula

$$R1\_calc = 2\,740 \times (U\_StateA - U\_StateB) / (U\_StateB - 0{,}7)$$

U_State A and U_state B are the two positive voltage values measured during the test of Table A.10 and VR2 is the value of the positive voltage across $R2_{test}$ in state B.

R1_calc shall be 1 000 Ω ± 3 %.

A.4.5 Duty cycle test

Duty cycle shall be tested at 5 % (if any), 10 % and the maximum current declared by the EV supply equipment manufacturer (in case the EV supply equipment cannot change the PWM it shall be tested only at the default duty cycle).

$R2_{test}$ (state Cx), $R2_{test}$ (state Dx) and $R3_{test}$ shall be at the nominal value for this test.

The measurement shall be carried out at the contacts CP and PE of the socket-outlet (in case A and case B) or of the vehicle connector (in case C).

The duty cycle shall be evaluated at 0 V crossing.

A.4.6 Pulse wave shape test

The PWM pulse shape shall be within the values indicated in Table A.11.

$R2_{test}$ (state Cx), $R2_{test}$ (state Dx) and $R3_{test}$ shall be at the nominal value for this test.

Table A.11 – Test parameters of control pilot signals

Parameter		Maximum value	Unit
Rise time (10 % to 90 %)	State B	10	µs
	State C	7	µs
	State D[a]	5	µs
Fall time (90 % to 10 %)	States B, C, D[a]	13	µs

NOTE Signals are evaluated for the values of the nominal resistance in the control pilot test circuit in Table A.9.

[a] In case ventilation is supported by the EV supply equipment.

A.4.7 Sequences test

A.4.7.1 General

This test checks the AC supply and the timing in order to test the operation at the maximum and minimum allowed voltage levels.

These tests verify the operation of the pilot control over a complete cycle using the resistance values defined in Table A.12.

In case the EV supply equipment cannot change the PWM duty cycle, there is no need to meet sequence 6.

For the EV supply all sequences need to be checked with the timing according to Table A.6. A minimum delay of 20 s shall separate the sequences unless a shorter delay is required by Table A.6.

The EV simulator shall wait for at least 20 s in case of a “no max” requirement for the EV before proceeding with the next step.

Unlatching / unlocking of the coupler in the EV supply equipment, if any, needs to take place according to Table A.5.

Four complete standard charging cycles shall be performed using the resistor values indicated in Table A.12. The EV supply equipment shall be deemed to have failed the test if the cycle is not completed.

Table A.12 – Parameters for sequence tests

	R3 test Ω	R2 test Ω State Cx	R2 test Ω State Dx	HF voltage
Test 1	4 610	1 723	448	Not present
Test 2	4 610	1 723	448	Present
Test 3	1 870	909	140	Not present
Test 4	1 870	909	140	Present

Resistances tolerance is better than or equal to ± 0,2 %.

HF voltage test is only required for EV supply equipment designed for digital communication. Lower voltages may apply for EV supply equipment not designed for digital communication systems.

NOTE HF voltage test is under consideration in ISO 15118-3.

A.4.7.2 Sequence test using the typical control pilot circuit

Figure A.5 shows a charging sequence using the typical control pilot circuit.

Test the EV supply equipment by simulating an EV using the typical control pilot circuit 1.1 --> 3.1 --> 4 --> 7 --> 8.1.--> 4 --> 6 --> 7 --> 8.1 --> 2.1.--> 9.3 as shown in Figure A.5.

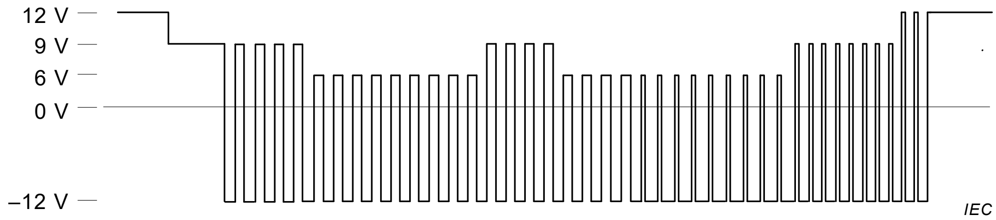


Figure A.5 – Test sequence using a typical control pilot circuit

A.4.7.3 Sequence test using the simplified control pilot circuit

Test the EV supply equipment by simulating an EV using the simplified control pilot circuit using sequences 1.2 --> 3.2 --> 5 --> 6 --> 2.2 as shown in Figure A.6.

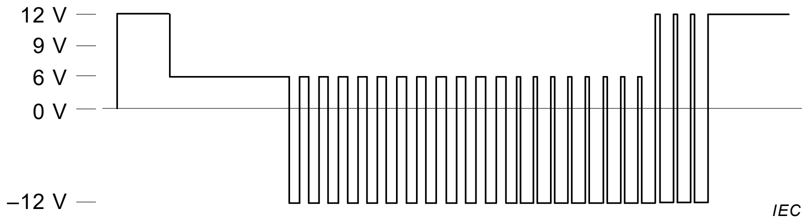


Figure A.6 – Test sequence using the simplified control pilot circuit

A.4.7.4 Optional testing the EV supply equipment that support grid

Optional testing the EV supply equipment that support grid management by simulating an EV using the typical control pilot circuit is shown in Figure A.7.

This test is done using the nominal values of R2 (state Cx), R2 (state Dx) and R3 given in Table A.12 using the sequences:

1.1 --> 3.1 --> 4 --> 9.1 --> 10.1 --> 8.2 --> 3.1 --> 4 --> 7 --> 8.1 --> 2.1--> 9.3

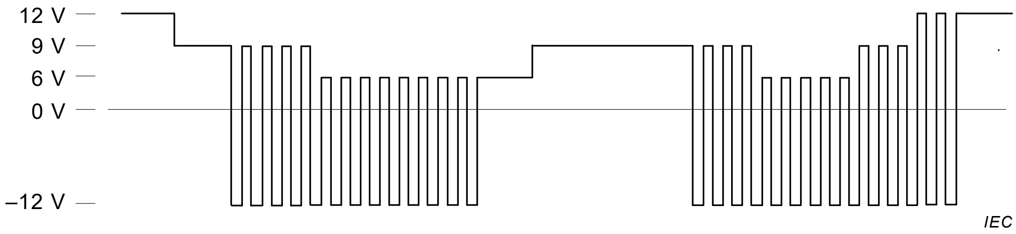


Figure A.7 – Optional test sequence with interruption by EV supply equipment

During sequence 4, go to state E and disconnect the power from the EV supply equipment.

A.4.8 Test of interruption of the protective conductor

The EV supply equipment shall cut off the power in max. 100 ms after the protective conductor is interrupted (test is also a part of Table A.13).

Test shall be initiated in state C or D that has been attained for at least 5 s. A supplementary switching device disconnects the protective conductor between the EV supply equipment and the EV or EV simulator.

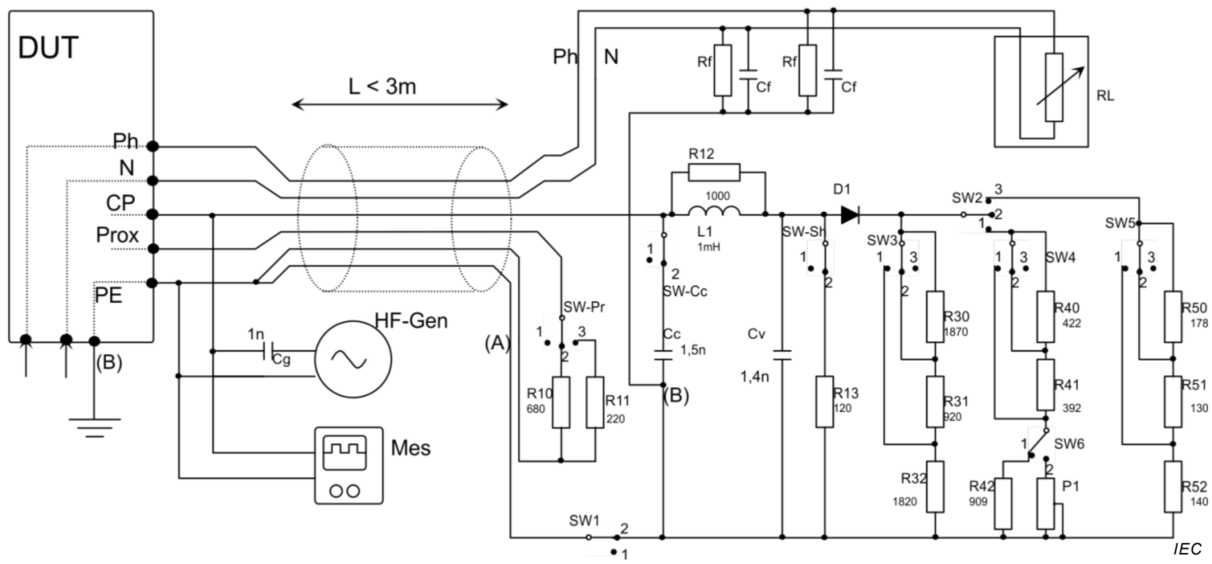
A.4.9 Test of short-circuit values of the voltage

This test verifies compliance with the timing requirement of Table A.6 sequence 12.

The test shall be initiated with $R2_{test}$ and $R3_{test}$ at the nominal value. When state C or D has been attained for at least 5 s, a supplementary resistance of 120 Ω is switched to connect between the control pilot conductor and the protective conductor.

A.4.10 Example of a test simulator of the vehicle (informative)

Figure A.8 gives an example of a possible test circuit that allows the simulation of the electric vehicle during charge. The switching of the resistor values (refer to Table A.12) allows the extreme voltage values to be tested according to Table A.9. The signal generator simulates the presence of an imposed high frequency data carrier.



Key

DUT	EV supply equipment to be tested
CP	Control Pilot contact
Prox	Proximity contact
HF-Gen	1 MHz – 30 MHz sine wave generator 50 Ω output
Mes	pulse width, voltage and HF voltage measurement
D1	fast turn off diode (e.g. 1N4934, [Irms = 1 A, Vr > 100 V, Tr = 200 ns])
R30,R31,R32,R40, R41,R42,R50,R51,R52	See Note 2
Cv,Cg	See Note 2
SW-Pr, R10, R11	Proximity and current coding for connector type 2, cases A and B (circuit for connector type 1 not shown). Other values can be added if required (See Clause B.2).
SW-Cc, Cc	Cable capacity. Position 2 is used for case A and case B.
SW-Sh,R13	Control Pilot short-circuit test
RL	Load representative of vehicle current
Cf, Rf	Y capacitors for HF filtering (47 nF) and resistors (1 MΩ)
L1, R12	Lse as defined in Table A.2 (max. value)
(A),(B)	See Note 3 and Note 4

NOTE 1 The HF-Gen and the measurement device are connected as close as possible to the EV socket-outlet (cases A and B) or the EV plug (case C).

NOTE 2 Usage of SW2,SW3,SW4, SW5and SW6:

The switch S2 in Figure A.1 is simulated by SW2 as follows:

- Position 2: state Bx (S2 is open);
- Position 1: state Cx (S2 is closed);
- Position 3: state Dx (S2 is closed).

The resistance R2 in Figure A.1 is simulated by SW4 and resistors R40 (422 Ω), R41 (392 Ω) and R42 (909 Ω) with SW6 in position 1, as follows:

– Position 2: R2 $_{test}$ has its nominal value;
– Position 1: R2 $_{test}$ has the minimum test value for state Cx;
– Position 3: R2 $_{test}$ has the maximum test value for state Cx;
– Position 2 of SW6 is used to test the hysteresis of the EV supply equipment detection device (see A.4.11).

The resistance R2 in Figure A.1 for state Dx is simulated by SW5 and resistors R50 (178 Ω), R51 (130 Ω) and R52 (140 Ω) as follows:

– Position 2: R2 $_{test}$ has its nominal value for state Dx;
– Position 1: R2 $_{test}$ has the minimum test valuefor state Dx;
– Position 3: R2 $_{test}$ has the maximum test value for state Dx.

The resistance R3 in Figure A.1 is simulated by SW3 and resistors R30 (1 870 Ω), R31 (920 Ω) and R32 (1 820 Ω) as follows:

– Position 2: R3 $_{test}$ has its nominal value;
– Position 1: R3 $_{test}$ has the minimum test value;
– Position 3: R3 $_{test}$ has the maximum test value.

NOTE 3 The proximity circuit is normally part of the EV plug. It can be included in the test equipment. Two earth lines are indicated on the figure for clarity but a single line is sufficient.

NOTE 4 The simulation circuit has no direct connection to the earthing terminal of the DUT and no direct connection to the protective earthing circuit of the test site.

NOTE 5 Charging cable length test equipment is less than 3 m.

NOTE 6 Metal film resistors are used with 0,2 % tolerance or better. Most of these resistances can be chosen from the E48 preferred resistance value table as indicated in Table 2, 4.2 of IEC 60063:2015. The resistors R31and R32 can be chosen from the E192 table. Their values can also be established using several resistors from the E48.

NOTE 7 High quality switches (e.g. gold or silver plated contacts) are used.

Figure A.8 – Example of a test circuit (EV simulator)

Table A.13 defines the switch positions for the different operation conditions. It allows the simulation of the complete test cycles using the nominal resistances, or the tolerance limit values of the EV resistances. Out of bound values can also be created.

For tests at nominal values, SW1 and SW2 are used to switch between states A, B, C and D. Nominal values of the resistance are obtained with SW3, SW4, SW5, in position 2 and SW6 in position 1.

Table A.13 – Position of switches

	State\switch		Pr	Sh	SW1	SW2	SW3	SW4	SW5	SW6
1	A	Unplugged	1	1	1	X	X	X	X	1
2	Earth Fault	Open earth wire	X	1	1	X	X	X	X	1
3	E		2,3	2	2	2	X	X	X	1
4	B	Nominal values	2,3	1	2	2	2	X	X	1
5	C		2,3	1	2	1	2	2	X	1
6	D		2,3	1	2	3	2	X	2	1
7	B	Upper values	2,3	1	2	2	3	X	X	1
8	C		2,3	1	2	1	3	3	X	1
9	D		2,3	1	2	3	3	X	3	1
10	B	Lower values	2,3	1	2	2	1	X	X	1
11	C		2,3	1	2	1	1	1	X	1
12	D		2,3	1	2	3	1	X	1	1
13	B – C	Hysteresis	2,3	1	2	1	2	1	X	2
14	C – D		2,3	1	2	1	2	1	X	2
15	C – E		2,3	1	2	1	2	1	X	2
16	D – E		2,3	1	2	1	2	1	X	2

SW-Cc is placed in position 1 for case C, and in position 2 for case A and case B

A.4.11 Optional hysteresis test

A.4.11.1 General

Slight hysteresis is generally used to improve the reliability of EV Supply Equipment. The test for hysteresis is done by modifying the value of R2 of Figure A.1 during the state C. The potentiometer P1 is used.

The test is done without the presence of a superimposed high frequency signal.

The voltage between the pilot wire terminals and the ground are monitored using a volt meter or similar.

It is not necessary to connect a load to the EV supply equipment during this test.

The initial value of the potentiometer P1 at the beginning of the test is set as indicated in Table A.14.

Table A.14 – Initial settings of the potentiometer at the beginning of each test

Hysteresis between states	Initial resistance
B-C	1 300 Ω
C-D	1 300 Ω
C-E	1 300 Ω
D-E	270 Ω

A.4.11.2 Test sequence for hysteresis between states B and C

P1 is set to 1 300 Ω.

The charging system is brought to state C2 that results in the closing of the supply switching device.

The value of P1 is increased slowly so that the voltage on the pilot wire increases at less than 0,01 V/s, until the switching device open. The voltage on the pilot wire at the instant of opening is noted.

The value of P1 is decreased slowly so that the voltage on the pilot wire increases at less than 0,01 V/s, until the switching device closes. The voltage on the pilot wire at the instant of closing is noted.

A.4.11.3 Test sequence for hysteresis between states C-E, D-E

P1 is set to 1 300 Ω for C-E and 270 Ω for D-E.

The charging system is brought to state C2 or D2 that results in the closing of the switching device.

The value of P1 is decreased slowly so that the voltage on the pilot wire decreases at less than 0,01 V/s, until the switching device opens. The voltage on the pilot wire at the instant of opening is noted.

The value of P1 is increased slowly so that the voltage on the pilot wire increases at less than 0,01 V/s, until the switching device closes. The voltage on the pilot wire at the instant of closing is noted.

A.4.11.4 Test sequence for hysteresis between states C-D.

P1 is set to 1 300 Ω.

The charging system is brought to state C2 that results in the closing of the switching device.

The value of P1 is decreased slowly so that the voltage on the pilot wire decreases at less than 0,01 V/s, until the ventilation switching device closes. The voltage on the pilot wire at the instant of closing is noted.

The value of P1 is increased slowly so that the voltage on the pilot wire increases at less than 0,01 V/s, until the ventilation switching device opens. The voltage on the pilot wire at the instant of opening is noted.

A.5 Implementation hints

A.5.1 Retaining a valid authentication until reaching CP State B

In case authentication (e.g. via RFID card) is needed, the EV supply equipment should implement two time-out supervision mechanisms as follows.

- Time-out supervision from successful authentication until inserting the plug into the socket-outlet of the EV supply equipment (if applicable, i.e. case A and case B). In case C, this time-out supervision is not necessary.
- Time-out value from successful authentication and plugging the plug into the socket-outlet of the EV supply equipment (if applicable, i.e. case A and case B) until inserting the vehicle connector into the vehicle inlet (if applicable, i.e. case B and case C).

Selecting the appropriate values for these two time-outs should be left to the charge point operator who should select these values based on local circumstances like:

- distance between location of the authentication module and socket-outlet;
- granting access to the vehicle connector or the socket-outlet, respectively, only after successful authentication: yes/no.
- etc.

If one of these time-out supervisions detects a time-out, the authentication should be invalidated.

A.5.2 Load control using transitions between state x1 and x2

Repeated transitions between states x1 and x2 can cause excessive wear on elements within the EV. It is therefore recommended to minimize the number of transitions between states x1 and x2.

Load control should preferably be implemented by adjustment of the duty cycle in state x2.

A.5.3 Information on difficulties encountered with some legacy EVs for wake-up after a long period of inactivity (informative)

This is a proposition that is written, only to resolve some difficulties encountered on legacy EVs.

No new EV shall implement a system that is dependent on this proposition which is only “a work-around”.

There are a number of EVs that exist that do not “wake-up” by detecting the transition B1/B2 as indicated in the sequence 3.1 or 3.2.

This is not considered as normal operation with regard to the text in this Annex A, and no new vehicle would encounter this difficulty if it conforms to this Annex A.

It would be possible for EV supply equipments to re-initialize the charging sequence on such legacy EVs by imposing a 0 V on the control wire (thus giving an equivalent of state E) for more than 4 s and thus re-initialize the charging sequence. Such re-initialisation would only be necessary if vehicles do not respond to the transition for 30 s. However note that such re-initialization may pose further problems on other EVs and the reproducibility of the resulting charging sequence cannot be guaranteed.

In case the EV supply equipment changes the duty cycle from 5 % to a duty cycle between 10 % and 96 %, or vice versa, this change may be done in the same way, as described in ISO 15118-3.

EV supply equipment could also implement alternative work-around based on, for example use of state F which is reserved for error signalling, to attain the same. The resultant reaction of the EV supply system to such a situation cannot be guaranteed.

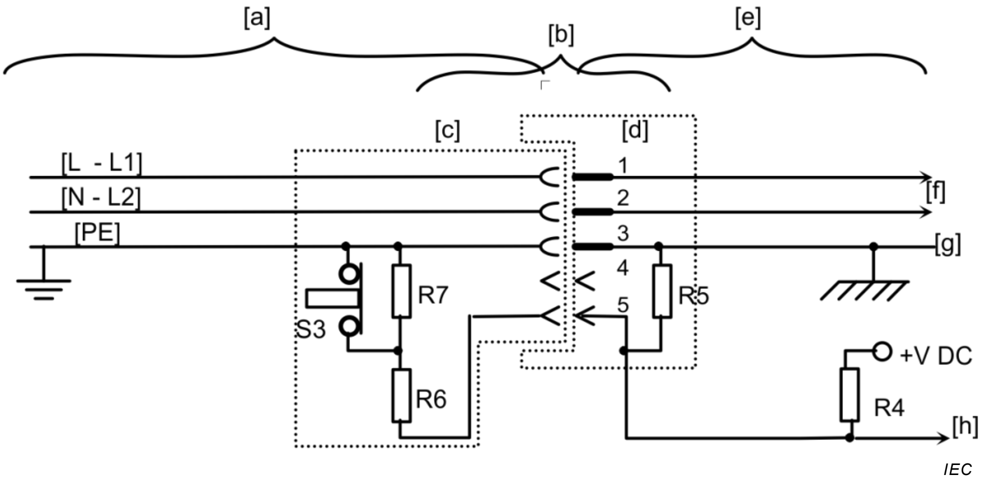
Annex B
(normative)

Proximity detection and cable current coding circuits for the basic interface

B.1 Circuit diagram for vehicle couplers using an auxiliary switch associated with the proximity detection contact

The vehicle couplers using the proximity contact with an auxiliary switch and without current capability coding of the cable assembly shall use the circuit diagram as indicated in Figure B.1 and Table B.1.

NOTE 1 The function of "proximity detection contact" is given in 3.3.5.



Key

a	cable assembly
b	vehicle coupler
c	vehicle connector
d	vehicle inlet
e	circuit on the vehicle
f	AC supply to the vehicle
g	vehicle chassis connection
h	to proximity detection circuit
S3	auxiliary switch

R4, R5, R6,R7 defined in Table B.1.

NOTE 1 The control pilot function is not indicated in this drawing.

NOTE 2 The switch S3 can be used for prevention of unintentional live disconnect.

NOTE 3 This scheme is used for the Type 1 vehicle connector as defined in IEC 62196-2.

Figure B.1 – Equivalent circuit diagram for proximity function using an auxiliary switch and no current coding

Table B.1 – Component values proximity circuit without current coding

	Value	Tolerance
R4 [a]	330 Ω	± 10 %
R5 [a]	2 700 Ω	± 10 %
R6	150 Ω	± 10 % ≥ 0,5 W
R7	330 Ω	± 10 % ≥ 0,5 W
+V DC [a,c]	5V	± 5 %
[a] These are recommended values.		

B.2 Circuit for simultaneous proximity detection and current coding

NOTE B.2 carried the number B.5 in IEC 61851-1:2010.

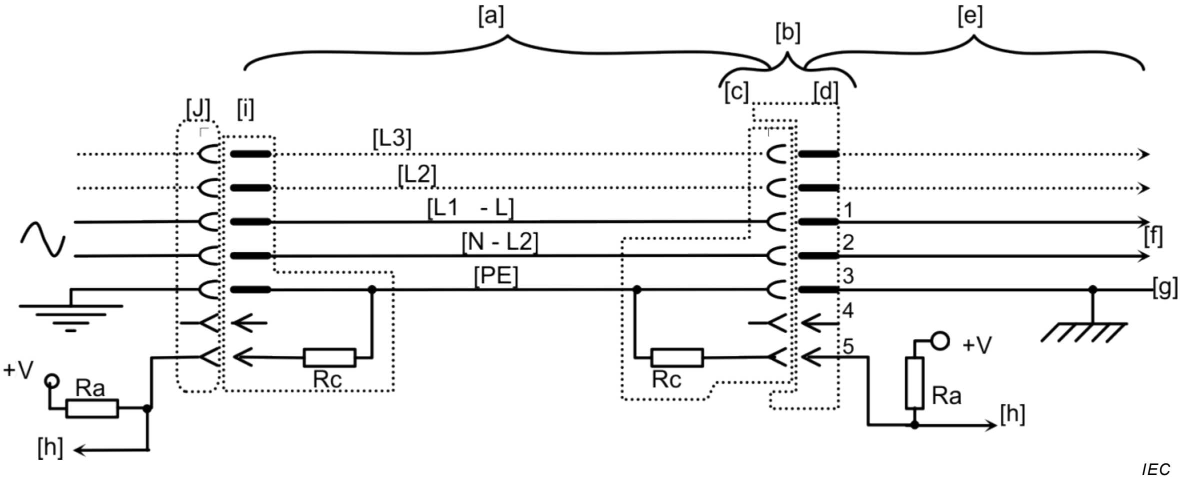
Vehicle connectors and plugs using the proximity contact for simultaneous proximity detection and current capability coding of the cable assembly shall have a resistor electrically connected between the proximity contact and the earthing contact (see Figure B.2) with a value as indicated in Table B.2.

The resistor shall be coded to the maximum current capability of the cable assembly.

The EV supply equipment shall interrupt the current supply if the current capability of the cable is exceeded as detected by the measurement of the Rc, as specified by the values for the recommended interpretation range in Table B.2.

The EV supply equipment shall detect the current coding by measurement of the Rc, as defined in Table B.2 and use the result to set the value of the maximum allowed current, if necessary, according to 6.3.1.6.

The resistor is also used for proximity detection.



Key

a cable assembly

b vehicle coupler

c vehicle connector

d vehicle inlet

e circuit on the vehicle

f AC supply to the vehicle

g vehicle chassis connection

h to proximity detection circuit

j EV socket-outlet

i EV plug

Ra, Rc defined in Table B.2.

The control pilot function may be necessary but is not indicated in this drawing.

NOTE 1 This circuit does not use an auxiliary switch.

NOTE 2 This scheme is used on Type 2 and type 3 vehicle connectors and plugs as defined in IEC 62196-2.

Figure B.2 – Equivalent circuit diagram for simultaneous proximity detection and current coding

Table B.2 – Current coding resistor for EV plug and vehicle connector

Current capability of the cable assembly (A)	Nominal resistance of Rc Tolerance ± 3 %[c] (Ω)	Minimum dissipation rating of resistances[a, b] (W)	Range of resistance Rc for interpretation by the EV supply equipment[e] (Ω)
	Error condition[d] or disconnection of plug		> 4 500
13	1 500	0,5	1 100 – 2 460
20	680	0,5	400 – 936
32	220	1	164 – 308
63 (3-phase) / 70 (1-phase)	100	1	80 – 140
	Error condition[d]		< 60

[a] The power dissipation of the resistor caused by the detection circuit shall not exceed the value given above. The value of the pull-up resistor Ra shall be chosen accordingly.

[b] Resistors used should preferably fail open circuit failure mode. Metal film resistors commonly show acceptable properties for this application. Dissipation ratings are chosen to avoid destruction in the case of a fault to +12 V supply.

[c] Tolerances to be maintained over the full useful life and under environmental conditions as specified by the manufacturer.

[d] EV supply equipment shall not provide power.

[e] The minimum and maximum values of each range shall be tested. The choice of the resistance value at the transition between current levels is at the discretion of the EV supply equipment designer.

Annex C
(informative)

Examples of circuit diagrams for a basic and universal vehicle couplers

C.1 General

This Annex C gives examples of circuit diagrams for the Mode 1, Mode 2 and Mode 3 charging methods using the basic interface (see Figures C.1 to C.5). It is included to give an overview and an historical record.

An example of Mode 4 charging is presented with the universal vehicle coupler (see Figure C.6).

C.2 Circuits diagrams for Mode 1, Mode 2 and Mode 3, using a basic single phase vehicle coupler

Figures C.1, C.2, C.3 and C.4 below show the application of a single phase basic interface fitted with a switch on the proximity circuits.

The auxiliary coupler contact indicated in the figures can be used for unintentional live disconnect avoidance using switch on vehicle connector. For example, for this function, the push button is linked to a mechanical locking device. The depressing S3 unlocks the coupler and opens the circuit. The opening of S3 stops charging operation and contributes to prevention of unintentional live disconnect.

This function can also be achieved using proximity switches or contacts on the vehicle inlet cover or on the locking device.

Figures C.1, C.2, C.3 and C.4 can also be realized with a connector that lacks a switch provided the switch S3 is not required.

Clause C.3 shows the application of a three-phase basic interface that is not fitted with a switch on the proximity circuit, used for single and three-phase supply.

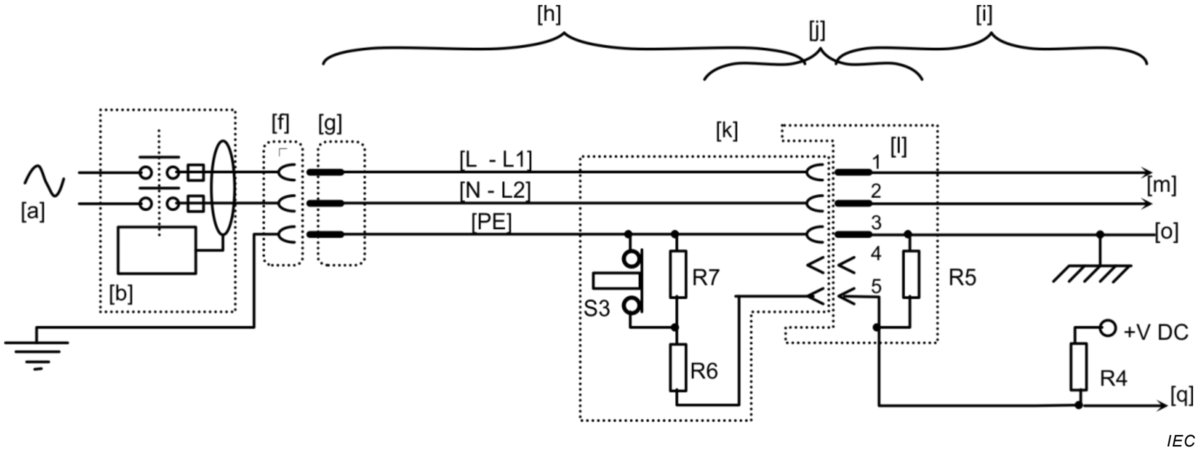
Components and functions in the circuit diagrams shown in Figures C.1 to C.5 are as follows.

The control pilot function controller is located on the AC supply network side.

This circuit realizes the basic functions described in Annex A. The circuit is normally supplied from an extra low voltage source that is isolated from the AC supply network by a transformer and contains a ± 12 V – 1 000 Hz pulse width modulated oscillator that indicates the power available from the socket-outlet.

Both Mode 2 diagram shown in Figure C.2 and Mode 3 diagram shown in Figures C.3 and C.4 have been drawn with a hard wired control pilot functions as described in Annex A. The basic functions described in Annex A are represented by R1, R2, R3, D and S2 (see Figure A.1). The values indicated in Annex A should be used (see Table A.3).

Values of components for the control pilot function are given in Annex A and values for the proximity function are given in Annex B.



Key

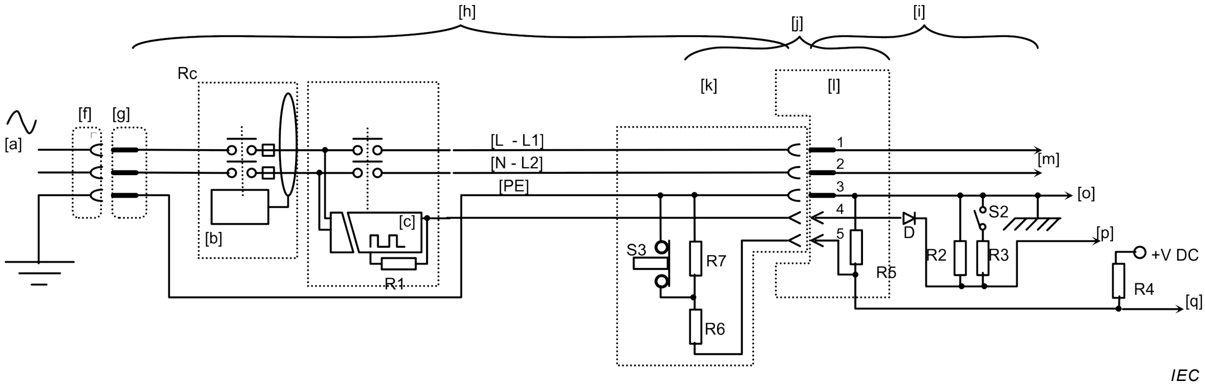
a	Supply network
b	RCD
c	Control circuit for pilot function
f,g	Standard plug and socket-outlet
h	Cable assembly
i	Onboard circuit
j	Vehicle coupler
k	Vehicle connector
l	Vehicle inlet
m	To charger and other AC loads
o	Electric chassis of EV
q	To the proximity control logic circuit
S3,R4,R5,R6,R7	Components for proximity function using an auxiliary switch and no current coding. Values are given in Table B.1.

Contact identification:

1,2	Phase and neutral contacts
3	Protective earthing contact
4	Control pilot function contact (not used)
5	Proximity contact

There is no control pilot function in Mode 1and pin 4 is not compulsory.

Figure C.1 – Example of Mode 1 case B using the proximity circuit as in B.1



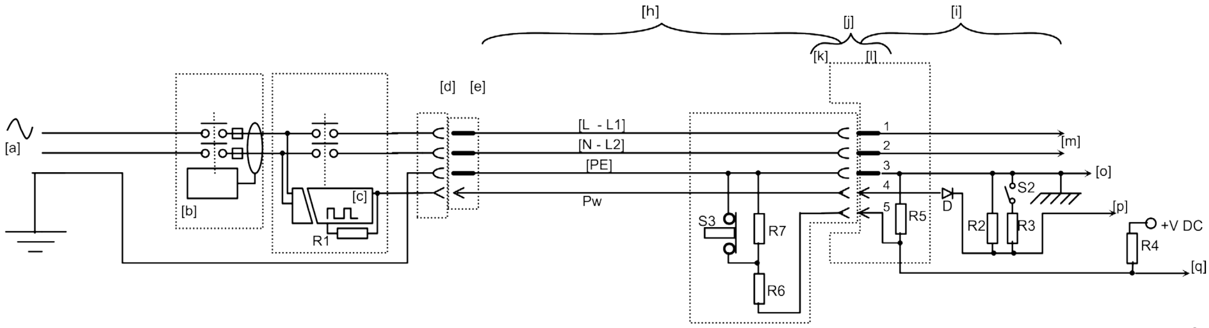
Key

a	Supply network
b	RCD
c	Control circuit for pilot function
f, g	Standard plug and socket-outlet
h	Cable assembly
i	Onboard circuit
j	Vehicle coupler
k	Vehicle connector
l	Vehicle inlet
m	To charger and other AC loads
o	Electric chassis of EV
p	To the vehicle pilot function logic circuit
q	To the proximity control logic circuit
Pw	Control pilot conductor
D,S2,R1,R2,R3	Components for control pilot function. Values are given in Tables A.2 and A.3.
S3,R4,R5,R6,R7	Components for proximity function using an auxiliary switch and no current coding.Values are given in Table B.1.

Contact identification:

1, 2	Phase and neutral contacts
3	Protective earthing contact
4	Control pilot function contact
5	Proximity contact

Figure C.2 – Example of Mode 2 case B using proximity detection as in B.1



Key

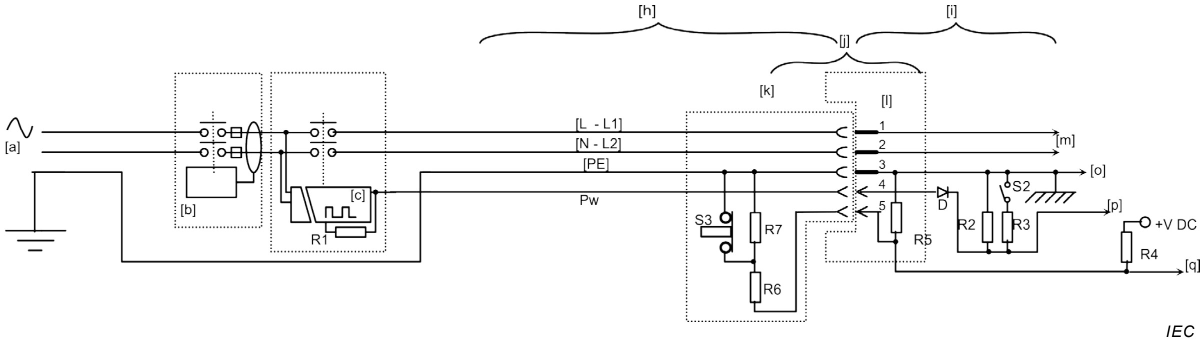
a	Supply network
b	RCD
c	Control circuit for pilot function
d, e	EV socket-outlet, EV plug
f,g	Standard plug and socket-outlet
h	Cable assembly
i	Onboard circuit
j	Vehicle coupler
k	Vehicle connector
l	Vehicle inlet
m	To charger and other AC loads
o	Electric chassis of EV
p	To the vehicle pilot function logic circuit
q	To the proximity control logic circuit

Pw	Control pilot conductor
D,S2,R1,R2,R3	Components for control pilot function. Values are given in Tables A.2 and A.3.
S3,R4,R5,R6,R7	Components for proximity function using an auxiliary switch and no current coding.Values are given in Table B.1.
Ra,Rc	Components for simultaneous proximity detection and current coding. Values are given in Table B.2.

Contact identification:

1,2	Phase and neutral contacts
3	Protective earthing contact
4	Control pilot function contact
5	Proximity contact

Figure C.3 – Example of Mode 3 case B using proximity detection as in B.1



Key

a	Supply network
b	RCD
c	Control circuit for pilot function
h	Cable assembly
i	Onboard circuit
j	Vehicle coupler
k	Vehicle connector
l	Vehicle inlet
m	To charger and other AC loads
o	Electric chassis of EV
p	To the vehicle pilot function logic circuit
q	To the proximity control logic circuit
Pw	Control pilot conductor
D,S2,R1,R2,R3	Components for control pilot function. Values are given in Tables A.2 and A.3.
S3,R4,R5,R6,R7	Components for proximity function using an auxiliary switch and no current coding. Values are given in Table B.1.

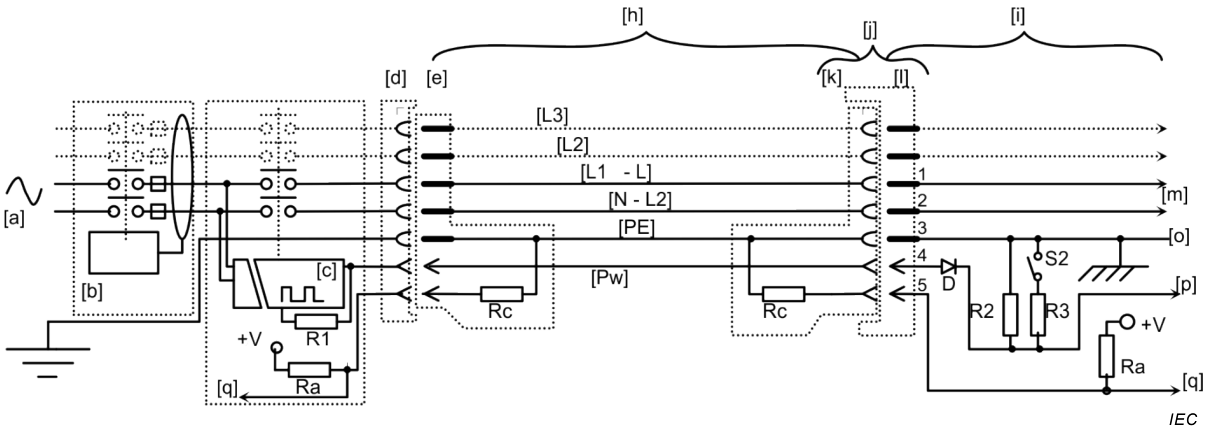
Contact identification:

1,2	Phase and neutral contacts
3	Protective earthing contact
4	Control pilot function contact
5	Proximity contact

Figure C.4 – Example of Mode 3 case C using proximity detection as in B.1

C.3 Circuits diagrams for Mode 3, using a basic single phase or three-phase accessory without proximity switch

Figure C.5 shows a three phase interface accessory that is used for either single phase or three phase supply. The same circuit diagram is also valid for single phase accessories. The current coding function described in Clause B.2 is indicated. Values of the pull-up resistances are given in Table B.2.



Key

a	Supply network
b	RCD
c	Control circuit for pilot function
d, e	EV socket-outlet, EV plug
h	Cable assembly
i	Onboard circuit
j	Vehicle coupler
k	Vehicle connector
l	Vehicle inlet
m	To charger and other AC loads
o	Electric chassis of EV
p	To the vehicle pilot function logic circuit
q	To the proximity control logic circuit
Pw	Control pilot conductor
D,S2,R1,R2,R3	Components for control pilot function. Values are given in Tables A.2 and A.3.
Ra,Rc	Components for simultaneous proximity detection and current coding. Values are given in Table B.2.

Contact identification:

1,2	Phase and neutral contacts
3	Protective earthing contact
4	Control pilot function contact
5	Proximity contact

NOTE Over-current and RCD protection devices [b] can be part of the fixed installation.

Figure C.5 – Example of Mode 3 case B using proximity detection as in B.2 (without proximity push button switch S3)

C.4 Example of circuit diagram for Mode 4 connection using universal coupler

NOTE This Clause C.4 is provided for information only and is copied from IEC 61851-1:2001. It is not to be considered for new designs. Recent information can be found in IEC 61851-23.

Parts list and function/characteristics in the circuit diagram for Mode 4 connection are shown in Table C.1 and Figure C.6.

Table C.1 – Component description for Figure C.6 Mode 4 case C

Reference	Parts list	Function/characteristics
A	Auxiliary contact	– detection of the connector – start for the DC equipment on the vehicle (option) – pilot circuit
PB	Locking release of the connector	– opens the pilot circuit to de-energize the system before the main contacts open: $t \leq 100$ ms
C1	Main contactor on the supply equipment	– closed on nominal operation if: $0{,}5\ \text{k}\Omega < R_o < 2\ \text{k}\Omega$
C2 (option)	Main contactor on the vehicle	– closed on normal operation
E1	Auxiliary supply	– extra-low DC voltage to energize the pilot circuit: earth protection connector + pilot + chassis
D1	Diode	– not used – prevent the energization of the vehicle computer by the supply equipment
D2	Diode	– prevent the energization of the auxiliary supply circuit E1 and M1, by the vehicle
D3	Diode	– prevent short-circuit between the auxiliary supply E1 and the earth, inside the charging station
FC (option)	Trap door close	– start for the DC equipment on the vehicle
G	Pilot contact (last closed during the connection)	– earth for detection of the connector – earth for the pilot circuit – clean data earth

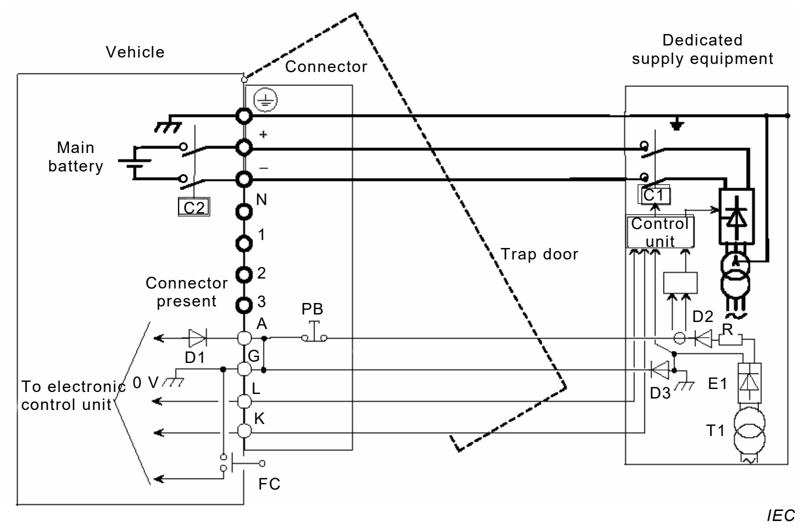


Figure C.6 – Example of Mode 4 case C using the universal vehicle coupler

Annex D
(informative)

Control pilot function that provides LIN communication using the control pilot circuit

D.1 Overview

D.1.1 General

This Annex D specifies a control pilot function that provides bidirectional communication between LIN nodes in the charging station and in the EV. Optionally, a third LIN node may be present in a cable assembly.

If the control pilot function, specified in this annex, is used with mode 2 charging, the requirements on the charging station apply to the ICCB.

In this annex the terms PWM-CP and LIN-CP are used for the control pilot functions specified in Annex A and in this annex respectively.

As specified in 6.3.1.1, charging stations that use accessories according to IEC 62196-2 shall implement PWM-CP according to Annex A. Clause D.9 provides requirements for charging stations that implement both LIN-CP and PWM-CP.

LIN-CP is based on the same control pilot circuit that is used for PWM-CP. This makes it easy to design backward compatible charging stations and EVs that implement both PWM-CP and LIN-CP.

D.1.2 LIN-CP features

Examples of features provided by LIN-CP:

– bi-directional digital communication using standard LIN protocol for local control between the EV and the charging station;
– single-phase and three-phase current limits can be separately controlled to control asymmetrical loading;
– the EV can indicate requested current to allow better coordination for shared power feeds and energy management systems;
– well specified sleep mode management;
– designed for low cost, low complexity, high reliability;
– default communication speed is 20 kbit/s nominal;
– the charging station and the EV can exchange diagnostic information.

D.1.3 Normative references

The following documents, in whole or in part, are normatively referenced in this annex and are indispensable for its application. For dated references, only the edition cited applies. For undated references, the latest edition of the referenced document (including any amendments) applies.

ISO 17987-1-7 [8], *Road vehicles – Local interconnect network (LIN) – Part 1: General information and use case definition*

[8] To be published.

ISO 17987-2:–[9], *Road vehicles – Local interconnect network (LIN) – Part 2: Transport protocol and network layer services*

ISO 17987-3:–[10], *Road vehicles – Local interconnect network (LIN) – Part 3: Protocol specification*

ISO 17987-4[11], *Road vehicles – Local interconnect network (LIN) – Part 4: Electrical Physical Layer (EPL) specification 12 V/24 V*

NOTE LIN Specification 2.2.A (2010) from the LIN consortium (http://www.lin-subbus.org/) has been discontinued and will be transcribed to the future ISO 17987-1.

D.1.4 Terms and abbreviations

For the purposes of this annex, the terms and definitions given in ISO 17987 series and the following apply.

D.1.4.1
LIN-CP
control pilot function using LIN communication and voltage level signalling on the control pilot circuit

Note 1 to entry: LIN-CP is specified in this Annex D.

D.1.4.2
PWM-CP
control pilot function using PWM signalling and voltage level signalling on the control pilot circuit

Note 1 to entry: PWM-CP is specified in Annex A.

D.1.4.3
recessive level
voltage level on a data bus when no data is sent

Note 1 to entry: For LIN the recessive level is the high level, nominally +9 V in CP voltage level B and +6 V in CP voltage level C. This is the logic “1” level during communication. This level is passively created by the control pilot circuit.

D.1.4.4
dominant level
voltage level on a data bus that has priority over the recessive level

Note 1 to entry: For LIN the dominant level is the low level, nominally +0 V in CP voltage level B and C. This is the logic “0” level during communication. This level is actively created by a LIN transceiver.

D.2 Scope and context

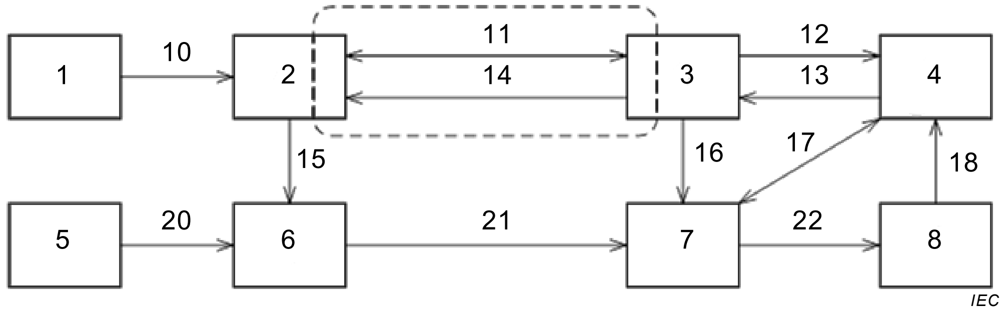
Figure D.1 shows an example of an EV charging system with a typical configuration of functions, information flow and power flow.

The scope of this annex is the information flow on the control pilot circuit between the communication controllers in the charging station and in the EV (inside the dashed area) and how this relates to the signals to and from other functions (outside the dashed area).

9 To be published.

10 To be published.

11 To be published.



Key

1	energy management system	11	LIN communication that provides information about status and properties of the charging station and the EV.
2	communication controller in the charging station	12	information about availability of power
3	communication controller in the EV	13	information about requested power
4	charge controller in the EV	14	CP voltage levels that provide information whether EV is connected and about readiness of EV to accept power.
5	supply network	15	commands to open/close switching device
6	switching device in the charging station	16	commands to switch on-board charger on/off, information regarding maximum AC charging current to the vehicle
7	on-board charger in the EV	17	information on charge controller status and charging parameters"
8	RESS in the EV	18	information regarding “status of the RESS”
10	information on maximum supply current from the electrical system to the charging station”	20,21,22	power flow

Figure D.1 – Example of an EV charging system with a typical configuration of functions, information flow and power flow

D.3 Overview of control pilot functions

Table D.1 lists control pilot functions which are implemented in both LIN-CP and PWM-CP and explains how these functions are implemented.

Table D.1 – Control pilot functions in LIN-CP and PWM-CP

Line	Control pilot function description	LIN-CP implementation	PWM-CP implementation See Annex A
1	Continuous continuity checking of the protective conductor Specified in 6.3.1.2	The charging station monitors the control pilot voltage and opens the switching device within the time limits.	The charging station monitors the control pilot voltage and opens the switching device within the time limits.
2	Verification that the EV is properly connected to the EV supply equipment Specified in 6.3.1.3	The charging station monitors the positive control pilot voltage and detects valid LIN signals from the EV.	The charging station monitors the positive and negative control pilot voltages.
3	Energization of the power supply to the EV Specified in 6.3.1.4	The charging station closes the switching device only if the EV has closed S2 and LIN signals verify that the EV is ready to receive power.	The charging station closes the switching device only if the EV has closed S2.
4	De-energization of the power supply to the EV Specified in 6.3.1.5	The charging station opens the switching device if the EV has opened S2 or if LIN signals indicate that the EV is no longer ready to receive power.	The charging station opens the switching device if the EV has opened S2.
5	Maximum allowable current Specified in 6.3.1.6	The charging station sets LIN signals to indicate between 0 A and 250 A individually on the connected phases.	The charging station sets PWM duty cycle to indicate between 6 A and 80 A on the connected phases.
6	The charging station indicates that it is ready to supply energy.	The charging station sends a LIN signal to indicate that it is ready.	The charging station changes the PWM duty cycle from 100 % to a value between 10 % and 96 %.
7	The charging station indicates that it is not ready to close the switching device, or it requests the EV to stop the vehicle load and open S2 to allow the charging station to open the switching device at no load.	The charging station sends LIN signals to indicate that it is not ready and the reason why".	The charging station changes the PWM duty cycle from a value between 10 % and 96 % to 100 %.
8	The charging station indicates that it is not available for charging (e.g. needs maintenance).	The charging station sends LIN signals to indicate that it is "not available" and the reason why.	The charging station sets the control pilot voltage to -12 V (nominal).
9	Control of socket-outlet latching for case A or case B charging stations which use IEC 62196-2 socket-outlets.	The charging station latches and unlatches the socket-outlet depending on the detected control pilot voltage.	The charging station latches and unlatches the socket-outlet depending on the detected control pilot voltage.
NOTE LIN-CP does not implement the following PWM-CP functions: indication that the EV needs ventilation, simplified control pilot function, selection of PLC and wireless communication according to ISO/IEC 15118 and use of the B-C-B toggle.			

Table D.2 gives an overview of additional LIN-CP control pilot functions, provided by use of LIN communication. See Clause D.8 for detailed specifications on LIN communication.

Table D.2 – Additional LIN-CP control pilot functions

Line	LIN-CP communication functions	Reference
1	Automatic selection of LIN-CP or PWM-CP after plug-in if the charging station implements both LIN-CP and PWM-CP	D.9.4 D.8.2.2
2	Communication version negotiation	D.8.4.3
3	Exchange of detailed status signals	D.8.4.5
4	The charging station sends maximum allowable current for each phase and for neutral. These values can be dynamically adjusted.	D.8.4.5
5	The charging station can send maximum current for each phase and for neutral. These values are static, but can vary between charging sessions depending on the cable used.	D.8.4.4
6	The EV can send min/max phase-to-phase and phase-to-neutral voltages	D.8.4.4
7	The charging station can send nominal supply network phase-to-phase and phase-to-neutral voltages	D.8.4.4
8	The EV can send min/requested/max current for each phase and for neutral	D.8.4.4 D.8.4.5
9	The EV can send measured/estimated currents for each phase and for neutral	D.8.4.5
10	An optional LIN node in the cable assembly can send rated current for each phase and for neutral and the rated voltage	D.4.6 D.8.4.4
11	The EV can request the LIN communication to go to “LIN sleep”. In “LIN sleep” both the charging station and the EV can wake up the communication.	D.8.2.6
12	The EV can go to “EV deep sleep” by powering down its communication controller. In “EV deep sleep”, the EV can resume the communication at any time.	D.8.2.6
NOTE This table shows only LIN communication functions specified in this Annex D. Additional functions are specified in SEK TS 481 05 16 (under development) and in SAE J3068 (under development).		

D.4 Control pilot circuit

D.4.1 General

The LIN-CP control pilot circuit is based on the PWM-CP control pilot circuit, described in Annex A. LIN-CP adds LIN nodes in the charging station and in the EV, shown as items 1 and 2 in Figure D.2. Optionally, a third LIN node may be present (not shown in Figure D.2), see D.4.6.

This Clause D.4 gives requirements for LIN-CP. See Clause D.9 for additional requirements when both LIN-CP and PWM-CP are implemented.

D.4.2 Control pilot circuit

For LIN-CP, the requirements on components value ranges are the same as given in Annex A, Tables A.2 and A.3, with the following modifications:

– The resistor R2 shall be 1 300 Ω (270 Ω is not used in LIN-CP) because requesting ventilation is not supported in LIN-CP.

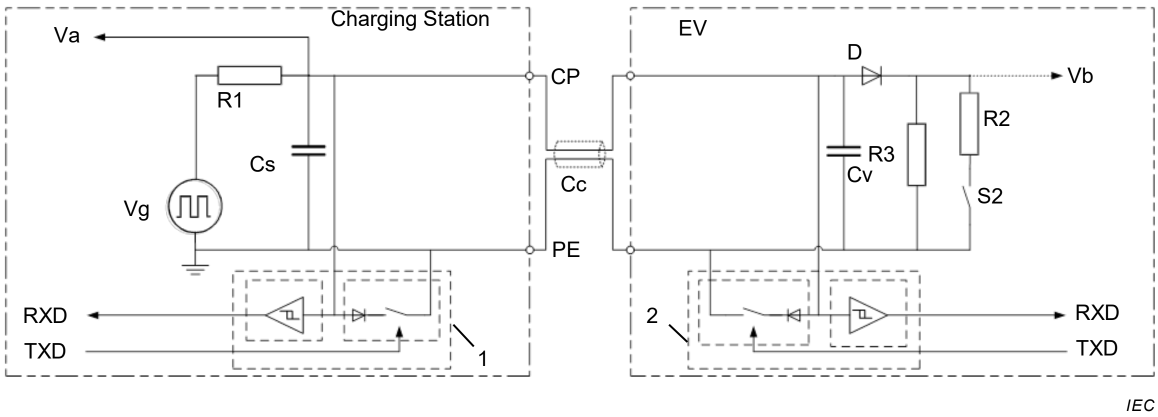
– The charging station capacitance Cs shall be between 1 000 pF and 1 600 pF (not between 300 pF and 1 600 pF).

– The EV capacitance Cv shall be between 1 500 pF and 2 400 pF (not between 0 pF and 2 400 pF).

– The cable capacitance Cc shall be between 0 pF and 6000 pF (not between 0 pF and 1 500 pF).

NOTE Requiring higher Cs and Cv capacitances improves EMC. Allowing higher cable capacitance allows longer cabling between the communication controllers.

The voltage Vg can be supplied from a +12 V supply or from an oscillator (as required in Annex A) if the oscillator duty-cycle is set to 100 %.



Key

Components		**Connections andsupplies**	
R1	resistor R = 1 kΩ	Va	voltage measurement to detect CP voltage levels
Cs	charging station capacitance, C = 1 000-1 600 pF	Vb	voltage measurement to detect control pilot circuit connected or not connected
Cc	cable capacitance, C = 0-6 000 pF	Vg	+12,0 V supply or
			+/-12,0 V oscillator with 100 % duty cycle
Cv	EV capacitance C = 1 500-2 400 pF	RXD	received data, LIN wake-up detection
D	diode	TXD	transmitted data
R3	resistor R = 2,74 kΩ	CP	control pilot conductor
R2	resistor R = 1,3 kΩ	PE	protective conductor
S2	switch	1, 2	LIN transceiver

Figure D.2 – Electrical equivalent circuit for connection of LIN nodes to the control pilot circuit

A LIN node typically consists of a LIN transceiver chip connected to a serial port of a microcontroller unit. The microcontroller unit has software that handles the LIN physical layer and the LIN protocol and normally also all other functions in a controller, including other communication ports (e.g. a CAN port to higher level controllers in an EV).

D.4.3 Charging station control pilot circuit interface

The charging station shall detect the voltage level on the control pilot circuit, between the CP conductor and the PE conductor. Table D.3 defines the meaning of the terms “CP voltage level A”, “CP voltage level B”, etc. which are used as control conditions throughout this Annex D.

During LIN communication, the CP voltage levels B or C shall be measured during the recessive LIN level. During CP voltage levels A, D and E there are normally no LIN communication signals.

To decide that the CP voltage level has changed, a filter shall be used to require that control pilot voltage measurements during at least 10 ms should indicate the new level.

CP voltage level D is not intentionally set by the EV in LIN-CP. As specified in Clause D.5, if the charging station detects CP voltage level D, then it handles this as if it detects CP voltage level E.

LIN-CP does not use negative voltage.

If the charging station implements both LIN-CP and PWM-CP,see Clause D.9.

Table D.3 – Generation and detection of CP voltage levels

Line	CP voltage level	Control pilot circuit status		Generation by the control pilot circuit			Detection by the charging station		
		Control pilot circuit connected	EV switch S2 closed	Generated range Va (V)			Minimum detection range Va (V)		Recommended detection threshold Va (V)
				min	nom	max	relative to Vg	nominal	
1	A	no	S2 not connected	11,4	12	12,6	> Vg*11/12	> 11	
2	A or B								Vg*10,5/12 ±hysteresis
3	B	yes	no	8,37	9	9,59	Vg*8/12 – Vg*10/12	8 – 10	
4	B or C								Vg*7,5/12 ±hysteresis
5	C	yes	yes	5,47	6	6,53	Vg*5/12 – Vg*7/12	5 – 7	
6	C or D								Vg*4,5/12 ±hysteresis
7	D	yes	yes	2,59	3	3,28	Vg*2/12 – Vg*4/12	2 – 4	
8	D or E								Vg*1,5/12 ±hysteresis
9	E	Undefined		Undefined			< Vg*1/12	< 1	

NOTE 1 The generated ranges are calculated from the component tolerances specified in Annex A, Tables A.2 and A.3. Leakage currents between the CP and PE conductors can cause real values to be lower.

NOTE 2 The relative detection ranges and the recommended detection thresholds are compliant with the tests specified in Clauses A.4 and D.10.

D.4.4 EV control pilot circuit interface

The EV control pilot circuit interface shall detect presence or absence of voltage between the CP conductor and the PE conductor to detect if the control pilot circuit is connected or not connected.

The recommended detection threshold voltage is Va = +1,5 V with max +/- 0,5 V hysteresis. The detection shall include a filter function to ensure that a transient level lasting less than 10 ms is not detected.

D.4.5 LIN communication transceiver

The LIN transceiver shall conform to ISO 17987-4, with the following additional requirements:

– operational with supply voltage down to +6,0 V;
– pulse forming on both rising and falling waveform edges of the LIN bus signal;
– operational with recessive level +5,0 V to +13,0 V, when supplied with +6,0 V to +7,0 V, without degrading the pulse forming;

– transmitter output dominant level < 1,4 V when supplied with +6,0 V to +7,0 V.

NOTE Figure D.2 does not include the diodes that ISO 17987-4 requires in series with the LIN transceiver supply and in series with the LIN bus master pull-up resistor. The diodes are not needed in a design that uses regulated supplies on all nodes.

Table D.4 – Generation and detection of LIN communication levels

Line	Parameter	Min	Typ	Max	Unit	Remark
1	Transceiver LIN output voltage when sending dominant bus level	0		1,4	V	
2	Transceiver LIN output voltage when sending recessive bus level					LIN output has high impedance to ground. Recessive bus levels are decided by the control pilot circuit.
3	Recessive bus level, generated by the control pilot circuit when in CP voltage level B	8,30	9	9,59	V	Based on max tolerances for Vg, R1, R3 and D.
4	Recessive bus level, generated by the control pilot circuit when in CP voltage level C	5,47	6	6,53	V	Based on max tolerances for Vg, R1, R2, R3 and D.
5	Transceiver supply voltage	6		7	V	[a]

[a] To allow LIN communication in CP voltage levels B and C, the supply voltage specified here is lower than required in ISO 17987-4. The low supply voltage is needed because the receiver input threshold voltage requirements of LIN transceivers are specified relative to the transceiver supply voltage.

Transceivers, with an extended supply voltage range, which include the range specified here, are commercially available.

D.4.6 Optional cable assembly node

Optionally, a third LIN node may be present in a cable assembly (not shown in Figure D.2).

As information about the capabilities of the cable assembly is only required during initialization of the system, the node in the cable assembly shall be designed to operate when CP voltage level B is present which provides nominal +9 V. The node may also be operational at other CP voltage levels.

The node in the cable assembly may be powered by harvesting power directly from the LIN bus if the node does not draw more than the following current from the control pilot conductor, except when it is actively sending the dominant voltage during communication:

– 12 mA inrush current in the first 5 ms after plugging the cable assembly into the charging station;
– 200 µA peak current and
– 100 µA average current.

Additional requirements apply if further LIN nodes are added to the system.

D.5 Control pilot circuit interaction

D.5.1 General

The control pilot circuit is designed to allow the charging station and the EV to interact as a common system, using CP voltage level signalling and LIN communication.

When the charging station and the EV are plugged-in, and there are no faults or exceptions, this interaction is possible using the control pilot circuit states and transitions shown in the box to the right in Figure D.3.

When the charging station and the EV are unplugged, or if there is a fault or exception, this interaction is not possible. EV states (name starting with Ev) and charging station states (name starting with St) are separated as shown in the box to the left in Figure D.3.

D.5.2 Control pilot circuit states and transitions

The state diagram in Figure D.3 gives an overview of all states and transitions used in LIN-CP.

The letters A, B, C and E, used in the state names, indicate the CP voltage level detected by the charging station, see Table D.3. The number "1" in B1 indicates that there is no active LIN signal or PWM signal on the control pilot circuit, as in Annex A. Ev0V indicates that the EV detects a voltage below the used threshold, see D.4.4.

The dotted arrows show how PWM-CP may be selected, if the charging station implements both LIN-CP and PWM-CP. See Clause D.9 for a combined LIN-CP and PWM-CP state diagram.

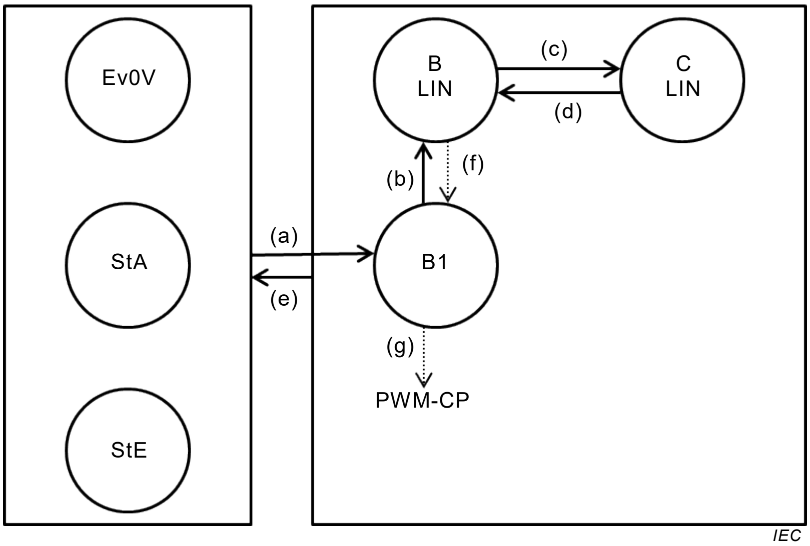


Figure D.3 – Control pilot circuit state diagram for LIN-CP (key list in Table D.5)

Table D.5 – Key list for Figure D.3 and Figure D.9

Line	Item	Description of states and transitions
States		
1	Ev0V	The EV is not plugged in,OR the control pilot is interrupted,OR the control pilot circuit is short-circuited,OR the charging station has control power outage,OR if PWM-CP is used,the charging station has set CP voltage level F. The EV does not detect a voltage above the used threshold, see D.4.4.
2	StA	The charging station is not plugged in OR the control pilot circuit is interrupted. The charging station detects CP voltage level A.
3	StE	The control pilot circuit is short-circuited,OR the charging station has a control power outage. The charging station detects CP voltage E, OR there is no voltage on the control pilot circuit.
4	B1	The control pilot circuit is normally connected. The EV detects a voltage above the used threshold, see D.4.4. The EV switch S2 is open. The charging station detects CP voltage level B.
5	B LIN	The control pilot circuit is normally connected. The EV detects a voltage above the used threshold, see D.4.4. The EV switch S2 is open. The charging station detects CP voltage level B. Periodic LIN communication, see Clause D.8.
6	C LIN	The control pilot circuit is normally connected. The EV detects a voltage above the used threshold, see D.4.4. The EV switch S2 is closed. The charging station detects CP voltage level C. Periodic LIN communication, see Clause D.8.
7	StF See Figure D.9	This state is used only in PWM-CP, see D.9.4. The charging station is not available for charging. The charging station sets CP voltage level F.
Transitions		
8	(a)	The user plugs in the EV, connecting the control pilot circuit.
9	(b)	The charging station starts sending LIN headers.
10	(c)	The EV closes the S2 switch.
11	(d)	The EV opens the S2 switch.
12	(e)	The user unplugs the EV,**OR** there is an exceptional situation preventing normal interaction on the control pilot circuit.
13	(f)	A charging station that implements both LIN-CP and PWM-CP stops sending LIN headers if the EV does not respond within time limit. See D.9.5 and Table D.5 line 5.
14	(g)	A charging station that implements both LIN-CP and PWM-CP starts sending PWM signals. See D.9.4.

D.6 System requirements

D.6.1 General

Table D.6, Table D.7, Table D.8 and Table D.9 specify system behaviour and system timing requirements. Refer to Clause D.10 for specifications of corresponding system tests.

D.6.2 Control of LIN signals

The sending of LIN signals shall be controlled as specified in Table D.6. See Clause D.8 for detailed specification of LIN communication.

Table D.6 – Control of LIN signals

Line		Time limit
Normal		
1	After the control pilot circuit is connected, and CP voltage level B is detected, the charging station shall start sending LIN headers.See D.8.2.2. For system behaviour if CP voltage level C is detected after the control pilot circuit is connected, see D.9.4.	min 10 ms / max 0,5 s after the control pilot circuit is connected
2	After the control pilot circuit is connected, the EV should start responding to LIN headers. See D.8.2.2.	max 1,0 s after first header
3	After the control pilot circuit is disconnected, the charging station shall stop sending LIN headers. The charging station and the EV reset information and signals to default, as specified in D.8.2.2.	max 2,0 s
4	After receiving indication of change in “maximum supply current”, e.g. from an energy management system, the charging stationshall adapt the applicable LIN signals.	max 10 s
Exception		
5	If the EV does not send valid LIN responses, as required in line 2, the charging station selects to use PWM-CP, if implemented, see Clause D.9.	min 1,1 s max 1,5 s after first header

D.6.3 Control of the S2 switch and the vehicle load current

The S2 switch in the EV and the vehicle load current shall be controlled as specified in Table D.7.

Table D.7 – Control of the S2 switchand the vehicle load

Line		Time limits
Normal		
1	The EV may close the S2 switch at any time if all of the following apply: – the EV is ready to receive energy; – applicable LIN signals indicate that the charging station is ready to supply energy, D.8.2.5; – closing is allowed according to line 2 in Table D.9, if applicable.	-
2	The EV may open the S2 switch at any time after first reducing the vehicle load current to <1 A.	-
3	The EV shall reduce the vehicle load current to <1 A and open the S2 switch if this is requested by applicable LIN signals, see D.8.2.5	max 3 s
4	The EV shall reduce the vehicle load current, if needed to respect the received “max maximum allowable current” LIN signals.	max 5 s
Exception		
5	If the EV detects no LIN bus activity while it is receiving energy, the EV shall reduce the vehicle load current to <1 A and open the S2 switch.	min 2 s, max 3 s after last bus activity

D.6.4 Control of the switching device in the charging station

The switching device in the charging station shall be controlled as specified in Table D.8.

Table D.8 – Control of the switching device

Line	Description	Time limits
Normal		
1	The charging station shall close the switching device if all of the following apply: – the CP voltage level is C – all applicable LIN signals confirm that closing is allowed (see D.8.2.5) – closing is allowed according to line 7 in Table D.9, if applicable.	max 3 s
2	The charging station shall open the switching device if any of the following applies: – the CP voltage level changes from C to B – any applicable LIN signals indicate that the switching device shall be opened (see D.8.2.5)	max 3 s
Exception		
3	The charging station shall open the switching device after the CP voltage level has changed from B or C to A.	max 100 ms
4	The charging station shall open the switching device after the CP voltage level has changed from any level to D or E.	max 3 s
5	The charging station shall open the switching device (under load) if the EV stops responding to LIN signals.	min 2 s, max 3 s after last response
6	The charging station may open the switching device (under load) if the EV does not stop the vehicle load current and does not open the S2 switch, although this is requested by the charging station using LIN signals.	min 6 s after request
7	The charging station may open the switching device (under load) if the vehicle load current is too high and the EV does not reduce the current although this is requested by the charging station using LIN signals.	min 6 safter request

D.6.5 Control of latching and unlatching of IEC 62196-2 type 2 socket-outlets and vehicle inlets

Latching and unlatching of IEC 62196-2 type 2 socket-outlets and vehicle inlets shall be controlled as specified in Table D.9.

Table D.9 – Control of latching and unlatching

Line	Description	Time limits
If an EV implements latching:		
1	The EV should latch the vehicle inlet when the vehicle connector is plugged in.	-
2	The EV shall confirm that the vehicle inlet is latched before closing the S2 switch.	-
3	The EV shall not unlatch the vehicle inlet while the S2 switch is closed.	-
If a case A or case B charging station implements latching:		
4	The charging station should latch its socket-outlet when the CP voltage level changes from A to B.	-
5	The charging station shall unlatch its socket-outlet when the CP voltage level changes from any level to A, unless unlatching can be done by adequate user interaction. In case B, if the cable belongs to the charging station owner, unlatching is under the owner’s decision.	max 5 s
6	The charging station shall unlatch its socket-outlet when power supply to the charging station is interrupted, unless unlatching can be done by adequate user interaction. In case B, if the cable belongs to the charging station owner, unlatching is under the owner’s decision.	max 30 s
7	The charging station shall confirm that the socket-outlet is latched before closing the switching device.	-
8	The charging station shall not unlatch its socket-outlet while the switching device is closed.	-
9	In case A (EV with attached cable), a switch to interrupt the control pilot circuit may be used on the EV side (cable, plug, vehicle) to simulate an EV unplug and trigger the charging station to unlatch the socket-outlet.The EV needs to make sure that the vehicle load current is below 1 A.	-

D.7 Charging sequences

D.7.1 General

Clause D.7 provides examples of typical charging sequences.

See Clause D.8 for information about LIN schedules, frames and signals that are used in these examples.

D.7.2 Start-up of normal AC charging sequence

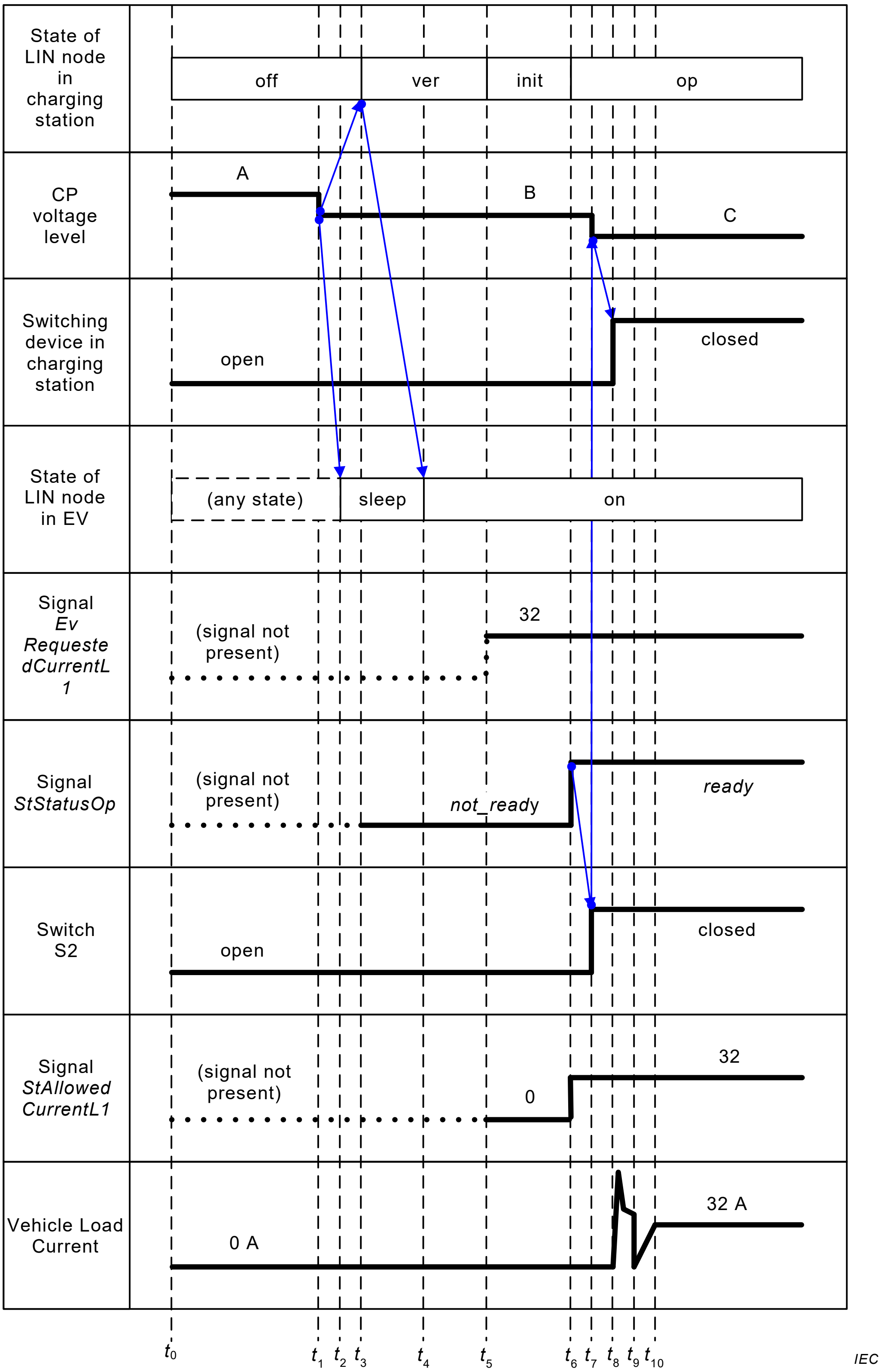


Figure D.4 – Example of timing diagram for start-up of normal AC charging sequence

Table D.10 – Timing for start-up of normal charging sequence

Time	Description	Associated time limits
t_0	Start of sequence. EV and charging station are not connected.	-
t_1	User inserts the connector into vehicle inlet which closes the control pilot circuit and changes the CP voltage level from A to B at the charging station side and from 0 V to CP voltage level B at the vehicle side.	-
t_2	EV detects that the control pilot voltage is no longer at 0 V.	-
t_3	Charging station starts triggering LIN frames according to schedule *ver*.	t_3-t_1: see Table D.6 line 1
t_4	The EV starts responding to LIN headers.	t_4-t_3: see Table D.6 line 2
t_5	Version selection complete. The EV has selected the communication version and the charging station changes to schedule *init*.	t_5-t_4: typ < 50 ms
t_6	Initialization complete. Charging station has sent and received all initialization info. Charging station changes to schedule *op*. In this example the charging station is ready to supply power, so the charging station sends the signal *StStatusOp* = *ready*. The figure shows only *EvRequestedCurrentL1,* not the signals for L2, L3 and N.	t_6-t_5: typ < 200 ms
t_7	EV closes S2, changing the CP voltage level from B to C.	t_7-t_6: typ < 100 ms
t_8	Charging station closes switching device.	t_8-t_7: see Table D.8 line 1
t_9	End of inrush current	t_9-t_8 (inrush current): see ISO 17409
t_{10}	EV has ramped up its load current, respecting the *StAllowableCurrent* signals. The figure shows only *StAllowableCurrentL1,* not the signals for L2, L3 and N.	t_{10}-t_9 (ramp up): vehicle-specific

D.7.3 Normal EV-triggered stop of charging

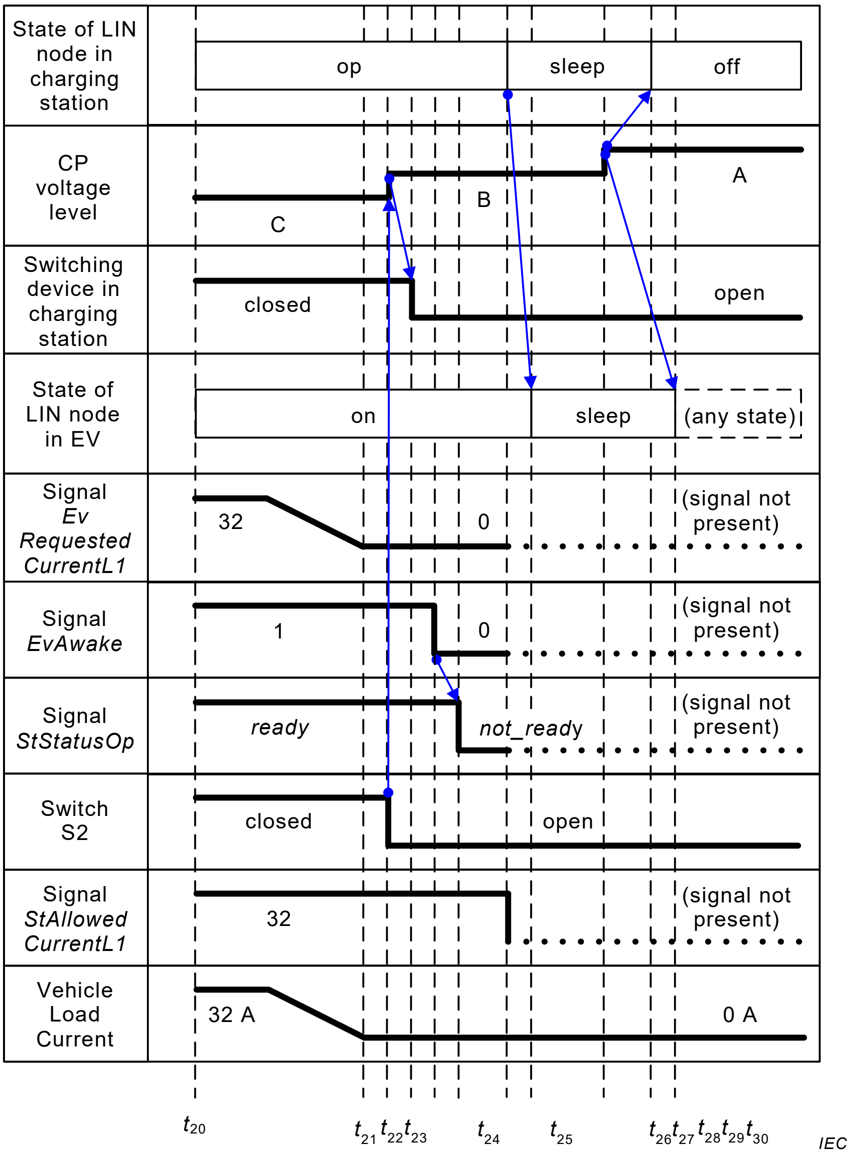


Figure D.5 – Timing diagram for normal EV-triggered stop of charging

Table D.11 – Timing for normal EV-triggered stop of charging

Time	Description	Limits
t_{20}	EV and charging station are connected, LIN communication is established using schedule *op*, S2 and the switching device are closed and the EV is drawing load current, respecting the *StAllowableCurrent* signals.	-
t_{20} .. t_{21}	As the battery nears full charge, the EV (in this example) reduces the vehicle load current and at the same time adjusts the *EvRequestedCurrent* signals accordingly.	-
t_{21}	EV has stopped drawing load current and is sending *EvRequestedCurrentL1* = 0 and *EvAwake = 0.* Note: The figure shows only *EvRequestedCurrentL1*, not the signals for L2, L3 and N.	t_{21}-t_{20}: vehicle-specific
t_{22}	EV opens S2, changing the CP voltage level from C to B.	t_{22}-t_{21}: vehicle-specific
t_{23}	Charging station opens switching device.	t_{23}-t_{22}: see Table D.8 line 2.
t_{24}	The EV sends*EvAwake* = 0.	t_{24}-t_{22}: vehicle-specific
t_{25}	The charging station sends *StStatusOp* = *not_ready*.	t_{25}-t_{24}: typ < 100 ms
t_{26}	Charging station sends "go to sleep" command.	t_{24}-t_{23}: typ < 100 ms
t_{27}	EV LIN node goes to sleep.	t_{27}-t_{26}: vehicle-specific
t_{28}	User unplugs EV, opening the control pilot circuit, which causes a transition to CP voltage level A at the charging station.	-
t_{29}	Charging station turns off its LIN master and resets all initialization and status values to default.	t_{29}-t_{28}: see Table D.6 line 3
t_{30}	EV sets all initialization and status values to default.	t_{30}-t_{28}: vehicle-specific

D.7.4 Normal stop of charging triggered by charging station

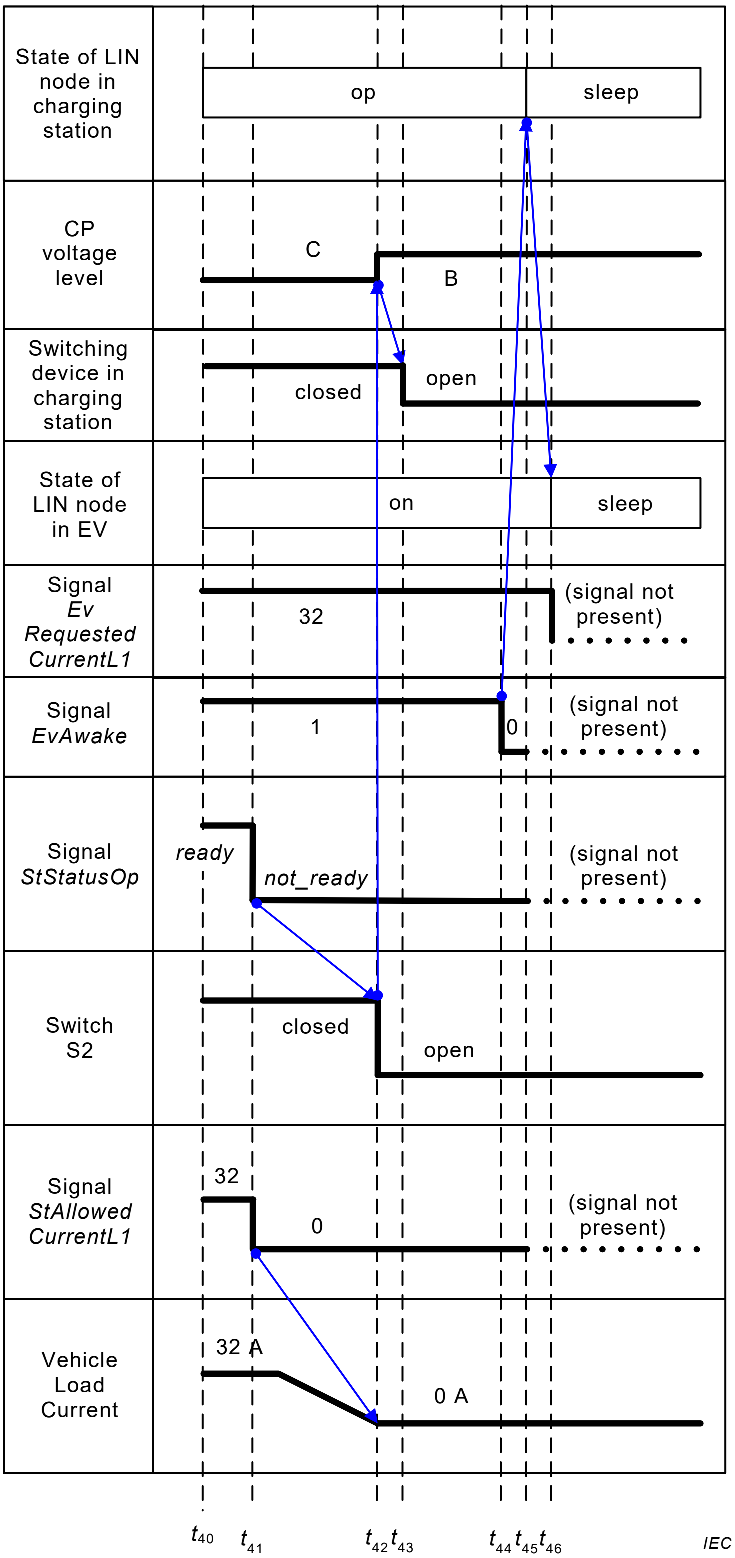


Figure D.6 – Example of timing diagram for normal stop of charging triggered by charging station

Table D.12 – Timing for normal stop of charging triggered by charging station

Time or duration	Description	Limits
t_{40}	EV and charging station are connected, LIN communication is established using schedule *op*, S2 and the switching device are closed and the EV is drawing load current, respecting the *StAllowableCurrent* signals.	-
t_{41}	To stop charging, the charging station sets *StStatusOp* = *not_ready* and all *StAllowableCurrent* signals to 0. Only the signal *StAllowableCurrentL1* is shown in this figure.	-
t_{42}	EV has reduced current to 0 and opens S2, creating a change of the CP voltage level from C to B.	t_{42}-t_{41}: see Table D.7 line 3.
t_{43}	Charging station opens switching device.	t_{43}-t_{42}: see Table D.8 line 2.
t_{44}	EV sends sleep request by sending signal *EvAwake* = 0.	t_{44}-t_{43}: vehicle-specific
t_{45}	Charging station sends go-to-sleep command.	t_{45}-t_{44}: typ<100 ms
t_{46}	EV LIN node goes to sleep. System sleeps until disconnect (see t_{28} and following) or wakeup.	t_{46}-t_{45}: vehicle-specific

D.8 LIN Communication

D.8.1 General

The charging station and the EV shall implement the LIN protocol according to ISO 17987-3.

The charging station shall act as the LIN master, when the CP voltage level is B or C, and control LIN communication using the schedules, frames and signals defined below.

The charging station shall communicate at a data rate of 20 kbit/s nominal, or, if specific EMC restrictions apply, at a reduced data rate of 10 kbit/s nominal. The EV shall auto-detect the data rate.

D.8.2 Schedules

D.8.2.1 General

The charging station shall implement LIN schedules as described in Figure D.7, Table D.13 and Table D.14. Additionally, the charging station may trigger frames that are not required.

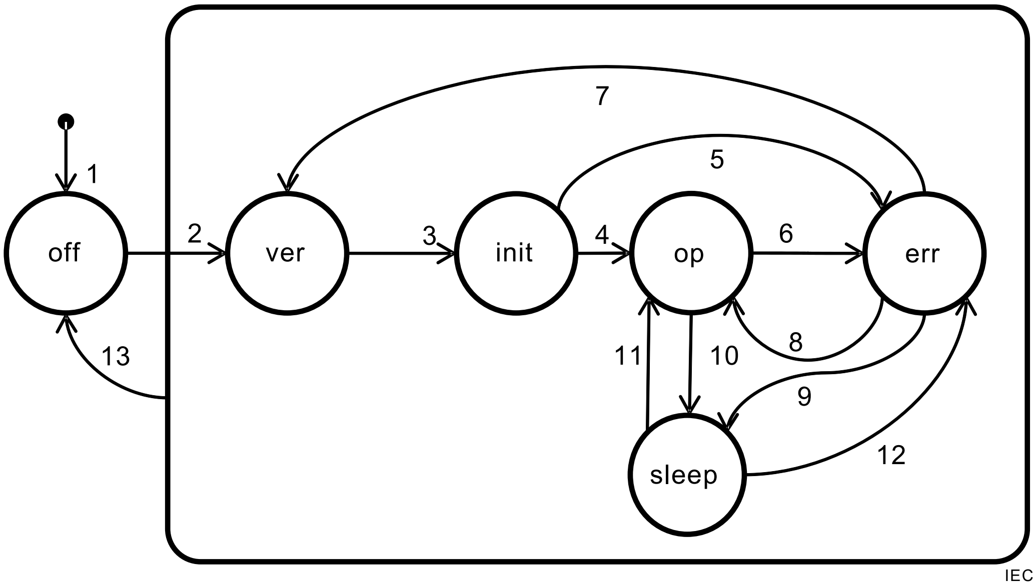


Figure D.7 – State diagram of the LIN node in the charging station

Table D.13 – States of the LIN node in the charging station and frame schedule description

State	Status signals			Switching Device	Required frames	Description
	StStatus Ver	*StStatus Init*	*StStatus Op*			
off	n/a	n/a	n/a	open	none	LIN communication is disabled.
ver	*incomplete* or *error*	any	any	open	*StVersionList, EvVersionList*	Charging station uses schedule *ver* to setup communication and exchange of information about supported communication versions. EV selects communication version.
init	*complete*	*incomplete*	any	open	*StStatus, EvStatus, StVoltages, StMaxCurrents, EvMaxVoltages, EvMinVoltages, EvMaxMinCurrents, CaProperties*	Charging station uses schedule *init* to initialize the charging system. Information is exchanged about the power supply and about the vehicle load to verify compatibility.
op	*complete*	*complete*	*ready* or *not_ready*	open or closed	*StStatus, EvStatus, StNotReadyList, EvS2openList, EvPresentCurrents*	The charging station uses schedule *op* when initialization is complete and no error has occurred.
err	*complete*	*complete*	*error*	open	*StStatus, EvStatus, StNotReadyList, EvS2openList, EvPresentCurrents, StErrorList, EvErrorList*	The charging station uses schedule *err* for error handling.
	complete	*error*	any			
sleep	n/a	n/a	n/a	open	none	To go to LIN sleep, the charging station sends the go-to-sleep command and sets its LIN transceiver to sleep mode. During LIN sleep, there is no bus activity.

Table D.14 – Transitions of the LIN node in the charging station

Transition	Event(s)	Action(s)
1	Startup or reset of charging station.	Default transition.
2	The charging station detects a transition of the CP voltage level from A to B or as defined in D.9.5 step 4.	The charging station starts sending LIN headers.
3	The charging station detects that EV has selected an *EvSupportedVersion* that is supported by the charging station.	-
4	The charging station detects that system initialization is complete (*StStatusInit=complete*).	-
5	The charging station detects an error. (*StStatusInit* = *error*)	-
6	The charging station has detected an error or has received an error signal from the EV. (*StStatusOp* = *error*)	-
7	Reset of system, e.g. the charging station has detected a reset of the EV (i.e. the EV has sent frames with *EvSelectedVersion* = FF_{16}).	-
8	All errors have been cleared AND *StStatusInit* = *complete*.	-
9, 10	The charging station has received *EvAwake* = 0 AND the switching device is open.	Charging station sends go-to-sleep command and sets its LIN node to sleep mode.
11	*StStatusOp* ≠ *error* AND *StStatusInit* = *complete* AND one of the following has occurred: – 10 minutes have passed OR – charging station detects wakeup signal from EV OR – charging station determines that a signal contained in the frames of schedule “*op*” has changed its value (this includes the signal *StStatusOp*).	The charging station wakes up and starts sending LIN headers.
12	(*StStatusOp* = *error* OR *StStatusInit* = *error*) AND one of the following has occurred: – 10 minutes have passed OR – charging station detects wakeup signal from EV OR – charging station determines that a signal contained in the frames of schedule *err* has changed its value (this includes the signal *StStatusOp*).	The charging station wakes up and starts sending LIN headers.
13	The EV is disconnected, opening the control pilot circuit. The charging station detects CP voltage level A.	The charging station stops LIN communication.

D.8.2.2 Start and stop of LIN communication

When the control pilot circuit is connected and the charging station starts sending LIN headers (see Table D.6 line 1), it shall use schedule *ver*.

The EV should respond to the LIN headers (see Table D.6 line 2).

When the control pilot circuit is disconnected and the charging station stops sending LIN headers (see Table D.6 line 3), it shall reset all information it has received from the EV to default values and reset all signals in *StStatus* to default values.

When the control pilot circuit is disconnected, the EV shall reset all information it has received from the charging station to default values and reset all signals in *EvStatus* to default values.

D.8.2.3 Version selection

The following sequence shall be used for version selection in schedule *ver*:

1) At the start of the charging session, the charging station sets *StPageNumber* = 0, *StStatusVer* = *incomplete,* and*StSelectedVersion* = FF_{16} (not available).
2) At the start of the charging session, the EV sets *EvPageNumber* = 0, *EvStatusVer* = *incomplete*, and *EvSelectedVersion* = FF_{16} (not available).
3) The frame *StVersionList* is triggered.
4) If *StPageNumber* = 0 and *EvStatusVer* = *error*, then the EV sets *EvStatusVer* = *incomplete*.
5) If the EV selects a value received in any *StSupportedVersion*, the EV shall set the signal *EvSelectedVersion* to that value. The EV may now set the signal *EvStatusVer* = *complete*, or the EV may wait until all non-empty *StVersionList* are received, for forensic purposes. If *EvStatusVer* = *error*, then the EV sets *EvStatusVer* = *incomplete*. Then go to step 7).
6) If the first *StPageNumber* received by the EV is not zero, then set *EvStatusVer* = *error*.

   If a subsequent *StPageNumber* received by the EV is not sequential, then set *EvStatusVer* = *error*.

   If the last *StVersionList* received by the EV before *StPageNumber* rolls back over to zero does not contain a blank entry (*StSupportedVersion* = FF_{16}), then the EV sets *EvStatusVer* = *error*.
7) The charging station triggers the frame *EvVersionList*.
8) If the signal *EvStatusVer* = error, the charging station shall set *StPageNumber* = 0 and go to step 3), for a limited number of retries.
9) If the signal *EvStatusVer* = *complete* and the signal *EvSelectedVersion* is equal to one of the *StSupportedVersion* signals from the charging station, the charging station sets the signals *StSelectedVersion* = *EvSelectedVersion*, *StStatusVer* = *complete.* The version negotiation is complete. The charging station may exit to schedule *init*, or the charging station may continue triggering *EvVersionList* if it wishes to collect forensic data, then exit to schedule *init* once all *EvVersionList* transmissions have been received successfully.
10) If the signal *EvStatusVer* = *complete* and the signal *EvSelectedVersion* is not equal to one of the *StSupportedVersion* signals from the charging station, then the charging station sets the signal *StStatusVer* = *error*, *StPageNumber* = 0, and go to step 3) for a limited number of retries.
11) If the signal *EvStatusVer* = *incomplete*, the station increments *StPageNumber*. If this next *StPageNumber* points to a blank page (all *StSupportedVersion* entries = FF_{16}), then set *StPageNumber* = 0 for a limited number of roll-overs. Then go to step 3).

D.8.2.4 System initialization

The following sequence is used for system initialization in schedule *init*:

1) After version selection is complete (see D.8.2.3), the charging station starts initialization of the system. At the start of initialization, the signals *StStatusInit* and *EvStatusInit* are set to *incomplete*.
2) During system initialization, the charging station triggers the frames *StStatus* and *EvStatus* within the max period stated in Table D.15. Between the frames *StStatus* and *EvStatus*, the charging station triggers the other frames that are members of schedule *init* (see Table D.13) as follows:
   a) first the frames *EvMaxVoltages, EvMinVoltages, EvMaxMinCurrents,*
   b) then the frame *CaProperties* and
   c) then the frames *StVoltages, StMaxCurrents* (possibly modified based on the information from *CaProperties*) and
   d) then it repeats triggering these frames, starting again from the frame *EvMaxVoltages*.
3) If the EV determines that the initialization information has been exchanged completely and that the parameters of the charging station and the EV are compatible, then it sets the signal *EvStatusInit* = *complete*.
4) If the charging station receives the signal *EvStatusInit* = *complete* and the charging station determines that the parameters of the charging station and the EV are compatible, then

the charging station sets *StStatusInit* = *complete*, initialization is complete, and it exits to schedule *op*.

5) If the charging station receives the signal *EvSelectedVersion*= FF_{16}, then it sets the first empty entry in the pages of *StErrorList* to 01_{16}, sets the signal*StStatusInit* = *error* and exits to schedule *err*.
6) If the EV determines that the parameters of the charging station and the EV are incompatible, then the EV sets the signal *EvStatusInit* = *error*.
7) If the EV determines that initialization takes longer than 5 s, then the EV sets the signal *EvStatusInit* = *error*.
8) If the charging station receives the signal *EvStatusInit* = *error*, then the charging station sets the first empty entry in the pages of *StErrorList* to 04_{16}, sets the signal *StStatusInit* = *error* and exits to schedule *err*.
9) If the charging station determines that the parameters of the charging station and the EV are incompatible, then the charging station sets the first empty entry in the pages of *StErrorList* to 05_{16}, sets the signal *StStatusInit* = *error* and exits to schedule *err*.
10) If initialization takes longer than 5 s, then the charging station sets the first empty entry in the pages of *StErrorList* to 05_{16}, sets the signal *StStatusInit* = *error* and exits to schedule *err*.

D.8.2.5 Operating

In schedule *op* the charging station controls LIN communication and the switching device as follows:

1) If the charging station detects CP voltage level B, then it opens the switching device (see Table D.8 line 2).
2) If the charging station detects an error, then it sets the first empty entry in the pages of *StErrorList* to 00_{16} (or a more specific error code, if applicable), sets the signal *StStatusOp* = *error* and exits to schedule *err*.
3) If the charging station receives a frame with *EvSelectedVersion*= FF_{16}, then it sets the first empty entry in the pages of *StErrorList* to 01_{16}, sets the signal *StStatusOp* = *error* and exits to schedule *err*.
4) If the charging station receives the signal *EvStatusOp* = *error*, then it sets the first empty entry in the pages of *StErrorList* to 02_{16}, sets the signal *StStatusOp* = *error* and exits to schedule *err*.
5) If power is not available, then the charging station sets the signal *StStatusOp* = *not_ready*. The charging station provides reasons for non-availability of power by keeping the entries in the pages of *StNotReadyList* updated.
6) If the charging station does not detect CP voltage level B within 3 seconds (see Table D.7 line 3) after setting *StStatusOp* = *not_ready*, then it sets the first empty entry in the pages of *StErrorList* to 03_{16}, sets the signal *StStatusOp* = *error* and exits to schedule *err*.
7) If power is available, then the charging station sets the signal *StStatusOp* = *ready* and clears all entries in the pages of *StNotReadyList*.
8) If *StStatusOp* = *ready* and the charging station detects CP voltage level C, then the charging station closes the switching device (see Table D.8 line 1).
9) The charging station triggers the frames*StStatus* and *EvStatus* within the max period stated in Table D.15. Between the frames *StStatus* and *EvStatus*, the charging station triggers the other frames that are members of the schedule *op* (see Table D.13).

If the EV receives the signal *StStatusOp* = *not_ready*, it stops drawing current and opens S2 within 3 seconds (see Table D.7 line 3).

The EV reduces the vehicle load current within 5 seconds, if needed to respect the *StAllowableCurrent* signals (see Table D.7 line 4).

D.8.2.6 Communication sleep modes

D.8.2.6.1 LIN sleep

The charging station and the EV shall implement LIN sleep as specified in ISO 17987-2:–, Clause 5, Network management.

To save power, the EV may set the signal *EvAwake* = 0 while S2 is open.

If the charging station receives the signal *EvAwake* = 0 while the switching device is open and it is in schedule *op* or *err*, the charging station should send the system to sleep mode. To send the system to sleep mode, the charging station:

– sets *StStatusOp* = *not_ready* and sets one entry in the pages of *StNotReadyList* to 01_{16} (EV is requesting to go to sleep);
– triggers the frame *StStatus* (to transmit the signal*StStatusOp*), triggers all pages of *StNotReadyList*, then triggers the frame *EvStatus* (to verify the signal *response_error*). If a *response_error* has occurred, the charging station repeats this step.
– Then sends a go-to-sleep command and stops all bus activity.

The charging station wakes up the system as specified for transitions 11 and 12 in Table D.14.

NOTE More information about wake up will be provided in the future ISO 17987-2:–.

D.8.2.6.2 EV deep sleep

The EV may turn off its LIN transceiver at any time and become unresponsive to LIN communication.

When the EV has become unresponsive, the charging station shall keep on triggering frames (in schedule *err* after detecting the communication error).

D.8.2.7 Error handling

In schedule *err*, the charging station controls LIN communication and the switching device as follows:

1) If the charging station detects CP voltage level B, then it opens the switching device (see Table D.8 line 2).
2) If the charging station does not detect CP voltage level B within 6 seconds (see Table D.7 line 3) after an entry in the pages of *StNotReadyList* was set, then it may open the switching device (see Table D.8 line 6).
3) If the charging station does not detect CP voltage level B within 6 seconds after an entry in the pages of *StErrorList* was set, then it may open the switching device (see Table D.8 line 6).
4) If the charging station receives a frame with *EvSelectedVersion*= FF_{16}, then it sets the first empty entry in the pages of *StErrorList* to 01_{16}, unless one entry in the pages of *StErrorList* is already set to 01_{16}.
5) If the charging station receives the signal *EvStatusOp* = *error*, then it sets the first empty entry in the pages of *StErrorList* to 02_{16}, unless one entry in the pages of *StErrorList* is already set to 02_{16}.
6) If the charging station does not detect CP voltage level B within 3 seconds (see Table D.7 line 3) after entering schedule *err*, then it sets the first empty entry in the pages of *StErrorList* to 03_{16}, unless one entry in the pages of *StErrorList* is already set to 03_{16}.
7) If the charging station detects an additional error, then it sets the first empty entry in the pages of *StErrorList* to 00_{16} (or the applicable more specific error code), unless one entry in the pages of *StErrorList* is already set to 00_{16} (or the applicable more specific error code).

8) As power becomes not available or available, the charging station keeps the entries in the pages of *StNotReadyList* updated.

9) The charging station triggers the frames *StStatus* and *EvStatus* with a period according to Table D.15. Between the frames *StStatus* and *EvStatus*, the charging station triggers the other frames that are members of the schedule *err* (see Table D.13). The charging station triggers all pages of *StErrorList* and all pages of*EvErrorList* at least once before it can leave schedule *err* as follows:

   a) If the charging station receives the signal *EvStatusOp* = *no_error*, then it clears the entry, if any, in the pages of *StErrorList* that is set to 02_{16}.

   b) If an additional error is resolved, the charging station clears the corresponding entry in pages of *StErrorList*.

   c) If the pages of *StErrorList* are empty and the pages of *StNotReadyList* are empty, the charging station sets the signal *StStatusOp* = *ready* and exits to schedule *op*.

   d) If the pages of *StErrorList* contain an entry that is set to 01_{16}, then the charging station resets all initialization values it has received from the EV to default values, resets the pages of *StErrorList* and the pages of *StNotReadyList* to default values, resets all signals in *StStatus* to default values and then exits to schedule *ver*.

If the EV receives the signal *StStatusOp* = *error*, it stops drawing current and opens S2 within 3 seconds (see Table D.7 line 3).

D.8.3 Frames

Table D.15 shows the defined LIN frames. All slave tasks shall provide the frames that are assigned to their node, providing at least default values for all contained signals.

The frame type of all frames is “unconditional frame”.

Frames marked with a star (*) shall be transmitted during the applicable schedule. Their timing is defined by the manufacturer of the charging station.

The prefix of a frame name indicates the publisher of the frame:

– *St*: charging station
– *Ca*: cable assembly
– *Ev*: EV

The frames *StVersionList*, *EvVersionList*, *StNotReadyList*, *EvS2openList*, *StErrorList* and *EvErrorList* can be used to transmit information that is organized in several pages as follows:

– The entries in the pages of *StVersionList* are sorted such that the *StSupportedVersion* values of versions that are most preferred by the charging station are provided first.
– The entries in the pages of *EvVersionList* are sorted such that the *EvSupportedVersion* values of versions that are most preferred by the EV are provided first.
– The entries in the pages of *StNotReadyList* can contain *StReasonCode* values. The entries in the pages of *EvS2openList* can contain *EvReasonCode* values. These values are set according to D.8.2.5, D.8.2.6 and D.8.2.7.
– The entries in the pages of *StErrorList* can contain *StErrorCode* values. The entries in the pages of *EvErrorList* can contain *EvErrorCode* values. These values are set according to D.8.2.7.
– For all signals, the remaining entries of their last page are filled with the code FF_{16} such that their last page contains at least one entry with this code.
– The first time each of the frames is triggered the publisher responds with *PageNumber* = 0 and provides the corresponding page. Each time the frame is triggered again, *PageNumber* is incremented and the corresponding page is provided. After the last page, the *PageNumber* is reset to 0.

Table D.15 – Frames for AC charging

Frame Identifier	Frame Name	Contained Signals	Max period in ms
0	*StVersionList*	Byte 0: *StSelectedVersion* Byte 1: Bit 0: reserved (typ 1) Bits 1-2: *StStatusVer* Bits 3-4: *StStatusInit* Bits 5-6: *StStatusOp* Bit 7: reserved (typ1) Byte 2: *StPageNumber* (typ00_{16}) Byte 3: *StSupportedVersion* (most preferred, typ 01_{16}) Byte 4: *StSupportedVersion* (next preference, typ FF_{16}) Byte 5: *StSupportedVersion* (next preference, typ FF_{16}) Byte 6: *StSupportedVersion* (next preference, typ FF_{16}) Byte 7: *StSupportedVersion* (least preferred, typ FF_{16})	100
1	*EvVersionList*	Byte 0: *EvSelectedVersion* (typ 01_{16}) Byte 1: Bit 0: *response_error* Bits 1-2: *EvStatusVer* Bits 3-4: *EvStatusInit* Bits 5-6: *EvStatusOp* Bit 7:*EvAwake* Byte 2: *EvPageNumber* (typ00_{16}) Byte 3: *EvSupportedVersion* (most preferred, typ 01_{16}) Byte 4: *EvSupportedVersion*(next preference, typ FF_{16}) Byte 5: *EvSupportedVersion*(next preference, typ FF_{16}) Byte 6: *EvSupportedVersion*(next preference, typ FF_{16}) Byte 7: *EvSupportedVersion*(least preferred, typFF_{16})	100
2	*StStatus*	Byte 0: *StSelectedVersion* (typ 01_{16}) Byte 1: Bit 0: reserved (1) Bits 1-2: *StStatusVer* Bits 3-4: *StStatusInit* Bits 5-6: *StStatusOp* Bit 7: reserved (1) Byte 2: *StAllowableCurrentL1* Byte 3: *StAllowableCurrentL2* Byte 4: *StAllowableCurrentL3* Byte 5: *StAllowableCurrentN* Bytes 6-7: reserved (FF_{16})	100
3	*EvStatus*	Byte 0: *EvSelectedVersion* (typ 01_{16}) Byte 1: Bit 0: *response_error* Bits 1-2: *EvStatusVer* Bits 3-4: *EvStatusInit* Bits 5-6: *EvStatusOp* Bit 7:*EvAwake* Byte 2: *EvRequestedCurrentL1* Byte 3: *EvRequestedCurrentL2* Byte 4: *EvRequestedCurrentL3* Byte 5: *EvRequestedCurrentN* Bytes 6-7: reserved (FF_{16})	100
4	*EvPresentCurrents*	Byte 0: *EvSelectedVersion* (typ 01_{16}) Byte 1: *EvPresentCurrentL1* Byte 2: *EvPresentCurrentL2* Byte 3: *EvPresentCurrentL3* Byte 4: *EvPresentCurrentN* Bytes 5-7: reserved (FF_{16})	*
5	*StVoltages*	Byte 0: *StSelectedVersion* (typ 01_{16}) Bytes 1-2: *StVoltageL1N* Bytes 3-4: *StVoltageLL* Byte 5: *StFrequency* Bytes 6-7: reserved (FF_{16})	1 000
6	*StMaxCurrents*	Byte 0: *StSelectedVersion* (typ 01_{16}) Byte 1: *StMaxCurrentL1* Byte 2: *StMaxCurrentL2* Byte 3: *StMaxCurrentL3* Byte 4: *StMaxCurrentN* Bytes 5-7: reserved (FF_{16})	1 000

Frame Identifier	Frame Name	Contained Signals	Max period in ms
7	*EvMaxVoltages*	Byte 0: *EvSelectedVersion* (typ 01_{16}) Bytes 1-2: *EvMaxVoltageL1N* Bytes 3-4: *EvMaxVoltageLL* Byte 5:*EvFrequencies* Bytes 6-7: reserved (FF_{16})	*
8	*EvMinVoltages*	Byte 0: *EvSelectedVersion* (typ 01_{16}) Bytes 1-2: *EvMinVoltageL1N* Bytes 3-4: *EvMinVoltageLL* Bytes 5-7:reserved (FF_{16})	*
9	*EvMaxMinCurrents*	Byte 0: *EvSelectedVersion* (typ 01_{16}) Byte 1: *EvMaxCurrentL1* Byte 2: *EvMaxCurrentL2* Byte 3: *EvMaxCurrentL3* Byte 4: *EvMaxCurrentN* Byte 5: *EvMinCurrentL1* Byte 6: *EvMinCurrentL2* Byte 7: *EvMinCurrentL3*	*
10	*CaProperties*	Byte 0: *CaVersion* (typ 01_{16}) Byte 1: Bit 0: response_error Bits 1-7: reserved (1) Byte 2-3: *CaMaxVoltage* Byte 4: *CaMaxCurrentL1* Byte 5: *CaMaxCurrentL2* Byte 6: *CaMaxCurrentL3* Byte 7: *CaMaxCurrentN*	*
11	*StNotReadyList*	Byte 0: *StSelectedVersion* (typ 01_{16}) Byte 1: *StPageNumber* Byte 2: First *StReasonCode* Byte 3: Next *StReasonCode* Byte 4: Next *StReasonCode* Byte 5: Next *StReasonCode* Byte 6: Next *StReasonCode* Byte 7: Last *StReasonCode*	1 000
12	*EvS2openList*	Byte 0: *EvSelectedVersion* Byte 1: *EvPageNumber* Byte 2: First *EvReasonCode* Byte 3: Next *EvReasonCode* Byte 4: Next *EvReasonCode* Byte 5: Next *EvReasonCode* Byte 6: Next *EvReasonCode* Byte 7: Last *EvReasonCode*	*
13	*StErrorList*	Byte 0: *StSelectedVersion* Byte 1: *StPageNumber* Byte 2: First *StErrorCode* Byte 3: Next *StErrorCode* Byte 4: Next *StErrorCode* Byte 5: Next *StErrorCode* Byte 6: Next *StErrorCode* Byte 7: Last *StErrorCode*	1 000
14	*EvErrorList*	Byte 0: *EvSelectedVersion* Byte 1: *EvPageNumber* Byte 2: First *EvErrorCode* Byte 3: Next *EvErrorCode* Byte 4: Next *EvErrorCode* Byte 5: Next *EvErrorCode* Byte 6: Next *EvErrorCode* Byte 7: Last *EvErrorCode*	*
15 to 49	Reserved, see SEK TS 481 05 16 (under development).		
50 to 59	Available for application specific frames.		
60 to 63	See ISO 17987 (under consideration) and SEK TS 481 05 16 (under development).		

D.8.4 Signals

D.8.4.1 General

The prefix of a signal indicates its publisher as follows:

– *Ev*: EV,
– *St*: charging station,
– *Ca*: cable assembly.

D.8.4.2 General signals

response_error

The *response_error* signal shall be set whenever a frame (except for event triggered frame responses) that is transmitted or received by the slave node contains an error in the frame response.

The *response_error* signal shall be cleared when the unconditional frame containing the *response_error* signal is successfully transmitted.

NOTE See 5.5.4 in ISO 17987-3:– for the LIN specification of *response_error.*

StPageNumber, EvPageNumber

These signals are enumerators for frames that can provide several pages of signals. Typically, St*PageNumber* or *EvPageNumber* is incremented each time the frame is sent. After the frame with the last page has been sent, *PageNumber* rolls over and the next frame will contain the first page. See D.8.3.

D.8.4.3 Signals for version negotiation

EvSelectedVersion, StSelectedVersion, CaVersion

These signals are used by the EV, the cable assembly and the charging station to uniquely identify the frame format and signal specification they are using. Future versions of this Annex D or SEK TS 481 05 16 (under development) can describe a new frame format and a new signal specification and then these signals will identify exactly which format is being used.

EvStatusVer

This signal is used by the EV to indicate the status of version negotiation, see D.8.2.3.

StStatusVer

This signal is used by the charging station to indicate the status of version negotiation, see D.8.2.3.

EvSupportedVersion, StSupportedVersion

These signals are used by the EV and the charging station to indicate which versions are supported. *SupportedVersion* = 1 describes this edition of Annex D (2016). *SupportedVersion* = 0 describes PWM, and is typically used only in case of unresolved errors. *SupportedVersion* = FF_{16} indicates a blank entry.

D.8.4.4 Signals for system initialization

EvStatusInit

This signal is used by the EV to indicate the status of system initialization, see D.8.2.4.

StStatusInit

This signal is used by the charging station to indicate the status of system initialization, see D.8.2.4.

EvMaxCurrentL1, EvMaxCurrentL2, EvMaxCurrentL3, EvMaxCurrentN

These signals are used by the EV to indicate its rated current at each corresponding contact of the vehicle inlet (or plug, in case A). For example, for *EvMaxCurrentL1* this is the contact marked L1.

If the vehicle is designed to draw only single-phase current between L1 and N, the EV shall set both *EvMaxCurrentL1* and *EvMaxCurrentN* to the same value.

EvMinCurrentL1, EvMinCurrentL2, EvMinCurrentL3

These signals are used by the EV to indicate the minimum current where it is fully operable. The information is given for each relevant contact of the vehicle inlet (or plug, in case A). For example for *EvMinCurrentL1* this is the contact marked L1.

Even if the *EvMinCurrent* signal is higher than the corresponding *StAllowableCurrent* signal, the EV may decide to close S2 and draw current (provided that the charging station is indicating *StStatusOp* = *ready*). In this case the EV will only be partly operable, e.g. it might only draw power for its communication controllers and not charge the RESS.

NOTE The signal *EvMinCurrentN* is not needed and does not exist.

EvMaxVoltageL1N

This signal is used by the EV to indicate its rated voltage between the contacts marked L1 and N of the vehicle inlet (or plug, in case A). If the EV has several rated voltages or a range of rated voltages, this signal is used to indicate the maximum of these voltages. Typically, the EV will accept an input voltage at least up to 10 % above this value, see ISO 17409.

EvMaxVoltageLL

This signal is used by the EV to indicate its rated voltage between any two of the contacts marked L1, L2 and L3 of the vehicle inlet (or plug, in case A). If the EV has several rated voltages or a range of rated voltages, this signal is used to indicate the maximum of these voltages. Typically, the EV will accept an input voltage at least up to 10 % above this value, see ISO 17409.

If this value is not applicable, as in the case of an EV where the contacts L2 and L3 are not wired, the EV shall set this signal to $FFFF_{16}$ (unknown).

EvMinVoltageL1N

This signal is used by the EV to indicate its rated voltage between any of the contacts marked L1 and N of the vehicle inlet (or plug, in case A). If the EV supports several rated voltages or a range of rated voltages, this signal is used to indicate the minimum of these voltages.

Typically, the EV will accept an input voltage at least down to 15 % below this value, see ISO 17409.

If the value is not available, the EV shall set this signal to 0.

EvFrequencies

This signal is used by the EV to indicate its rated frequencies. The EV may be rated for one or more frequencies.

StFrequency

This signal is used by the charging station to indicate the nominal frequency of the supply network.

EvMinVoltageLL

This signal is used by the EV to indicate its rated voltage between any two of the contacts marked L1, L2 and L3 of the vehicle inlet (or plug, in case A). If the EV supports several rated voltages or a range of rated voltages, this signal is used to indicate the minimum of these voltages. Typically, the EV will accept an input voltage at least down to 15 % below this value, see ISO 17409.

If this value is not applicable, as in the case of an EV where the contacts L2 and L3 are not wired or not available, the EV shall set this signal to "0".

StMaxCurrentL1, StMaxCurrentL2, StMaxCurrentL3, StMaxCurrentN

These signals are used by the charging station to indicate the maximum current at the corresponding contact of the vehicle connector (in case C) or socket-outlet. For example, for *StMaxCurrentL1* the corresponding contact is marked L1. This value is the minimum of the rated current of the cable assembly (as indicated by the coding resistor in case A or B or the *CaMaxCurrent* signals), the rated current of the charging station and the rated supply current.

StVoltageL1N

This signal is used by the charging station to indicate the nominal voltage between the contacts marked L1 and N of the vehicle connector (in case C) or the socket-outlet of the charging station. This is not the rated voltage of the charging station but rather the nominal voltage provided by the supply network. The actual voltage typically varies between +10 % and -15 % of this value.

Typical values are 120 V, 230 V and 240 V.

StVoltageLL

This signal is used by the charging station to indicate the nominal voltage between any of the contacts marked L1, L2 and L3 of the vehicle connector (in case C) or the socket-outlet of the charging station. This is not the rated voltage of the charging station but rather the nominal voltage provided by the supply network. The actual voltage typically varies between +10 % and -15 % of the nominal voltage.

Typical values are 208 V, 400 V and 480 V.

CaMaxCurrentL1, CaMaxCurrentL2, CaMaxCurrentL3, CaMaxCurrentN

These signals are used by a cable assembly to indicate the rated current of the conductor that is wired to the corresponding contact.

NOTE These values are static and not changed during the charging session.

CaMaxVoltage

This signal is used by the cable assembly to indicate its rated voltage.

D.8.4.5 Signals for status information

EvStatusOp

This signal is used by the EV to indicate its status while the system is operating, see D.8.2.5.

StStatusOp

This signal is used by the charging station to indicate its status while the system is operating, see D.8.2.5.

EvAwake

The EV clears this signal when it wishes to save power, see D.8.2.

EvRequestedCurrentL1, EvRequestedCurrentL2, EvRequestedCurrentL3, EvRequestedCurrentN

The EV uses these signals to indicate the current that it would like to draw at the corresponding contact of the vehicle inlet (or plug, in case A). For example, for *EvRequestedCurrentL1* the corresponding contact is marked L1.

To indicate that the EV could use more current, the values of the *EvRequestedCurrent* signals may be higher than the limits indicated by the corresponding *StAllowableCurrent* signals. The signals should be lower than the cable ratings. The EV should adjust these values as needed to follow the actual current needed by vehicle loads. This signal may be used by the charging station to dynamically adjust the corresponding *StAllowableCurrent* signals.

EvPresentCurrentL1, EvPresentCurrentL2, EvPresentCurrentL3, EvPresentCurrentN

The EV uses these signals to provide information about the measured or estimated load current that is drawn by the EV at the corresponding contact of the vehicle inlet (or plug, in case A). For example, for *EvPresentCurrentL1* this is the contact marked L1.

The EV shall adjust these values dynamically in accordance with the current that it is drawing.

The signal *EvPresentCurrentN* is used to indicate asymmetric current drawn by the vehicle. If this value is set to 0, the vehicle is operating as a balanced three-phase load. When drawing only single-phase power between L1 and neutral, the EV shall set both *EvPresentCurrentL1* and *EvPresentCurrentN* to the same value.

If the EV does not measure or estimate its load current, it shall set the signal for each connected contact to FF_{16} (unknown) and the signal for each unconnected contact to “0”. For example, an EV with a single-phase on-board charger which does not measure or estimate its load current will set *EvPresentCurrentL1* and *EvPresentCurrentN* to “unknown” and *EvPresentCurrentL2* and *EvPresentCurrentL3* to “0”.

StAllowableCurrentL1, StAllowableCurrentL2, StAllowableCurrentL3, StAllowableCurrentN

NOTE These signals implement the requirements of the "maximum allowable current" as described in 6.3.1.6.

The charging station uses these signals to provide information about the current that is available at the charging station as defined by its physical limits and energy management (see Table D.6 line 4), at the corresponding contact of the vehicle connector (or plug, in case A). For example, for *StAllowableCurrentL1* this is the contact marked L1.

The charging station should adjust these values dynamically in accordance with the current requested by the EV via the signal *EvRequestedCurrent*, if possible.

The signal *StAllowableCurrentN* may be used to limit asymmetric current drawn by the vehicle. If this value is set to 0, it will not be possible to operate single-phase chargers connected between L1 and neutral. Therefore, when providing only single-phase power, the charging station shall typically set both *StAllowableCurrentL1* and *StAllowableCurrentN* to the same value.

To indicate short interruptions of the availability of electric power (max. 15 minutes), the charging station may set all four *StAllowableCurrent* signals to 0 while at the same time indicating *StStatusOp* = *ready*. In this case, the EV shall stop drawing current but neither the switching device in the charging station nor the battery contactor on the EV have to be opened. If the interruption lasts too long, the EV is likely to open S2, the switching device shall be opened and the EV is likely to go to sleep mode until the charging station sets the *StAllowableCurrent* signals to values that allow charging.

StReasonCode

This signal is used by the charging station to populate the entries of *StNotReadyList* as appropriate, see Table D.21.

EvReasonCode

This signal is used by the EV to populate the entries of *EvS2openList* as appropriate, see Table D.22.

StErrorCode

This signal is used by the charging station to populate the entries of *StErrorList* as appropriate, see Table D.23.

EvErrorCode

This signal is used by the EV to populate the entries of *EvErrorList* as appropriate, see Table D.24.

D.8.4.6 Signal tables

All signal values that are not described in Table D.16 to Table D.24 are reserved.

Table D.16 – General signals

Signal	Data Type	Values	Description, see D.8.4.2
response_error	BOOL	0 (default)	signal cleared
		1	signal set
StPageNumber, EvPageNumber	UINT8	0 to *max_page_number*	page number The maximum page number is calculated from *x*, the number of UINT8 entries per page, as follows: *max_page_number* = ceiling($256_{10}/x$)-1 E.g., for $x = 6_{10}$ the valid page numbers are 0 to 42_{10}, for $x = 5_{10}$ the valid page numbers are 0 to 51_{10}.

Table D.17 – Signals for version negotiation

Signal	Data Type	Values	Description, see D.8.4.3
EvStatusVer	BOOL[2]	00_2	*incomplete*: Version selection incomplete. (default at EV)
		01_2	*complete*: Version selection complete.
		10_2	*error*: Error during version selection.
		11_2	*not_available*: Signal not available. (default at charging station)
StStatusVer	BOOL[2]	00_2	*incomplete*: Version selection incomplete. (default at charging station)
		01_2	*complete*: Version selection complete.
		10_2	*error*: Error during version selection.
		11_2	*not_available*: Signal not available. (default at EV)
EvSupportedVersion StSupportedVersion	UINT8	01_{16}	The EV or charging station supports the frame format and signal specification described in this Annex D.
		FF_{16}	Empty entry (default)
EvSelectedVersion StSelectedVersion CaVersion	UINT8	01_{16}	The EV or charging station or cable assembly is using the frame format and signal specification described in this Annex D.
		FF_{16}	unknown, not selected (default)

Table D.18 – Signals for system initialization

Signal	Data Type	Values	Description, see D.8.4.4
EvStatusInit	BOOL[2]	00_2	*incomplete*: Initialization incomplete. (default at EV)
		01_2	*complete*: Initialization complete.
		10_2	*error*: Error during initialization.
		11_2	*not_available*: Signal not available. (default at charging station)
StStatusInit	BOOL[2]	00_2	*incomplete*: Initialization incomplete. (default at charging station)
		01_2	*complete*: Initialization complete.
		10_2	*error*: Error during initialization.
		11_2	*not_available*: Signal not available. (default at EV)
EvMaxCurrentL1 *EvMaxCurrentL2* *EvMaxCurrentL3* *EvMaxCurrentN*	UINT8	0 to 250_{10}	Maximum current in A
		FF_{16}	unknown (default)
EvMinCurrentL1 *EvMinCurrentL2* *EvMinCurrentL3*	UINT8	0 to 250_{10}	Minimum current in A (default at EV: 0)
		FF_{16}	(default at charging station)
EvMaxVoltageL1N *EvMaxVoltageLL*	UINT16	0 to 10 000_{10}	Maximum rated voltage in 0,1 V
		$FFFF_{16}$	unknown (default)
EvMinVoltageL1N *EvMinVoltageLL*	UINT16	0 to 10 000_{10}	Minimum rated voltage in 0,1 V
		$FFFF_{16}$	unknown (default at charging station)
EvFrequencies	UINT8	01_{16}	The EV can operate at 50 Hz supply frequency.
		02_{16}	The EV can operate at 60 Hz supply frequency.
		03_{16}	The EV can operate at both 50 Hz and 60 Hz supply frequency.
		FF_{16}	unknown (default)
StMaxCurrentL1, *StMaxCurrentL2,* *StMaxCurrentL3,* *StMaxCurrentN*	UINT8	0 to 250_{10}	Maximum current in A
		FF_{16}	unknown (default)
StVoltageL1N, *StVoltageLL*	UINT16	0 to 10 000_{10}	Nominal voltage in 0,1 V.
		$FFFF_{16}$	unknown (default)
StFrequency	UINT8	01_{16}	The charging station provides 50 Hz supply frequency.
		02_{16}	The charging station provides 60 Hz supply frequency.
		FF_{16}	unknown (default)
CaMaxCurrentL1 *CaMaxCurrentL2* *CaMaxCurrentL3* *CaMaxCurrentN*	UINT8	0 to 250_{10}	Rated current in A
		FF_{16}	unknown (default)
CaMaxVoltage	UINT16	0 to 10 000_{10}	Rated voltage in 0,1 V.
		$FFFF_{16}$	unknown (default)

Table D.19 – Signals for EV status information

Signal	Data Type	Values	Description, see D.8.4.5
EvStatusOp	BOOL[2]	00_2	*no_error*: No error in EV – other than those reported by *EvStatusVer* and *EvStatusInit*. (default at EV)
		01_2	(reserved)
		10_2	*error*: Error in EV (other than those reported by *EvStatusVer*and *EvStatusInit*).
		11_2	*not_available*: Signal not available. (default at charging station)
EvAwake	BOOL	0_2	EV requests to go to sleep.
		1_2	EV does not request to go to sleep. (default)
EvRequestedCurrentL1 *EvRequestedCurrentL2* *EvRequestedCurrentL3* *EvRequestedCurrentN* *EvPresentCurrentL1* *EvPresentCurrentL2* *EvPresentCurrentL3* *EvPresentCurrentN*	UINT8	0 to 250_{10}	Requested current in A (default at charging station: 0)
		FF_{16}	unknown (default at EV)
EvReasonCode	UINT8	See Table D.22	
EvErrorCode	UINT8	See Table D.24	

Table D.20 – Signals for charging station status information

Signal	Data Type	Values	Description, see D.8.4.5
StStatusOp	BOOL[2]	00_2	*not_ready*: The charging station is not ready to supply power. (default at charging station)
		01_2	*ready*: The charging station is ready to supply power.
		10_2	*error*: Error in charging station – other than those reported by *StStatusVer* and *StStatusInit*.
		11_2	*not_available*: Signal not available. (default at EV)
StAllowableCurrentL1 *StAllowableCurrentL2* *StAllowableCurrentL3* *StAllowableCurrentN*	UINT8	0 to 250_{10}	Maximum allowable current in A (default: 0)
StReasonCode	UINT8	See Table D.21	
StErrorCode	UINT8	See Table D.23	

Table D.21 – Codes for the frame *StNotReadyList*

StReasonCode	**Description, see D.8.4.5**
00_{16}	Unspecified reason or other reason
01_{16}	EV is requesting to go to sleep (*EvAwake* = 0).
02_{16}	Energy management reason. Simultaneously, all *StAllowableCurrent* signals will be set to 0. Station will become available again automatically after some time.
03_{16}	Charging stopped by user, e.g. stop button on charging station pressed.
FF_{16}	Empty (default)
NOTE Additional reason codes are defined in SEK TS 481 05 16 (under development).	

Table D.22 – Codes for frame *EvS2openList*

EvReasonCode	**Description, see D.8.4.5**
00_{16}	Unspecified reason or other reason
01_{16}	Charging station is signalling *StStatusOp* = *not_ready*. For further information see which *StReasonCode* signals are set.
02_{16}	Energy management reason at EV, e.g. battery is full or charging is scheduled to start at a later point in time. Simultaneously, *EvRequestedCurrent* will be set to 0.
03_{16}	Charging stopped by user, e.g. S3 button or unlock button pressed.
04_{16}	Maximum allowable current too low.
FF_{16}	Empty (default)
NOTE Additional reason codes are defined in SEK TS 481 05 16 (under development).	

Table D.23 – Codes for frame *StErrorList*

StErrorCode	**Description, see D.8.4.5**
00_{16}	Unspecified error
01_{16}	EV reset detected.
02_{16}	EV reports an error, see *EvErrorList* for details.
03_{16}	EV does not open S2 when charging station is not ready to supply power.
04_{16}	EV reports an initialization error, see *EvErrorList* for details.
05_{16}	Charging station has detected incompatible parameters.
06_{16}	Initialization timeout at charging station.
FF_{16}	Empty (default)
NOTE Additional error codes are defined in SEK TS 481 05 16 (under development).	

Table D.24 – Codes for frame *EvErrorList*

EvErrorCode	**Description, see D.8.4.5**
00_{16}	Unspecified error
FF_{16}	Empty (default)
NOTE Additional error codes are defined in SEK TS 481 05 16 (under development).	

D.9 Requirements for charging stations and EVs that implement both LIN-CP and PWM-CP

D.9.1 General

Clause D.9 specifies implementation of PWM-CP together with LIN-CP.

As specified in 6.3.1.1, charging stations that use accessories according to IEC 62196-2 implement PWM-CP according to Annex A. This Clause D.9 provides requirements for charging stations that implement both LIN-CP and PWM-CP.

D.9.2 Interoperability between charging stations and EVs

Figure D.8 shows possible charging situations with charging stations and EVs using different combinations of LIN-CP and PWM-CP. The arrows indicate the possible energy transfer from a charging station, to the left, to an EV, to the right.

Charging stations and EVs which implement both LIN-CP and PWM-CP are compatible in all situations.

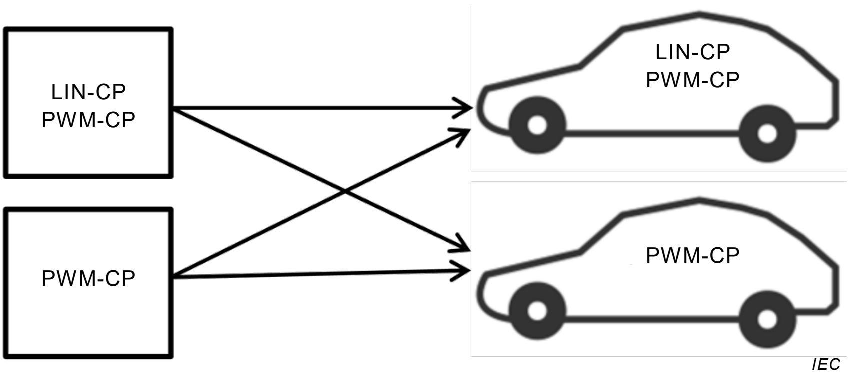


Figure D.8 – Energy transfer between different charging stations and EVs that are equipped with accessories according to IEC 62196-2

D.9.3 Control pilot circuit hardware

– As described in Clause D.4, LIN-CP is designed to facilitate design of implementations that meet the requirements in both Annex A and in this Annex D. This is done by using a common control pilot circuit.

– When implementing both LIN-CP and PWM-CP all requirements in Annex A and in this Annex D apply, with the following modifications:

- The capacitance Cc shall follow the specification in Annex A, not the specification in D.4.2.
- The capacitances Cs and Cv shall follow the specification in D.4.2, not the specification in Annex A.
- The resistor R2 value specified in D.4.2 shall be used.
- The EV shall be able to detect if PWM signals or LIN signals are received after plug-in.
- When the charging station sends PWM signals, the negative part of the PWM signal shall not be pulled out of the tolerance, specified in Annex A for CP voltage level F, by excessive source current, e.g. from the LIN transceivers.

D.9.4 Control pilot circuit functionality

Figure D.9 shows a state diagram for a common LIN-CP and PWM-CP design. See D.5.1 for introduction to the diagram.

For LIN-CP the states Ev0V, StA, StE, B1, B LIN and C LIN are used. The states and the transitions between these states are described in D.5.2.

For PWM-CP the states EvE, StA, StE, StF, B1, B2, C1 and C2 are used. The states and the transitions between these states are described in D.5.2 and in Annex A.

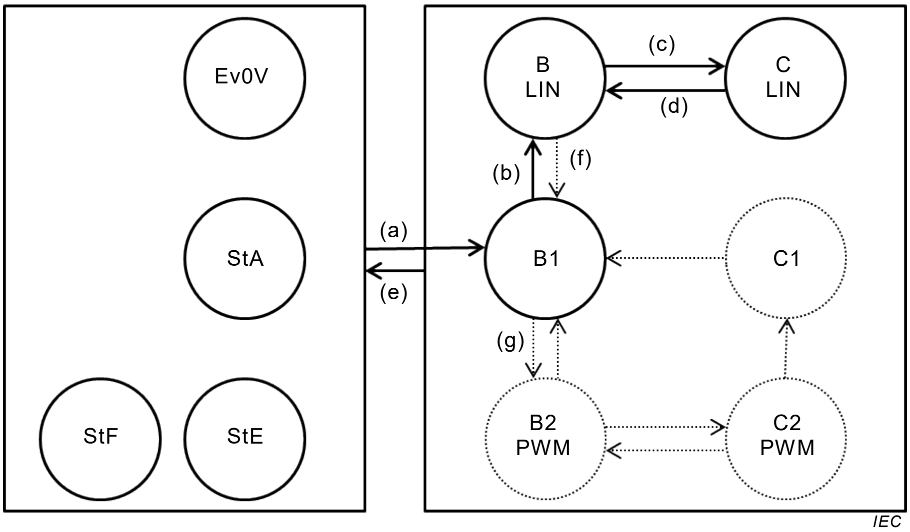


Figure D.9 – Control pilot circuit state diagram for LIN-CP and PWM-CP (See key list in Table D.5)

D.9.5 Sequence to select LIN-CP or PWM-CP after plug-in

If the charging station detects CP voltage level C after plug-in (not shown in Figure D.9), this indicates that the EV has simplified control pilot function. If this is supported by the charging station, it shall select to use PWM-CP. If this is not supported, the charging station shall not supply power.

If the charging station detects CP voltage level B after plug-in, the charging station shall transition to state B LIN and start sending LIN headers. See Table D.6 line 1 and D.8.2.2.

In state B LIN, if the EV responds to the LIN headers by sending LIN responses (see Table D.6 line 2 and D.8.2.2) the charging station shall continue to use LIN communication, as described in Clause D.8.

In state B LIN, if the EV does not respond to the LIN headers (see Table D.6 line 5), the charging station shall repeat the following steps, until the EV either closes S2 (in step 3b below) or sends LIN responses (in step 5 below):

1) stop sending LIN headers to transition to state B1;
2) wait in state B1 for min 100 ms and max 500 ms;
3) if power is available, then
   a) start sending PWM signals to transition to state B2,
   b) stay in state B2 for min 30 s and max 60 s,
   c) stop sending PWM signals to transition to state B1,
   d) wait in state B1 for min 4 s and max 10 s.
4) start sending LIN headersto transition to state B LIN;
5) stay in state B LIN for min 5 s and max 10 s;
6) stop sending LIN headers to transition to state B1;
7) wait in state B1 for min 4 s and max 10 s;
8) repeat from 3).

D.10 Procedures for test of charging stations

D.10.1 General

The following tests verify the required system behaviour and timing according to Table D.6, Table D.7 and Table D.8.

Test equipment that can emulate the EV behaviour in LIN-CP and also complies with applicable requirements in A.4.2 "Constructional requirements of the EV simulator" shall be used for the tests. An example of a circuit for an EV simulator is shown in A.4.10.

D.10.2 Test of normal use

This test covers Table D.6 lines 1, 2 and 4, Table D.7 lines 3 and 4, Table D.8 lines 1 and 2.

Cycle steps:

1) Plug-in → Charging station starts LIN (Table D.6 line 1) → EV sends LIN response (Table D.6 line 2) →
2) Initialization → *StStatusOp* = *ready* and S2 closed (Table D.7 line 1)→ Switching device closed (Table D.8 line 1)→ S2 opened (Table D.7 line 2) → Switching device opened (Table D.8 line 2) → S2 closed (Table D.7 line 1) → Switching device closed (Table D.8 line 1) →
3) Charging station reduces the signals for "maximum allowable current" (Table D.6 line 4), the EV reduces the vehicle load current (Table D.7 line 4) →*StStatusOp* = *not_ready*: S2 opened (Table D.7 line 3) → Switching device opened (Table D.8 line 2) →
4) Unplug

Three complete normal cycles including all steps 1 to 4 shall be performed using the resistor values for R3 and R2 indicated for Tests 1, 2 and 3 in Table D.25. R_leak shall be a (11 ± 1 %) kΩ resistor connected between the control pilot conductor and the protective conductor in such a way that it is still connected to the charging station when the EV is unplugged. The charging station shall be deemed to have failed the test if a cycle is not completed.

Table D.25 – Normal charge cycle test

	R3_test (Ω)	**R2_test** (Ω)	**R_leak** (Ω)
Test 1	4 610	1 723	not connected
Test 2	1 870	909	not connected
Test 3	2 740	1 300	1 1000
Resistances tolerance is at least ± 0,2 %.			

D.10.3 Test of disconnection under load

This test covers Table D.8 line 3.

The EV simulator is used with setting according to Test 3 in Table D.25. The R_leak shall still be connected to the charging station when the control pilot conductor or the protective conductor is interrupted.

The steps 1 and 2 of the sequence defined in D.10.2 are carried out twice, followed by:

1) Interruption of control pilot conductor, or
2) Interruption of protective conductor.

D.10.4 Overcurrent test

This test covers Table D.9 Lines 6 and 7.

The EV simulator is used. Steps 1 and 2 of sequence defined in D.10.2 are carried out.

The following conditions are tested:

1) (Table D.8 line 7) Vehicle load current exceeds indicated current by less than 10 %; the charging station shall not shut off power.
2) (Table D.8 line 7, optional) Vehicle load current exceeds indicated current by more than 10 % ; the charging station shall shut off power.
3) (Table D.8 line 6) EV does not respond to signal *StStatusOp* = *not_ready*; EV supply equipment shall shut off power.

Test 2 is only carried out if the charging station is designed to detect overcurrent.

D.10.5 Test of interruption of LIN communication

This test covers Table D.7 line 5 and Table D.8 line 5.

Charging station simulator and EV simulator are used. Steps 1 and 2 of the sequence defined in D.10.2 are carried out.

The following conditions are tested:

1) Charging station stops sending LIN headers;
2) EV stops sending LIN responses.

D.10.6 Test of short circuit between the control pilot conductor and the protective conductor

This test covers Table D.8 line 4.

EV simulator is used. Steps 1 and 2 of sequence defined in D.10.2 are carried out.

A supplementary resistance of 120 Ω is switched to connect the control pilot conductor and the protective conductor.

D.10.7 Test of options

A combined controller which implements both LIN-CP and PWM-CP shall also be tested according to Clause A.4.

Annex E
(informative)

Charging station designed with a standard socket-outlet – Minimum gap for connection of Modes 1 and 2 cable assembly

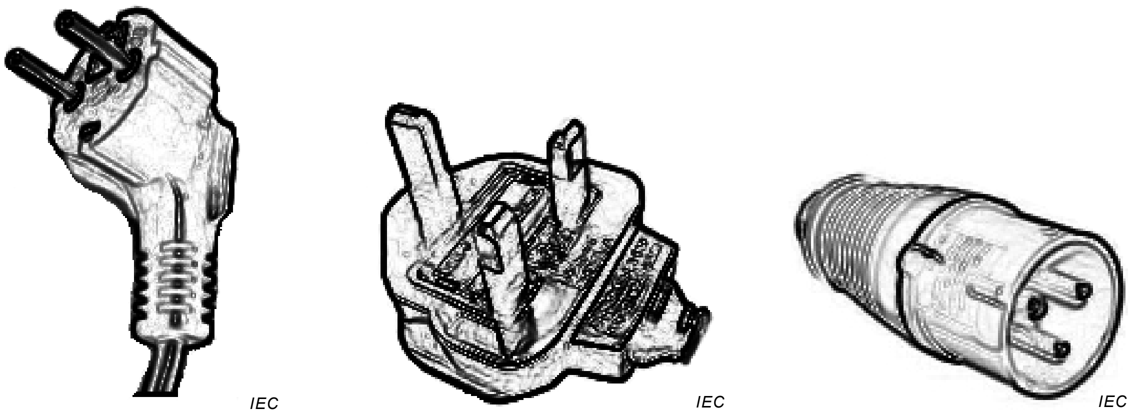
E.1 Overview

Easy connection of the Mode 1 and Mode 2 cable assembly equipped with any standard plug of the market cannot be guaranteed on an EV charging station, due to the absence of standardized sheets which allow the design of enclosure dimensions around the Mode 1 or Mode 2 connecting point for all standard plugs and socket-outlets.

This Annex E is therefore aimed at providing a recommended minimum space around the socket-outlet of an EV charging station enclosure in order to allow Mode 1 and Mode 2 cable assembly manufacturers to design their products.

This Annex E is informative, but should be used as a basis for interoperability, as it is already defined in IEC 62196-2 for Mode 3 couplers, with the "packaging rooms".

Figure E.2 shows different packaging configurations that allow the use of a large part of the common products for standard plugs and socket-outlets. There is no guarantee that all possible standard plugs and sockets are covered.



a) CEE (Type E/F) **b) BS 1363 (Type G)** **c) IEC 60309-2 (16 A single Phase)**

NOTE For Type E/F and G see IEC website:http://www.iec.ch/worldplugs/

Figure E.1 – Examples of standard plugs that are considered for this Annex E

E.2 General

The dimensions and volumes have been defined to permit plugging in straight plugs as well as 90°angle plugs. According to the national standards in force, either both cases shall be considered or just one case.

The free volume applies to each Mode 2 or Mode 1 connection point of the charging station and when the design is provided as follows:

– a recess is implemented,
– a recess with a covering system is implemented,
– a recess is not implemented.

The depth is defined as a minimum, if a covering system (e.g. flap, lid) is implemented. If this is not the case, the depth is just given for information and has no functional role.

When the charging station enclosure is equipped with a covering system (e.g. flap, lid) a pathway for the cable shall be provided and so designed that its position, when the covering system is closed, does not create undue strain on the interface contacts independent of the type of plug used.

The cable passage shall allow for cables of up to 20 mm in diameter.

For a Mode 2 cable assembly, the cable length between the ICCB and the plug shall be at least 250 mm.

A “clear zone” shall be prepared to provide enough room in order to let the “ICCB” fit into without obstacle, independently of the free volume.

If the reference plane is not vertical by design, the complete drawing shall be rotated by the imposed angle.

E.3 Minimum gap for connection of Mode 2 cables with type E/F plug and socket-outlet systems

Dimensions of the free volume are specified in the Figure E.2 with the following precisions:

– height: 50 mm minimum from the main longitudinal axe of the plug to any obstacle above, and 60 mm minimum from the main longitudinal axe of the plug to any obstacle below;
– depth: 80 mm minimum from the engagement face of the plug and the internal face of the covering system when it is closed, if there is not a closing system, depth can be less than 80 mm;
– width: 60 mm minimum from the main longitudinal axe of the plug to any obstacles on sides.

E.4 Minimum gap for connection of Mode 2 cables with type BS1363 plug and socket-outlet systems

Dimensions of the free volume are specified in the Figure E.2 with the following precisions:

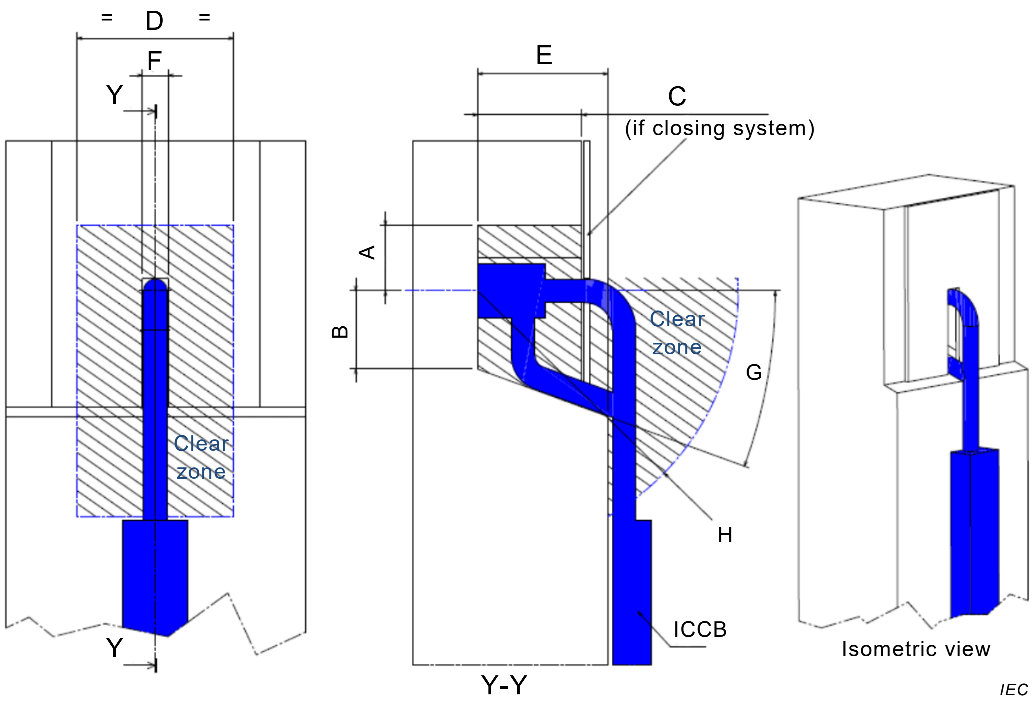
– height: 50 mm minimum from the main longitudinal axe of the plug to any obstacle above, and 60 mm minimum from the main longitudinal axe of the plug to any obstacle below;
– depth: 35 mm minimum from the engagement face of the plug and the internal face of the covering system when it is closed, if there is not a closing system, depth can be less of 35 mm;
– width: 60 mm minimum from the main longitudinal axe of the plug to any obstacles on sides.

E.5 Minimum gap for connection of Mode 2 cables with IEC 60309-2 straight plug and socket-outlet systems

Dimensions of the free volume are specified in the Figure E.2 with the following precisions:

– only the straight IEC 60309-2 plug rated up to 16/20 A is considered;

– height: 50 mm minimum from the main longitudinal axe of the plug to any obstacle above, and 60 mm minimum from the main longitudinal axe of the plug to any obstacle below;
– depth: 120 mm minimum from the engagement face of the plug and the internal face of the covering system when it is closed, if there is not a closing system, depth can be less of 35 mm;
– width: 60 mm minimum from the main longitudinal axe of the plug to any obstacles on sides.



Key dimension	Type E/F	Type G	Type IEC 60309-2 16A, 20A (straight)
A	50 mm minimum	50 mm minimum	50 mm minimum
B	60 mm minimum	60 mm minimum	60 mm minimum
C	80 mm minimum	35 mm minimum	120 mm minimum
D	120 mm minimum	120 mm minimum	120 mm minimum
E	100 mm MAXIMUM	100 mm MAXIMUM	100 mm MAXIMUM
F	20 mm minimum	20 mm minimum	20 mm minimum
G	20° minimum	20° minimum	20° minimum
H	R200 mm minimum	R200 mm minimum	R200 mm minimum

Figure E.2 – Packaging configurations allowing the use of a large part of the common products for standard plugs and socket-outlets

Figure E.2 shows a horizontal cable connection as an example only. Such a configuration can exert excessive strain on the cable and the contacts. Other configurations may be considered (e.g. placing the socket-outlet at an angle).

Bibliography

Standards directly applicable to this document

IEC 62053-21:2003, *Electricity metering equipment (a.c.) – Particular requirements – Part 21: Static meters for active energy (classes 1 and 2)*

ISO 4628-3:2016, *Paints and varnishes – Evaluation of degradation of coatings – Designation of quantity and size of defects, and of intensity of uniform changes in appearance – Part 3: Assessment of degree of rusting*

EN 50065-1:2001, *Signalling on low-voltage electrical installations in the frequency range 3 kHz to 148,5 kHz – Part 1: General requirements, frequency bands and electromagnetic disturbances*

EN 50470-1:2006, *Electricity metering equipment (AC) – Part 1: General requirements, tests and test conditions – Metering equipment (class indexes A, B and C)*

EN 50470-3:2006, *Electricity metering equipment (AC) – Part 3: Particular requirements – Static meters for active energy (class indexes A, B and C)*

EN 50557:2011, *Requirements for automatic reclosing devices (ARDs) for circuit breakers-RCBOs-RCCBs for household and similar uses*

Other standards that can be of use as references

IEC 60050-151:2001, *International Electrotechnical Vocabulary – Part 151: Electrical and magnetic* devices
IEC 60050-151:2001/AMD1:2013
IEC 60050-151:2001/AMD2:2014

IEC 60050-195:1998, *International Electrotechnical Vocabulary – Part 195: Earthing and protection against electric shock*
IEC 60050-195:1998/AMD1:2001

IEC 60050-441:1984, *International Electrotechnical Vocabulary – Part 441: Switchgear, controlgear and fuses*
IEC 60050-441:1984/AMD1:2000

IEC 60050-442:1998, *International Electrotechnical Vocabulary – Part 442: Electrical* accessories
IEC 60050-442:1998/AMD1:2015
IEC 60050-442:1998/AMD2:2015

IEC 60050-826:2004, *International Electrotechnical Vocabulary – Part 826: Electrical installations*

IEC 60063:2015, *Preferred number series for resistors and capacitors*

IEC 60068-2-2, *Environmental testing – Part 2-2: Tests – Test B: Dry heat*

IEC 60068-2-5:2010, *Environmental testing – Part 2-5: Tests – Test Sa: Simulated solar radiation at ground level and guidance for solar radiation testing*

IEC 60068-2-6:2007, *Environmental testing – Part 2-6: Tests – Test Fc: Vibration (sinusoidal)*

IEC 60068-2-14:2009, *Environmental testing – Part 2-14: Tests – Test N: Change of temperature*

IEC 60068-2-27:2008, *Environmental testing – Part 2-27: Tests – Test Ea and guidance: Shock*

IEC 60068-2-52:1996, *Environmental testing – Part 2: Tests – Test Kb: Salt mist, cyclic (sodium, chloride solution)*

IEC 60068-2-53:2010, *Environmental testing – Part 2-53: Tests and guidance – Combined climatic (temperature/humidity) and dynamic (vibration/shock) tests*

IEC 60068-2-75, *Environmental testing – Part 2-75: Tests – Test Eh: Hammer tests*

IEC TR 60083, *Plugs and socket-outlets for domestic and similar general use standardized in member countries of IEC*

IEC 60245-1, *Rubber insulated cables – Rated voltages up to and including 450/750 V – Part 1: General requirements*

IEC 60245-2, *Rubber insulated cables – Rated voltages up to and including 450/750 V – Part 2: Test methods*

IEC 60245-3, *Rubber insulated cables – Rated voltages up to and including 450/750 V – Part 3: Heat resistant silicone insulated cables*

IEC 60245-4, *Rubber insulated cables – Rated voltages up to and including 450/750 V – Part 4: Cords and flexible cables*

IEC 60245-6:1994, *Rubber insulated cables – Rated voltages up to and including 450/750 V – Part 6: Arc welding electrode cables*
IEC 60245-6:1994/AMD1:1997
IEC 60245-6:1994/AMD2:2003

IEC 60364-5-53:2001, *Electrical installations of buildings – Part 5-53: Selection and erection of electrical equipment – Isolation, switching and control*
IEC 60364-5-53:2001/AMD1:2002
IEC 60364-5-53:2001/AMD2:2015

IEC 60364-6:2016, *Low-voltage electrical installations – Part 6 Verification*

IEC TS 60479-1:2005, *Effects of current on human beings and livestock – Part 1: General aspects*

IEC/TR 60755:2008, *General requirements for residual current operated protective devices*

IEC 60884-2-5, *Plugs and socket-outlets for household and similar purposes – Part 2 particular requirements for adaptors*

IEC 60947-1:2007, *Low-voltage switchgear and controlgear – Part 1: General rules*
IEC 60947-1:2007/AMD1:2010
IEC 60947-1:2007/AMD2:2014

IEC 60947-6-1:2005, *Low-voltage switchgear and controlgear – Part 6-1: Multiple function equipment – Transfer switching equipment*

IEC 61140, *Protection against electric shock – Common aspects for installations and equipment*

IEC 61439-1:2011, *Low-voltage switchgear and controlgear assemblies – Part 1: General rules*

IEC 61540, *Electrical accessories – Portable residual current devices without integral overcurrent protection for household and similar use (PRCDs)*

IEC 61558-1:2005, *Safety of power transformers, power supplies, reactors and similar products – Part 1: General requirements and tests*
IEC 61558-1:2005/AMD1:2009

IEC 61558-2-4:2009, *Safety of transformers, reactors, power supply units and similar products for supply voltages up to 1 100 V – Part 2-4: Particular requirements and tests for isolating transformers and power supply units incorporating isolating transformers*

IEC 61558-2-12:2011, *Safety of transformers, reactors, power supply units and combination thereof – Part 2-12: Particular requirements and tests for constant voltage transformers and power supply units for constant voltage*

IEC 61558-2-16:2009, *Safety of transformers, reactors, power supply units and similar products for supply voltages up to 1 100 V – Part 2-16: Particular requirements and tests for switch mode power supply units and transformers for switch mode power supply units*
IEC 61558-2-16:2009/AMD1:2013

IEC 61851-3 (all parts) [12], *Electric vehicles conductive power supply system*

IEC 61851-21-2 [13] , *Electric vehicle conductive charging system – Part 21-2: EMC requirements for OFF board electric vehicle charging systems*

IEC 62893 [14], *Charging cables for electric vehicles- Part 3: Cables for AC charging according to modes 1, 2 and 3 of IEC 61851-1*

IEC 61980-1, *Electric vehicle wireless power transfer (WPT) systems – Part 1: General requirements*

IEC 62262:2002, *Degrees of protection provided by enclosures for electrical equipment against external mechanical impacts (IK code)*

IEC 62335, *Circuit breakers – Switched protective earth portable residual current devices for class I and battery powered vehicle applications*

ISO/IEC 15118 (all parts), *Road Vehicles – Vehicle to grid communication interface*

ISO 6469-2:2009, *Electrically propelled road vehicles – Safety specifications – Part 2: Vehicle operational safety means and protection against failures*

ISO 13849-1:2015, *Safety of machinery – Safety-related parts of control systems – Part 1: General principles for design*

---

[12] Under consideration.

[13] Under consideration.

[14] Under consideration.

ISO 15118-3, *Road vehicles – Vehicle to grid communication interface – Part 3: Physical and data link layer requirements*

ISO 16750-3:2012, *Road vehicles – Environmental conditions and testing for electrical and electronic equipment – Part 3: Mechanical loads*

ISO 16750-4:2010, *Road vehicles – Environmental conditions and testing for electrical and electronic equipment – Part 4: Climatic loads*

ISO 17987-5 [15], *Road vehicles – Local interconnect network (LIN) – Part 5: Application Programmers Interface (API)*

ISO 17987-6 [16], *Road vehicles – Local interconnect network (LIN) – Part 6: Protocol conformance test specification*

ISO 17987-7 [17], R*oad vehicles – Local interconnect network (LIN) – Part 7: Electrical Physical Layer (EPL) conformance test specification*

SAE J1772:2016, *SAE Electric Vehicle and Plug-In Hybrid Electric Vehicle Conductive Charge Coupler*

LIN Specification2.2A:2010. LIN consortium (http://www.lin-subbus.org/)

NOTE LIN Specification 2.2.A (2010) from the LIN consortium (http://www.lin-subbus.org/) will be transcribed to the future ISO 17987-1.

SEK TS 481 05 16, *Control pilot function that provides LIN communication using the control pilot circuit, with additional annexes* (under development)

SAE J3068, *Electric Vehicle Power Transfer System Using a Three-phase Capable Coupler* (under development)

[15] To be published.

[16] To be published.

[17] To be published.

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